

Computer Systems

Martin

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1 Data Representation and Manipulation

1.1 Binary

Computers consist of a collection of switches called transistors, which can either be on or off. This leads to binary representation, where off is 0 and on is 1. All data and instructions are stored in binary. Binary digits are known as 'bits'. It is possible to build computers that use other number systems, but it is less expensive and more reliable to use basic components with only two states.

A binary string is a string of any length that consists only of bits. A 1-bit string is called a bit. An 8-bit string is called a byte. (Less common, but very cute: a 4-bit string is called a nibble.)

How many possible byte strings are there? Possible 1-bit strings = 2 = 2^1 Possible 2-bit strings = 4 = 2^2 Possible 3-bit strings = 8 = 2^3 Possible 4-bit strings = 16 = 2^4 Possible byte strings = 2^8 = 256

So, if we are limited to a single byte to represent something, we can represent 256 different things.

1.2 Characters

Characters are represented as binary values using text encoding schemes such as ASCII and Unicode. Modern schemes support internationalisation and tend to use 8 bit, 16 bit or 32 bit to encode characters. Strings are sequences of characters.

1.3 Number Bases

1.3.1 Decimal to Binary, Octal and Hexadecimal

$$\begin{array}{l} 234_{10} : \\ \left. \begin{array}{l} 2 \overline{) 234} \quad 0 \\ 2 \overline{) 117} \quad 1 \\ 2 \overline{) 58} \quad 0 \\ 2 \overline{) 29} \quad 1 \\ 2 \overline{) 14} \quad 0 \\ 2 \overline{) 7} \quad 1 \\ 2 \overline{) 3} \quad 1 \\ 2 \overline{) 1} \quad 1 \end{array} \right\} = 11101010_2 \end{array} \qquad \begin{array}{l} 234_{10} : \\ \left. \begin{array}{l} 8 \overline{) 234} \quad 2 \\ 8 \overline{) 29} \quad 5 \\ 8 \overline{) 3} \quad 3 \end{array} \right\} = 352_8 \end{array} \qquad \begin{array}{l} 234_{10} : \\ \left. \begin{array}{l} 16 \overline{) 234} \quad 10 \\ 16 \overline{) 14} \quad 14 \end{array} \right\} = EA_{16} \end{array}$$

1.3.2 Binary, Octal and Hexadecimal to Decimal

Method 1: Left-to-right subtraction

To convert to binary, we repeatedly subtract powers of two from the number, from left to right, until we reach 0. 214 in binary is 011010110 256 128 64 32 16 8 4 2 1 0 1 1 0 1 0 1 1 0

Method 2: Right-to-left division.

$101011_2 :$

$$\begin{aligned}1 \times 2^0 &= 1 \\1 \times 2^1 &= 2 \\0 \times 2^2 &= 0 \\1 \times 2^3 &= 8 \\0 \times 2^4 &= 0 \\1 \times 2^5 &= \underline{32} \\&= 43_{10}\end{aligned}$$

$724_8 :$

$$\begin{aligned}4 \times 8^0 &= 4 \\2 \times 8^1 &= 16 \\7 \times 8^2 &= \underline{448} \\&= 468_{10}\end{aligned}$$

$ABC_{16} :$

$$\begin{aligned}C \times 16^0 &= 12 \\B \times 16^1 &= 176 \\A \times 16^2 &= \underline{2560} \\&= 2748_{10}\end{aligned}$$

1.3.3 Binary to Octal and Octal to Binary

$$1011010111_2 = 1327_8 :$$

1 011 010 111

1 3 2 7

$$705_8 = 111000101_2 :$$

7 0 5

111 000 101

1.3.4 Binary to Hexadecimal and Hexadecimal to Binary

$$1010111011_2 = 2BB_{16} :$$

10 1011 1011

2 B B

$$10AF_{16} = 1000010101111_2 :$$

1 0 A F

0001 0000 1010 1111

1.3.5 Octal to Hexadecimal and Hexadecimal to Octal

Conversions between octal and hexadecimal can be performed by first converting to binary.

1.4 Integers in Binary

Counting in powers of 2.

Prefix	Power of 2	Value
kibi	2^{10}	1024
mebi	2^{20}	1048576
gibi	2^{30}	1073741824
tebi	2^{40}	1099511628000

1.4.1 Overflow

In a computer, an integer is represented by a fixed number of bits. The maximum value that can be stored in an unsigned integer of n bits is $2^n - 1$. It is possible that the addition of two n bit numbers yields a result that requires $n + 1$ bits.

In Java, the 'overflow' bits are lost with no error. The remaining bits give the wrong answer. It is important to make sure the data type used is big enough for the values it will represent.

1.4.2 Two's Complement

In 8 bit arithmetic, $255 + 1$ appears to be 0 due to overflow. In this case, 255 is behaving like -1 . This leads to the 'two's complement' representation of negative integers in binary.

The two's complement representation of $-x$ is equivalent to the unsigned binary representation of $2^n - x$. Another method to find the representation is to flip all the bits of the binary representation of x and add 1 to the least significant bit.

Using two's complement, a signed binary integer of n bits can hold any value from -2^{n-1} to $2^{n-1}-1$, inclusive. The most significant bit represents -2^{n-1} . If the number of bits in a signed number is increased, the new bits are given the same value as the most significant bit. This is known as sign extension.

In Java, all integers are signed.

1.5 Representing Colour

A mixture of primary colours! In additive (RGB) colour models, every colour can be modelled as a mixture of red, green and blue, which is added to black. In subtractive (CMY) colour models, every colour can be modelled as a mixture of cyan, magenta and yellow, which is subtracted from white.

So how should a given 24-bit binary word relate to a colour, which is a mixture of red, green and blue? Let's split the 24 bits into three bytes. The first byte represents red, the next represents green and the last represents blue. Graph-80

1.6 Representing Text

ASCII is still used today, and using only 7 bits means that text files remain very lightweight. 8 bits in a byte, 1000 bytes in a Kilobyte, 1000 Kilobytes in a Megabyte.

1.7 Real Numbers in Binary

1.7.1 Fixed Point Decimal to Binary

The integral part is converted as usual. The fractional part is converted by doubling as follows.

$$0.537_{10} = 0.100010_2 :$$

$$0.537 \times 2 = \underline{1}.074$$

$$0.074 \times 2 = \underline{0}.148$$

$$0.148 \times 2 = \underline{0}.296$$

$$0.296 \times 2 = \underline{0}.592$$

$$0.592 \times 2 = \underline{1}.184$$

$$0.184 \times 2 = \underline{0}.368$$

1.7.2 Fixed Point Binary to Decimal

1101.0101₂ :

$$1 \times 2^{-4} = 0.0625$$

$$0 \times 2^{-3} = 0$$

$$1 \times 2^{-2} = 0.25$$

$$0 \times 2^{-1} = 0$$

$$1 \times 2^0 = 1$$

$$0 \times 2^1 = 0$$

$$1 \times 2^2 = 4$$

$$1 \times 2^3 = 8$$

$$= 13.3125_{10}$$

1.7.3 Floating Point Numbers

Fixed point is convenient and intuitive, but has two major problems.

1. Numerical precision — only values that are multiples of the smallest used power of two can be represented.
2. Numerical range — fractional precision comes at the expense of numerical range.

Floating point representation in binary is similar to scientific notation in decimal. In binary, real numbers can be represented in the form $\pm m \times 2^e$. Floating point numbers consist of a sign bit ($\pm$, 0 for positive or 1 for negative), a mantissa (m , the significant bits) and an exponent (e , two's complement signed binary).

S	Offset exponent, e'	Normalised mantissa, m'
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The stored representation of a floating point number.

Since the mantissa is normalised (unless the value is small enough to represent without an exponent), the leading bit is almost always 1. It is therefore omitted from the stored representation. The exponent has a bias (offset) of $2^{n-1} - 1$ added to it for engineering purposes.

Floating point data types in Java.

Type	Bits				Bytes	Exponent bias
	Sign	Mantissa	Exponent	Total		
float	1	23	8	32	4	127
double	1	52	11	64	8	1023

With the 52 bit mantissa of a double (and the additional hidden bit), $2^{53} \approx 8 \times 10^{15}$. Hence, the double data type can be used to represent 15 significant decimal digits.

In Java, special values are returned for floating point overflow and other unusual circumstances. For example, if the result is too large for the double data type, `Double.POSITIVE_INFINITY` or `Double.NEGATIVE_INFINITY` are returned. If the result is indistinguishable from zero but known to be negative, `-0.0` is returned. If the result is not a real number, `Double.NaN` is returned.

1.7.4 Numerical Precision

Floating point arithmetic loses accuracy in its less significant digits. This is an issue even in values of type double.

It is a bad idea to use floating point representation for money — i.e. with integral pounds and fractional pence — as this will result in rounding errors. Taking £0.10 as an example, $0.10_{10} = 0.0001\overline{1}_2$.

Since currency is inherently integral, integer types with suitable range, such as `int`, `long` or `java.math.BigInteger`, should be used to represent multiples of the smallest denomination.

2 Introduction to Computer Architecture

2.1 Anatomy of a Computer System

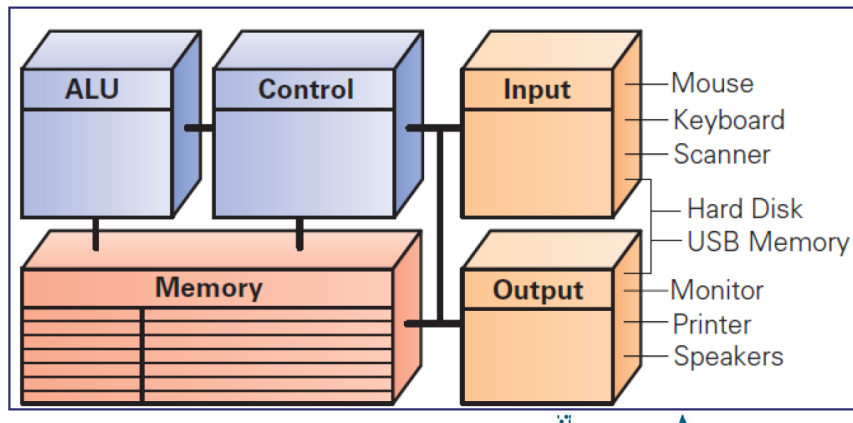
A computer is a complex system (machine) that can be instructed to carry out sequences of arithmetic or logical operations automatically via computer programming.

A computing device must be able to

- load a program (input interface),
- process instructions in the correct order (track progress, storage and decoding of instructions),
- access data according to its instructions (local storage),
- perform computations (calculation 'engine'),
- make decisions according to its computations (control mechanism), and
- send results to an external device (output interface).

Thus, all computers, regardless of their implementing technology, have five basic subsystems.

1. Memory
2. Control unit
3. Arithmetic logic unit (ALU)
4. Input unit
5. Output unit



Anatomy of a Computer System

2.1.1 Memory

Memory stores both the program while it is running and the data on which the program operates: Memory is organized as a sequence of discrete locations. In modern memory, each location is composed of 1 byte(8 bits).

Memory locations have finite capacity. Data may not fit in a memory location. Blocks of four or eight bytes are used so often as a unit that they are known as memory 'words'.

Computer memory is known as random access memory (RAM). This simply means that the computer can refer to memory locations in any order.

2.1.2 Control unit

The control unit of a computer is where the fetch/execute cycle occurs. It fetches a machine instruction from memory and performs other operations of the fetch/execute cycle accordingly. A typical machine instruction has the form: ADD 4000, 2000, 2080 What it really means is that whatever numbers are stored in memory locations 2000 and 2080 be added together, and the result be stored in location 4000

2.1.3 Arithmetic Logic Unit (ALU)

The arithmetic logic unit carries out each machine instruction with a separate circuit. It contains various circuits for arithmetic and logic, such as addition, subtraction, multiplication, comparison and logic gates.

2.1.4 Input and Output Units

These components are the wires and circuits through which information travels into and out of a computer. Without input or output, a computer is useless.

Peripherals connect to the computer through input/output (I/O) ports. They are not considered parts of the computer.

2.2 The Fetch-Execute Cycle

A program is loaded into memory and the address of the first instruction is placed in the program counter (PC).

The fetch-execute cycle proceeds as follows.

1. Instruction fetch (IF)
2. Instruction decode (ID)
3. Data fetch (DF) / operand fetch (OF)
4. Instruction execution (EX)
5. Result return (RR) / store (ST)

2.2.1 Instruction Fetch (IF)

The instruction at the memory address given by the PC is copied to the instruction register of the control unit. The PC is incremented to point at the next instruction to be fetched.

2.2.2 Instruction Decode (ID)

The ALU is prepared for the operation specified by the instruction. Decoder determines what operation the ALU will perform (ADD), and sets up the ALU. The decoder finds the addresses of the instruction operands. These addresses are passed to the ALU circuit that fetches the operands from memory in the next stage. The decoder also finds the destination address for the result. This is passed to the RR circuit.

2.2.3 Data fetch (DF)

The operands are copied from the specified memory addresses into the ALU circuits. The data remains in memory and is not destroyed. They are not destroyed).

2.2.4 Instruction Execute (EX)

The ALU performs the operation on its operands. The result is held in its circuitry.

2.2.5 Result Return (RR)

The result of the EX stage is stored at the specified destination memory address. The cycle begins again.

2.3 Machine Instructions

A computer “knows” very few instructions. The decoder hardware in the control unit recognises, and the ALU performs, of the order of 100 instructions. Everything that a computer does must be reduced to some combination of these primitive instructions. Computers can carry out millions of these instructions per second.

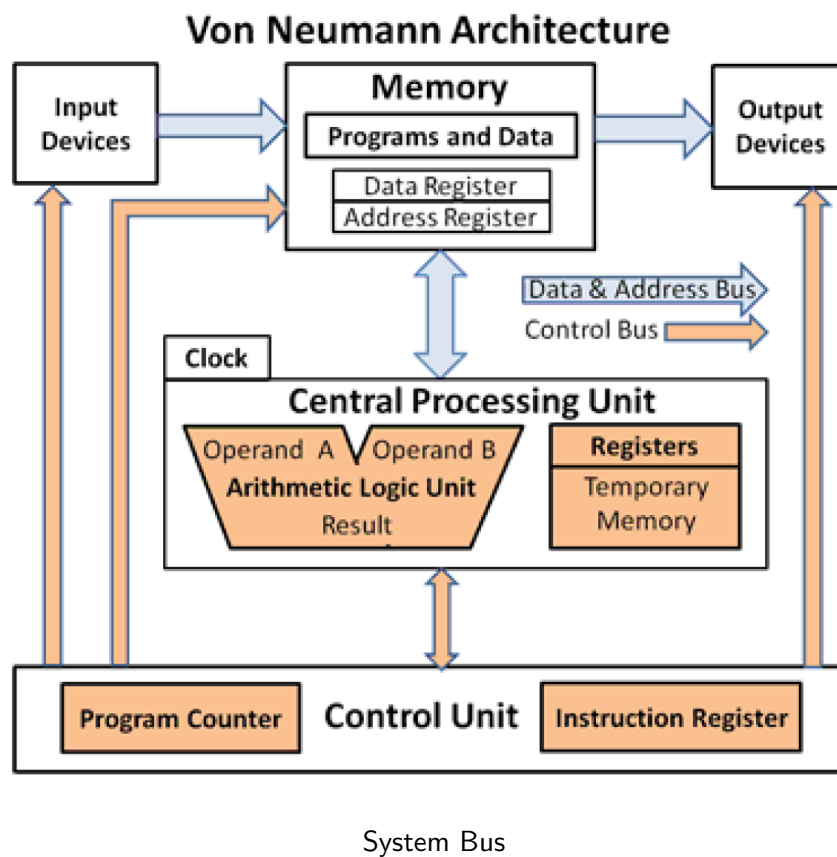
2.4 Summary

All computers, regardless of their implementing technology, have five basic parts or subsystems: The memory, a very long sequence of bytes, each with an address, stores the program and data while the program is running. The ALU does the actual computing. The Control Unit controls the activity of all other units in the computer system, giving them directions. The input and output units are the interfaces for the peripheral devices connected to the computer.

A program is executed according to the Fetch/Execute Cycle: The repeating process fetches each instruction (indicated by the PC), decodes the operation, retrieves the data, performs the operation, and stores the result back into the memory.

3 Instructions, Assembly Language and Machine Code

3.1 The von Neumann Architecture



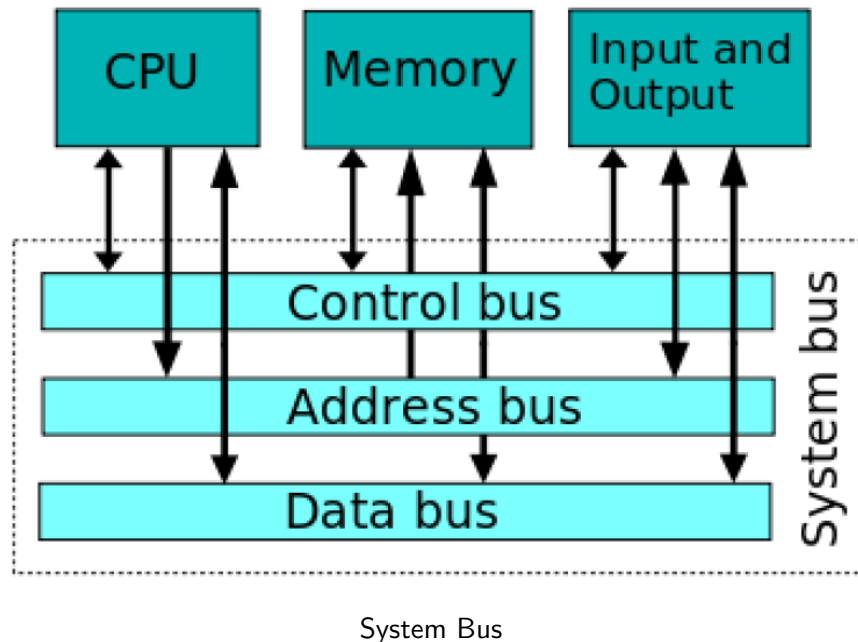
3.1.1 Central Processing Unit (CPU)

The central processing unit (CPU) consists of a **control unit**, **ALU** and **registers**. These registers include

- the program counter (PC) (holds the address of the next instruction),
- the instruction register (holds the instruction currently being decoded or executed),

- the address register (Either stores the memory address from which data will be fetched, or the address to which data will be sent and stored.)
- the accumulator (holds temporarily the results of ALU computations).

3.1.2 System Bus



3.1.3 Memory

The system bus includes

- a control bus (carries commands from the CPU and returns status signals from devices),
- an address bus (carries information about the device with which the CPU is communicating, namely physical memory addresses), and
- a data bus (carries the data being processed).

3.1.4 Memory

In the von Neumann architecture, the memory contains both program instructions and data.

3.1.5 Clock

A clock cycle is a signal that oscillates between high and low. It is used to coordinate actions. The rate of the fetch-execute cycle is determined by the clock. Instructions begin execution on the rising edge of the signal. The time between rising edges is known as the 'clock period'. The time between the rising and falling edges of the signal is known as the 'clock width'. The number of clock cycles per second is used to determine the speed of a CPU.

Modern computers attempt to start an instruction on each clock tick. Since each stage of the fetch-execute cycle is handled by separate circuitry, the fetch unit is freed to start the next instruction

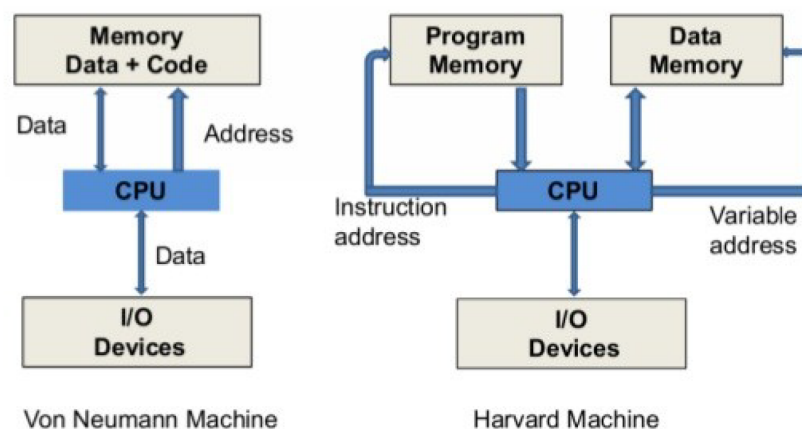
before the previous instruction is complete. This process is known as 'pipelining'. When the pipeline is filled, five instructions are in process at a time, each at a different stage of the fetch-execute cycle. Additionally, one instruction is finished on each clock tick, making it appear as though the computer is running one instruction per tick.

3.2 The Harvard Architecture

The main difference between the von Neumann and Harvard architectures is their storage of instructions and data in memory.

A von Neumann (or 'stored-program') machine stores its instructions and data in a shared memory. This means that an instruction fetch and a data operation cannot occur at the same time because they share a common bus. This is known as the 'von Neumann bottleneck' and can impinge on the performance of the system. However, the architecture allows for self-modifying code that can tune its performance at runtime.

A Harvard machine stores its instructions and data in separate memories. This makes the machine more complex, but allows instruction fetch and data operation to occur at the same time.



System Bus

3.3 MIPS R4000

MIPS R4000 uses the modified Harvard architecture. Instructions and data are stored in the same memory (as in von Neumann), but they have separate interfaces (as in Harvard).

MIPS R4000 was one of the first 64 bit microprocessors. It features integrated caches (primary on-chip and secondary off-chip), an integrated floating point unit and a deep pipeline. It uses MIPS III — a compact instruction set with a regular format — which makes it easy to convert from high-level code to machine code.

3.3.1 Registers

The CPU contains 32 registers denoted \$0–\$31 (or r0–r31). The registers can be either 32 bit or 64 bit wide, depending on the mode of operation. \$0 (r0) always stores zero. \$31 (r31) is the link register used by jump and link instructions. It should not be used by other instructions.

The ALU takes input from up to two registers and sends output to registers only.

3.3.2 Deep Pipeline

The typical MIPS pipeline consists of the following five stages.

1. Instruction fetch (IF)
2. Instruction decode (ID) and register fetch (RF)
3. Execute (EX)
4. Memory access (MEM)
5. Write back (WB)

MIPS R4000 is super-pipelined; its pipeline contains eight stages rather than the regular five. The additional stages exist due to the extension of the IF and MEM stages to account for cache overheads. When the pipeline is full, eight instructions are in process at a time — i.e. the pipeline is eight-deep.

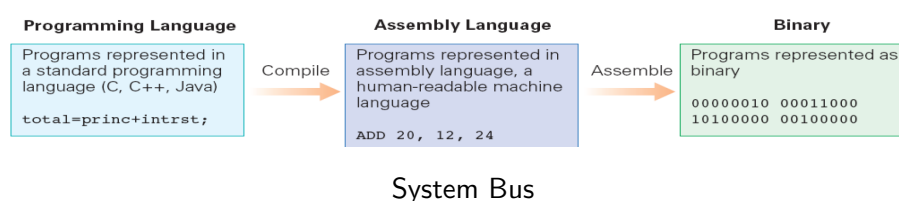
3.4 Instruction Sets

Programs are composed of complex combinations of simple “instructions”. Every machine has a unique instruction set architecture (ISA). This is the set of primitive instructions that the CPU can execute.

3.5 Assembly Language

Binary is the only form in which a computer can be given instructions. Assembly language is a human-readable alternative form of machine language that uses letters and normal numbers so people can understand it. Computer scans assembly code. As it encounters words it looks them up in a table to convert to binary. Converts numbers to binary, then assembles (puts together) the binary pieces into an instruction.

Most modern software is written in a high-level programming language. High-level languages are compiled (translated) into assembly language, which is then assembled into binary. These languages are preferred because they have special statement forms that help programmers describe the complicated tasks they want done.



MIPS is a RISC (Reduced Instruction Set Computer) architecture. In MIPS, there are only 32 registers. MIPS R4000 uses the MIPS III ISA. Each MIPS III instruction has a 32 bit representation. It can be represented as human-readable assembly code, or binary machine code. There are three types of MIPS III instruction.

I-type Immediate: requires one or two registers and an operand.

R-type Register: requires three registers.

J-type Jump: requires an operand.

Commonly used variables kept in registers. Then use a combination of registers and memory to access a large amount of data quickly.

4 Compilation, Interpretation and an Overview of the Java Virtual Machine (JVM)

4.1 Levels of Programming Languages

High-level programming languages, such as Java, C, C++ and C# are close to the English language and, therefore, easy for humans to read and write. Code written in these languages is more concerned with problem-solving than implementation. Programmers are less likely to make errors when programming in these languages and do not need to worry about memory management and other complexities.

Low-level languages, such as machine code, use instructions stored in memory (opcodes) and refer to specific locations in memory. Code written in these languages reflects the processes being used rather than the problem being solved. Machine language is difficult for humans to read and write. Since it runs directly on hardware, it is also hardware-specific.

Assembly language is slightly higher than machine code. It uses mnemonics so that it can be more easily read or written by humans. This is used to translate high-level languages to machine code.

4.2 Language Processing Systems

To run on a computer, a program must be supplied in machine code. Programs written in high-level languages are converted to binary through a series of phases.

1. Preprocessor (prepares high-level code for the compiler)
2. Compiler/interpreter (produces assembly/intermediate code)
3. Assembler (produces relocatable machine code)
4. Linker/loader (produces object code)

The preprocessor organises and prepares the source code for the compiler. This includes importing packages, macro processing and file inclusion.

The compiler reads the entire source code in one go and creates tokens. It checks the meaning of these tokens and generates intermediate assembly code. Alternatively, the interpreter reads the source statement by statement, converting each statement to intermediate code that is executed immediately.

The assembler calculates relative addresses for jumps and produces relocatable machine code.

The linker combines assembled parts into a whole.

Alternatively, the loader places the code directly into memory.

4.3 High-Level Program Execution

A program written in a high-level language can be run in two different ways.

1. Compiled into a program in the native machine language of the target machine
2. Directly interpreted and executed via simulation on the target machine

4.4 Compilation

A compiler converts source code into assembly, object or machine code that does the same thing as the original. Object code is usually relocatable, so it can later be linked or loaded.

This is done just once for each program. Hardware features can be exploited to optimise object code so that it will run more quickly.

However, this is more difficult than interpretation, as the compiler must understand the entire program in order to convert it. Compilers are also hardware-dependent, so cannot run on different platforms.

A compiler runs on the same platform as the target machine. A cross-compiler can produce code that will run only on a separate platform.

1. Lexical analysis
2. Syntax analysis
3. Semantic analysis
4. Intermediate code generation
5. Optimisation
6. Code generation

4.4.1 Lexical Analysis

4.4.2 Intermediate Code Generation

During intermediate code generation, the compiler produces code in an intermediate language between that of the source code and machine language.

4.4.3 Optimisation

During optimisation, the compiler simplifies or removes unnecessary code. This allows the program to run more quickly or use fewer resources.

4.4.4 Code Generation

During code generation, the compiler maps the optimised intermediate code to the target machine language.

4.5 Interpretation

An interpreter is a program that follows the source code and performs appropriate actions accordingly. A CPU can be viewed as a hardware implementation of an interpreter for machine code.

Interpreters begin execution immediately and facilitate interactive debugging and testing. A user can read and modify the values of variables and invoke procedures from the command line. Interpreted languages are not hardware-dependent. However, interpreted programs have slower execution than compiled programs.

Compilers are compute-intensive and require more preparation time. Additionally, compiled programs are hardware-dependent. However, they can run very quickly.

4.6 Combined Compilation and Interpretation

Combined compilation and interpretation is a method by which programs are compiled to an intermediate language that can be interpreted efficiently. Execution of programs produced by this method is slower than pure compilation, but quicker than pure interpretation.

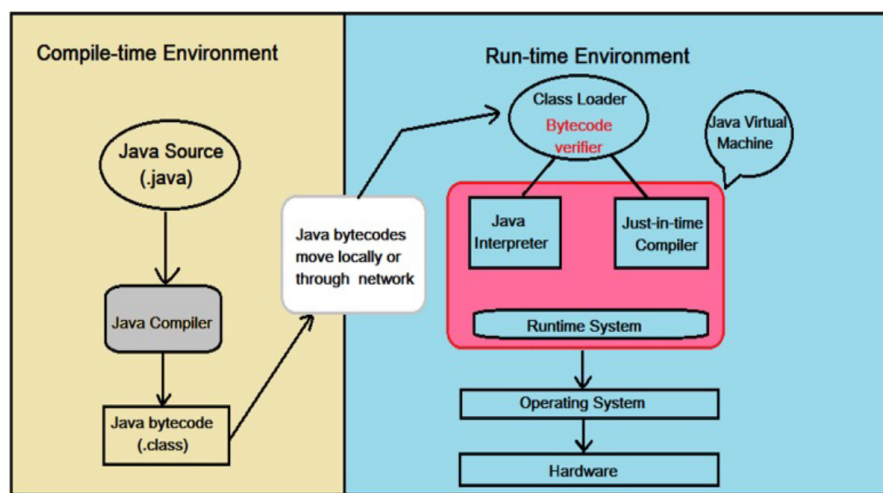
Compilation for any platform requires only a single compiler that is independent of the target CPU. Interpretation of the intermediate language is delegated to the target CPU.

Java was conceived as a language that uses combined compilation and interpretation. The `javac` compiler converts `.java` source code to `.class` bytecode, which is interpreted by the Java Runtime Environment (JRE) on the Java Virtual Machine (JVM).

4.7 Compilation and Execution on Virtual Machines

A virtual machine executes an instruction stream using software rather than hardware. Virtual machines emulate hardware using software and are, therefore, hardware-independent. Virtual machines are used in languages such as Java, Pascal and C#.

Java compilers generate bytecode that is interpreted by the JVM. This requires a JVM to exist on the target machine. The JVM may translate portions of bytecode to machine code at runtime via just-in-time (JIT) compilation if it finds those portions are used frequently.



Java

4.8 Java Virtual Machines

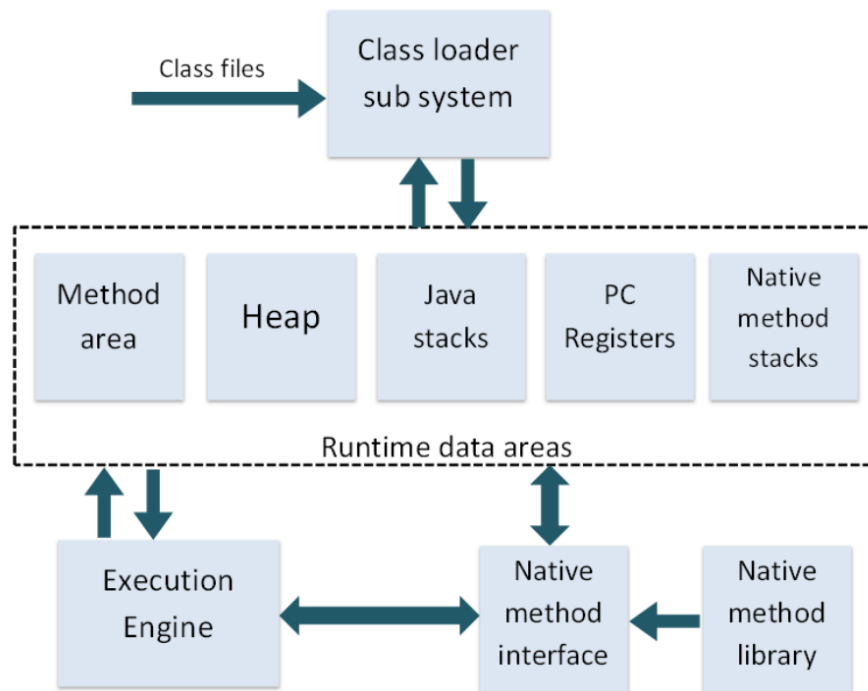
Portability of Java why not compile Java to machine code?

1. need to generate code for each target machine
2. cannot exchange executable code

The Sun Java solution

1. design machine architecture (JVM) specifically for the Java language
2. translate Java source code into JVM code (bytecode)
3. write software interpreter for JVM in C (widely available)
4. Thus bytecode can be exchanged ,remote execution is possible

The architecture



The JVM Architecture

1. It is a stack machine! Closer to modern high-level languages than the von Neumann machine (Register machines).
2. Memory: 32-bit words (=4 bytes)

5 Subroutines and Stacks

5.1 How Subroutines Work

5.1.1 Call and Return Instructions

When execution jumps from line n to a subroutine, there needs to be a way to instruct the computer to continue execution on line $n + 1$ once the subroutine is complete.

Two hypothetical instructions that could be used to achieve this are `call`, which jumps to the subroutine and stores the return address (the current PC value) somewhere suitable, and `ret`, which reads the return address from where it was stored and loads it into the PC.

5.1.2 Method 1 — Storing Return Address as First Byte

Using this method, the return address is stored as the first byte of the subroutine. The instruction `call N` would store the return address at location N and begin executing the subroutine at location $N + 1$. The instruction `ret N` would load the return address stored at location N into the PC.

The disadvantage of this method is that only one return address can be stored for each subroutine. Consequently, a subroutine could not call itself, as this would require two return addresses (one for the original call and another for the recursive call).

FORTRAN — the first commercially available high-level programming language — disallowed recursion in order to implement this method.

5.1.3 Introduction to Stacks

A stack is a data structure that can flexibly store a variable number of bytes. Stacks employ 'last in, first out' (LIFO) semantics. Elements are 'pushed' to or 'popped' from the top of the stack. A stack pointer (SP) is a CPU register that points to the top of the stack. It is not necessary to know where the bottom of the stack is, but programmers must make sure they do not attempt to pop from an empty stack.

When a value is pushed to the stack, it is written to memory at the address given by the SP. The SP is then incremented.

When a value is popped from the stack, the SP is decremented, then the value is read from memory at the address given by SP. Popped values remain in memory and are later overwritten by pushed values.

5.1.4 Method 2 — Storing Return Address on a Stack

Using this method, the return address is pushed to a stack when the subroutine is entered. The instruction `call N` would push the return address to the stack and begin executing the subroutine at location *N*. The instruction `ret` would pop the return address from the stack and load it into the PC.

This method allows multiple return addresses to be stored and, therefore, makes recursion possible.

5.1.5 Saving Registers

A subroutine may need to use CPU registers in its calculations. However, the previous register values will need to be restored when execution returns to the caller.

A common solution to this problem is to push all register values to the stack at the beginning of a subroutine (after the return address has been pushed) and to pop the register values from the stack at the end of the subroutine (before the return address is popped). Register values are popped in reverse order.

The return address and saved registers are known as a 'stack frame'.

5.2 Stacks for Calculations

5.2.1 Reverse Polish Notation (RPN)

Reverse Polish notation (RPN) or postfix notation is a method of writing calculations in which operators follow their operands and the order of operations is as written (no brackets are needed). This differs from standard infix notation, in which operators are written between or around their operands and the order of operations is determined by precedence.

RPN is a useful way of implementing calculations with stacks. If the next item in the input stream is a value or variable, it is pushed to the operand stack. If the next item is an operation, the relevant number of operands is popped from the operand stack, the operation is performed, and the result is pushed to the operand stack.

The following infix calculation and postfix calculation are equivalent.

$$(5 + 2) \times \sqrt{x \times x + y \times y} + 8$$

$$5\ 2\ +\ x\ x\ \times\ y\ y\ \times\ +\ \sqrt{\ }\ \times\ 8\ +$$

One notation that shows the effect of an operation, such as subtraction, on a stack is as follows.

$$\dots, \text{val1}, \text{val2} \rightarrow \dots, \text{val1} - \text{val2}$$

Java bytecode uses operand stacks for calculations. In Java, each method call has its own operand stack.

5.2.2 Stack Machines

Stack machines use a return stack for subroutine return, and an operand stack for RPN calculations. Registers are not needed for calculations. This allows more space for calculations, and machine instructions do not need to specify registers. However, it is difficult to know where things are on the stack.

The term 'operand' could refer either to the byte(s) after the instruction opcode in memory, or the entries in the operand stack.

5.3 Bitwise Boolean Operations

The XOR operation is an exclusive OR. It returns true only if exactly one operand is true.

Truth tables.

A	B	A AND B	A OR B	A XOR B
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

5.4 Conditional and Unconditional Jumps

No stacks are used when an unconditional jump is executed. Conditional jumps are only executed if a condition is true; if the condition is false, execution continues to the next instruction.

Java bytecode contains six comparisons that are used in conditional jump instructions. There are two types of conditional jump instruction. Jump instructions of the form `ifXX` pop one value from the stack and compare it to zero. Jump instructions of the form `if_cmpXX` pop two values from the stack and compare them to each other.

For example, `ifeq N` pops a value from the stack and jumps to location *N* if the value is zero. The instruction `if_cmpeq N` pops two values from the stack and jumps to location *N* if they are equal.

Conditional jump instructions.

Comparison		Instruction	
Symbol	Mnemonic	<code>ifXX</code>	<code>if_cmpXX</code>
=	eq	<code>ifeq</code>	<code>if_cmpeq</code>
<	lt	<code>iflt</code>	<code>if_cmplt</code>
≤	le	<code>ifle</code>	<code>if_cmple</code>
≠	ne	<code>ifne</code>	<code>if_cmpne</code>
>	gt	<code>ifgt</code>	<code>if_cmpgt</code>
≥	ge	<code>ifge</code>	<code>if_cmpge</code>

6 Introduction to Operating Systems

6.1 Early Computers

Early computers used magnetic tape decks for storage and used paper tape for input and output. Each program was keyed in by hand to control the required system function. There was a distinct lack of monitors. The ‘operators’ controlled the loading of magnetic tapes, fed in cards and paper tape, and maintained the system. This ‘operation’ was tedious, expensive and error-prone.

As computers became faster and more powerful, the role of the operators became a limiting factor. An automated system was required in order to free the operators from mundane tasks. Such a system is known as an ‘operating system’.

6.2 Operating System (OS)

An operating system (OS) is a layer of system software that acts as an intermediary between a computer user and the underlying hardware. The main function of an OS is to dynamically allocate the shared system resources to executing programs.

From the perspective of a user, the OS is designed for ease of use. It provides a convenient interface for executing user programs, and hides the complexity of the hardware implementation.

From the perspective of the system, the OS is intimately involved with the hardware and acts as a resource manager for CPU time, memory, file storage, I/O devices etc. Requests for these resources may be numerous and conflicting.

It also acts as a control system between I/O devices and user programs. It manages execution to prevent errors and improper use, such as one program attempting to view the memory of another.

A program that controls the execution of application programs. An interface between applications and hardware.

Main Objectives of an OS:

- Convenience
- Efficiency
- Ability to Evolve

6.3 Components of a Computer System

The main components of a computer system are the

- hardware (the basic computing resources),
- operating system (controls and coordinates use of hardware),
- application programs (use the system resources to solve computing problems): compilers, text editors, email clients, webs browsers, games
- users (people, machines and other computers).

Services provided by the OS include

- program development,
- program execution,
- access to I/O devices,
- access to files,
- system access,
- error detection and response, and
- accounting.

The most frequently used functions of the OS are stored in its 'kernel'. Role of an Operating System
A computer is a set of resources for the movement, storage, and processing of data and for the control of these functions. The OS is responsible for managing these resources

6.4 Evolution of Operating Systems

6.4.1 Serial Processing

Serial processing was used in the earliest computers. This relied on human operators rather than an OS. Users had access to the computer in 'series'.

One problem with serial processing concerned scheduling. Most installations used a sign-up sheet for the reservation of computer time. Mistakes in the estimations of time allocations would result in wasted computer time. A considerable amount of time was also spent simply setting up the program to run.

6.4.2 Simple Batch Systems

Since early computers were very expensive, it was important to maximise processor utilisation. In simple batch systems, the user no longer had direct access to the processor. Computer operators would batch jobs together and place them on an input device. The loading of programs and management of jobs was handled by software known as a 'monitor'.

The resident monitor always resides in memory. After it reads in a job, the processor executes instructions from the memory containing the monitor until it reaches an error or ending condition. Control then passes back to the monitor.

In this system, processor execution alternates between user programs and the monitor. As a result, main memory and processor time is given to the monitor. Despite this overhead, simple batch systems improved the utilisation of the processor.

6.4.3 Multiprogrammed Batch Systems

In uniprogrammed systems, the processor executes a program until it reaches an I/O instruction. It must then wait for the I/O instruction to complete before it continues.

In multiprogrammed systems, when one job must wait for I/O to complete, the processor can switch to another job.

6.4.4 Time Sharing Systems

Time sharing systems are used to handle multiple interactive jobs. Multiple users access the system simultaneously through terminals. The OS interleaves the execution of each user program in short bursts or quanta of computation.

Comparison of multiprogrammed batch and time sharing systems.

Operating system	Principal objective	Source of directives
Multiprogrammed batch	Maximise processor use	Commands provided with job
Time sharing	Minimise response time	Commands entered at terminal

6.5 Operating System Start and Firmware

6.5.1 Firmware

A small 'bootstrap' program is loaded when a computer is started. This program is stored in read-only memory (ROM) or electrically erasable programmable read-only memory (EEPROM) so that it cannot easily be altered, either accidentally or maliciously. The program is, therefore, known as 'firmware'.

The bootstrap program initialises all CPU registers, device controllers and memory contents, and loads the OS kernel into memory. Once the firmware is loaded, some services are provided outside the kernel. These are loaded at boot time to become system processes or system 'daemons', and run the entire time the kernel is running. On Unix systems, the first system process is 'init'. Once all these processes are loaded, the OS waits for events to occur.

Events such as a user clicking a mouse button or a program attempting to access a file are known as 'interrupts'. Hardware may trigger an interrupt by sending a signal to the CPU. Software triggers an interrupt by executing a special operation known as a 'system call'.

The CPU reacts to an interrupt by suspending its current execution, immediately transferring execution to a fixed address (usually the starting address of the service routine for the interrupt), executes the interrupt service routine, and finally resumes execution where it was originally interrupted.

6.6 Storage Structure and Memory Hierarchy

A CPU can load instructions only from memory. Main memory is rewritable RAM — commonly implemented with DRAM semiconductor technology.

Memory is an array of bytes. Each byte has its own address. Bytes are moved between memory and CPU registers with load and store instructions. The CPU automatically loads instructions from main memory for execution.

The fetch instruction loads an instruction from memory. When the instruction is decoded, operands may also be loaded from memory. After the instruction is executed, the result is stored to memory. The memory unit only sees a stream of addresses.

Ideally, both program instructions and data would reside in the main memory. However, this is difficult as main memory is small and volatile (erased after loss of power). The solution is to store

programs and data in secondary storage. When programs are run, their instructions are loaded from secondary storage to main memory.

There are many forms of storage available. They differ in speed, cost, size and volatility.

Typical properties of storage types.

Level	Name	Size	Access time / ns	Bandwidth / MiB s^{-1}	Managed by
1	Registers	< 1 KiB	0.25 to 0.5	20000 to 100000	Compiler
2	Cache	< 16 MiB	0.5 to 25	5000 to 10000	Hardware
3	Main memory	< 64 GiB	80 to 250	1000 to 5000	OS
4	Solid-state disk	< 1 TiB	25000 to 50000	500	OS
5	Magnetic disk	< 10 TiB	5000000	20 to 150	OS

6.6.1 Memory

Memory is an array of bytes, in which each byte is addressable. ROM cannot be modified. EEPROM can be modified, but only infrequently. The CPU requires memory that can be read and written. It uses dynamic RAM (DRAM) and registers.

Main memory is erased when a machine powers off. Secondary storage is used as a more permanent solution. This includes magnetic and optical disks.

Data that is used frequently is stored in cache memory for faster access. If the CPU requires data, the cache is checked first.

Level	1	2	3	4	5
Name	registers	cache	main memory	solid state disk	magnetic disk
Typical size	< 1 KB	< 16MB	< 64GB	< 1 TB	< 10 TB
Implementation technology	custom memory with multiple ports CMOS	on-chip or off-chip CMOS SRAM	CMOS SRAM	flash memory	magnetic disk
Access time (ns)	0.25 - 0.5	0.5 - 25	80 - 250	25,000 - 50,000	5,000,000
Bandwidth (MB/sec)	20,000 - 100,000	5,000 - 10,000	1,000 - 5,000	500	20 - 150
Managed by	compiler	hardware	operating system	operating system	operating system
Backed by	cache	main memory	disk	disk	disk or tape

Storage Comparison

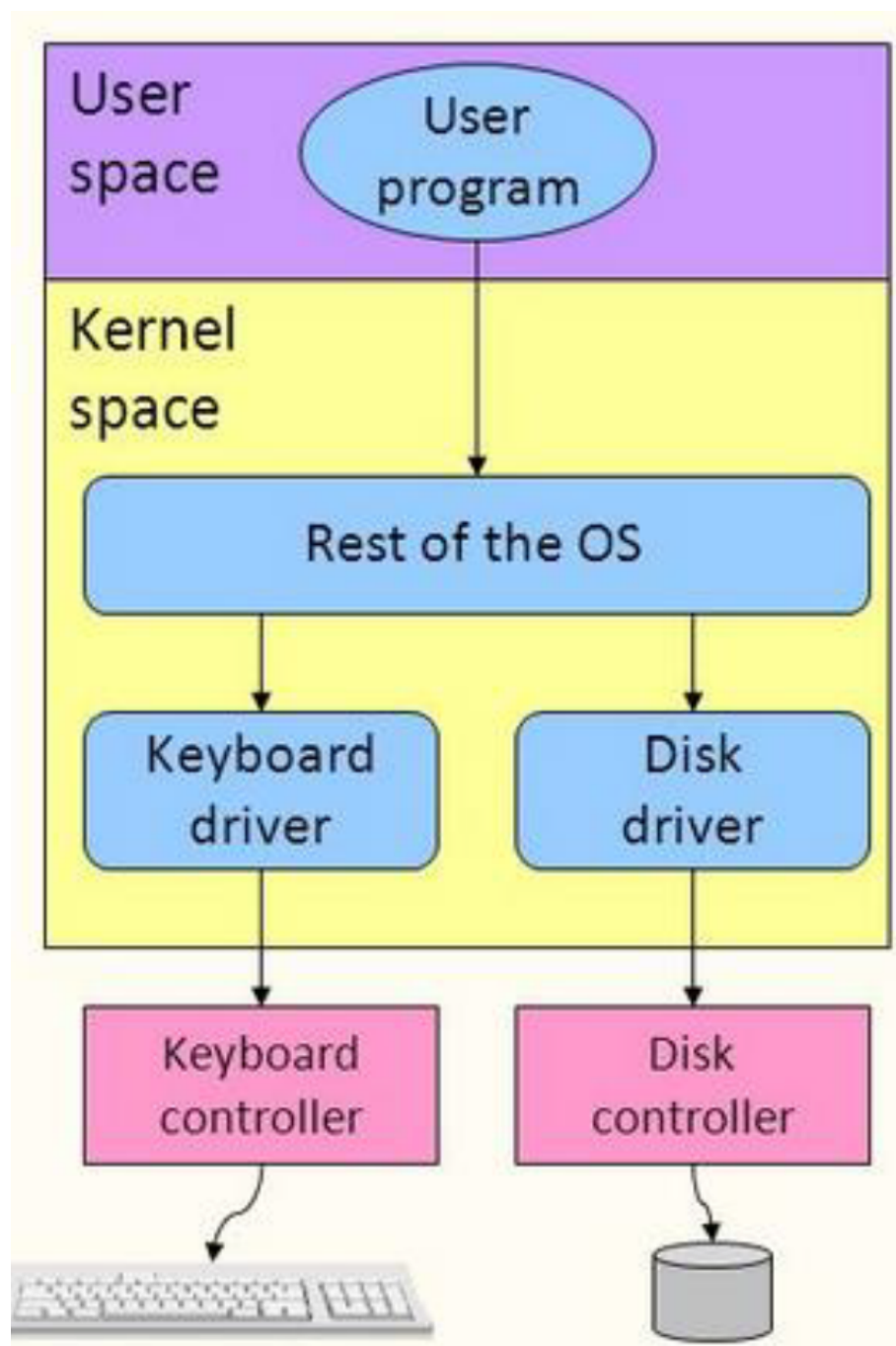
6.7 I/O Structure

A large portion of OS code is dedicated to the management of I/O. Without I/O, computers would be useless. The hardware may trigger an interrupt at any time by sending a signal to the CPU. An I/O device interacts with the system via a 'device controller' connected through a common bus to the CPU.

A small computer system interface (SCSI) controller is a piece of hardware that allows an SCSI storage device to communicate with the operating system via an SCSI bus. A device controller

maintains some local buffer storage and a set of special-purpose registers. The device controller moves data between the peripheral device it controls and its local buffer storage.

An OS has a device driver for each device controller. These are typically downloaded. The device driver is able to interact with its corresponding device controller and provides the OS with a uniform interface to the device.



Storage Comparison

6.8 I/O Mechanisms

There are three types of I/O mechanism.

1. Programmed I/O (also known as 'polling')
2. Interrupt-driven I/O
3. Direct memory access (DMA)

6.8.1 Programmed I/O

Programmed I/O requires the CPU to 'poll' the status of the controller by repeatedly checking its status until it is ready to accept an I/O request. It then continues to check the status of the controller until the I/O operation is complete.

This mechanism is very fast due to the direct involvement of the CPU. However, this also prevents the CPU from working on other tasks that may be more important.

The processor has special instructions that control I/O device, test status and transfer of data (read/write). However, the performance issue is the processor is busy checking the I/O status of the I/O module.

6.8.2 Interrupt-Driven I/O

Interrupt driven I/O works as follows.

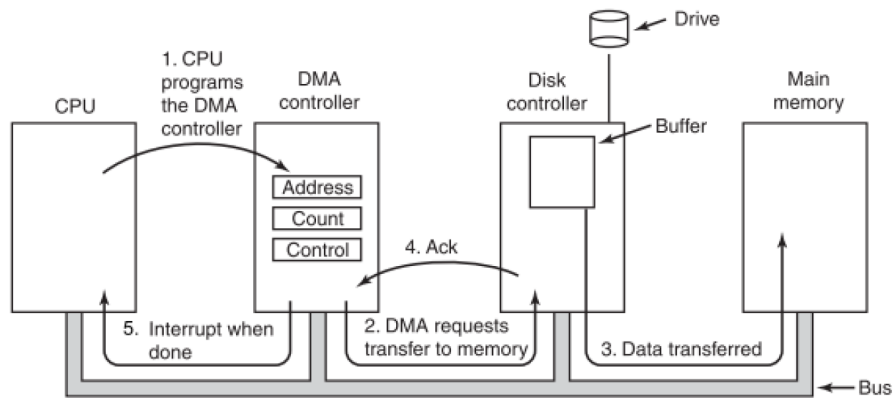
1. The CPU initiates I/O through the device driver.
2. The CPU checks for interrupts between executing other instructions.
3. The device controller begins the I/O operation.
4. When input is ready, output is complete or there is an error, the device controller sends an interrupt signal to the CPU.
5. When the CPU detects the interrupt, control is transferred to the interrupt handler, which will process the data before returning from the interrupt.
6. The CPU resumes the interrupted task.

Interrupt driven I/O is suitable for transferring small amounts of data. The CPU is less involved in the I/O operation and is free to execute other instructions. However, this reduces the speed of the I/O operation.

6.8.3 Direct Memory Access (DMA)

The DMA controller transfers data directly between the I/O device and memory. This requires additional architecture for the device, including buffers, pointers and counters. The device controller transfers entire blocks of data to and from its own buffer storage and memory. There is no direct involvement of the CPU during transfer.

1. The CPU programs the DMA controller.
2. The DMA controller requests transfer to memory.
3. The disk controller transfers the data to memory.
4. The disk controller checks with the DMA controller that there is no more data to transfer.
5. The DMA controller sends an interrupt signal to the CPU when transfer is complete.



Direct Memory Access

Only a single interrupt is generated. This is much better than the one interrupt per byte generated by low-speed devices. The CPU is completely free to execute other instructions. However, both the DMA controller and the CPU share the bus that connects different hardware components. Thus, the CPU will experience slowdown during DMA transfer as it must wait for access to the bus.

6.8.4 Computer System Architecture

Single-processor and multiprocessor systems. Single-processor systems have one CPU, sometimes including special-purpose processors. In contrast, multiprocessor systems, which include parallel and multicore setups, share the bus, clock, memory, and peripherals. These systems range from servers to smartphones and offer benefits such as increased throughput, improved cost efficiency, and better reliability through graceful degradation.

6.9 General Computer System Architectures

6.9.1 Multiprocessor Systems

Multiprocessor systems have multiple processors in parallel or multiple cores. The processors share the bus, clock, memory and peripherals. This increases throughput, although this does not increase linearly with the number of processors. There is also an economy of scale, as additional processors share the same peripherals, storage and power. Reliability is also increased, as a failed processor will not stop the entire system. This is known as 'graceful degradation'.

Asymmetric multiprocessor systems follow a boss-worker relationship. Each processor is assigned a specific task.

Symmetric multiprocessor systems follow a peer-to-peer relationship. Each processor is utilised and has an equal chance of being used.

6.9.2 Non-Uniform Memory Access (NUMA)

~~In non-uniform memory access (NUMA) systems, multiple processors are interconnected and each processor has its own memory. Each processor can access the other memories through their corresponding processors, but this is slower than direct access to its own memory.~~

6.9.3 Multicore Systems

~~Multicore systems have multiple cores on a single chip. This is useful because on-chip communication is faster and consumes less power. This is used in blade servers to minimise physical space and power consumption.~~

6.9.4 Clustered Systems

~~Clustered systems incorporate multiple interconnected computers with a shared network storage. This provides a high availability of computers and processors, which is useful if one computer fails. It also facilitates standby or backup servers, high-performance computing and parallelisation of applications.~~

6.10 Operating System Services

Operating system installations typically include

- system programs (program loader, command interpreter),
- language processors (C compiler, assembler, linker),
- utilities (text editor, terminal emulator), and
- subroutine libraries (standard C library, JVM).

6.11 System Calls

System calls provide OS services to the user via an application programming interface (API). System calls APIs are typically written in C or C++, though some are written in assembly language. System calls are mostly accessed by programs via a high-level Application Programming Interface (API) rather than direct system call use. Three most common APIs are Win32 API for Windows, POSIX API for POSIX-based systems (including virtually all versions of UNIX, Linux, and Mac OS X), and Java API for the Java virtual machine (JVM).

Typically, there is a number associated with each system call. The system call interface maintains a table indexed according to these numbers. The system call interface invokes the requested system call in the OS kernel and returns the status of the system call and any return values.

The system call for reading data from one file and writing it to another is `cp <file1> <file2>`. This uses the following services.

- Open `file1`
- Handle possible error
- Create `file2`
- Start read and write
- Handle possible errors
- Close `file1` and `file2`

The services are not accessed directly. They are accessed via an API.

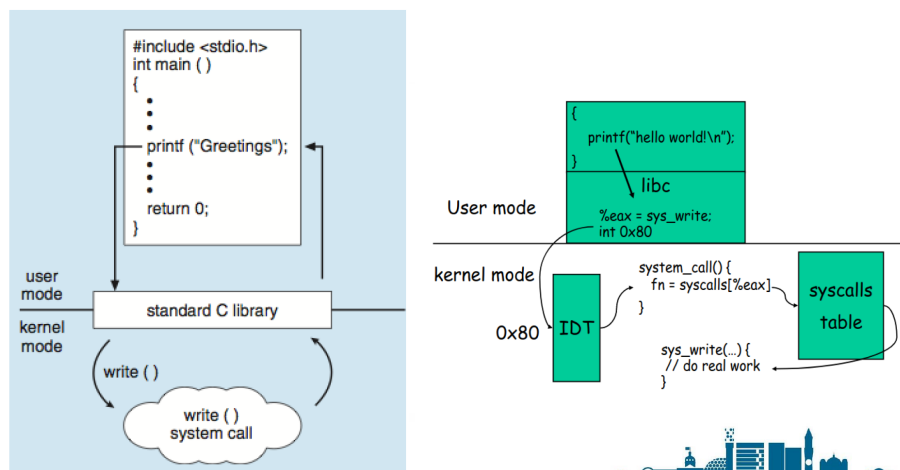
Examples of Windows and Unix system call APIs.

Type of call	Windows	Unix
Process Control	CreateProcess()	fork()
	ExitProcess()	exit()
	WaitForSingleObject()	wait()
File Manipulation	CreateFile()	open()
	ReadFile()	read()
	WriteFile()	write()
	CloseHandle()	close()
Device manipulation	SetConsoleMode()	ioctl()
	ReadConsole()	read()
	WriteConsole()	write()
Information maintenance	GetCurrentProcessID()	getpid()
	SetTimer()	alarm()
	Sleep()	sleep()
Communication	CreatePipe()	pipe()
	CreateFileMapping()	shmget()
	MapViewOfFile()	mmap()
Protection	SetFileSecurity()	chmod()
	InitializeSecurityDescriptor()	umask()
	SetSecurityDescriptorGroup()	chown()

System call APIs specify a set of functions that are available to an application programmer. They describe the parameters to pass to each function, and the values that are returned. The programmer accesses the API via a library provided by the OS.

A Java API exists for programs that run on the JVM. The JVM itself uses the system calls of its particular platform.

System call APIs are used for program portability. A program using the API can be compiled and run on any system that supports the API. Invoking system calls directly is usually more difficult. Using a system call API is similar to implementing an interface.



Standard C Library Example

6.12 Dual Mode

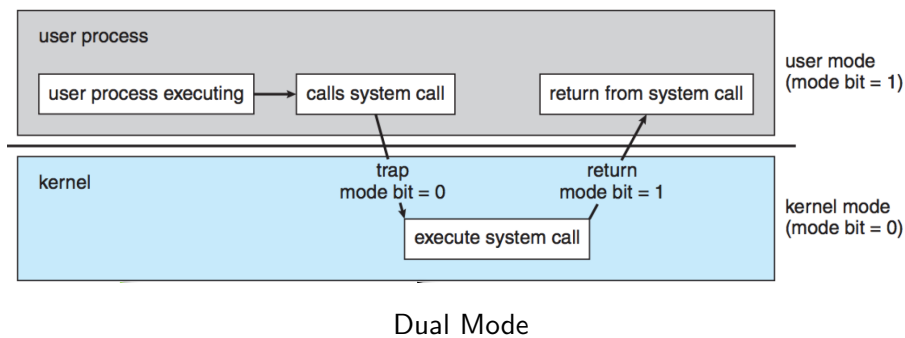
Operating system execution is split into user mode (mode bit of value 1) and kernel mode (mode bit of value 0). In user mode, certain areas of memory are protected from user access, and certain instructions cannot be executed. In kernel mode, protected areas of memory may be accessed, and privileged instructions may be executed.

The OS allows user programs to execute system calls by resetting the mode bit to 0. Once the kernel has executed the requested system call, the mode bit is set to 1. This prevents crashes in user mode from affecting the kernel.

The MS-DOS architecture has no mode bit. Hence, user programs were able to wipe the entire OS, write to devices without permission, execute illegal instructions and access the memory of other users.

With dual mode, hardware can stop user program attempting to execute an illegal instruction or to access memory of other users.

When a mode violation is detected, the OS must terminate the program, give an error message and produce memory dumps by writing to a file.



6.13 OS Design and Implementation

The internal structures of operating systems can vary widely. The goals and specifications of an OS can be affected by hardware and the type of system. User goals are that the OS should be convenient to use, easy to learn, reliable, safe and fast. System goals are that the OS should be easy to design, implement and maintain, as well as flexible, reliable, error-free and efficient.

Early operating systems were produced in assembly language, then in system programming languages such as Algol and PL/I. Nowadays they are written in C and C++.

Modern operating systems actually use a combination of different languages. The lowest levels may be written in assembly, the main body in C, and system programs in C, C++, PERL, Python and shell scripts. Operating systems written in higher-level languages are easier to port to other hardware, but they are also slower. Emulation allows an OS to run on non-native hardware.

6.14 Operating System Structure

A general-purpose OS is a very large program. There are various ways to structure one.

- Simple structure (e.g. MS-DOS)
- More complex or non-simple structure (e.g. Unix)
- Layered approach (an abstraction)
- Microkernel structure (e.g. Mach OS)
- Modular approach
- Hybrid approach

6.14.1 Simple Structure

MS-DOS is an OS with a simple structure. It was written to provide the most functionality in the least space. It is not divided into modules. Although it has some structure, its interfaces and levels of functionality are not well separated. This is dangerous, as application programs have the ability to access the lowest levels of the system.

6.14.2 More Complex or Non-Simple Structure

The Unix OS has a more complex structure. The Unix OS is limited by hardware functionality, has a limited structure, and consists of two separate parts. These are the system programs and the kernel.

The kernel consists of everything below the system call interface and above the physical hardware. It provides the file system, CPU scheduling, memory management and other OS functions. This is a large number of functions for one level, and makes the operating kernel difficult to debug, since the kernel services rely on each other. It is also dangerous since there is a lot of functionality running in kernel mode.

6.14.3 Layered Approach

In the layered approach, the OS is divided into a number of layers or levels. Each layer is built on top of a lower layer. The lowest layer (layer 0) is the hardware. The highest layer (layer N) is the user interface.

Using modularity, functions are split into layers such that the functions of one layer call upon only the functions of the layer directly beneath. While this is more structured and less dangerous, it prevents higher layers from communicating with lower levels.

6.14.4 Microkernel Structure

Mach is an example of a microkernel OS. Microkernel structure moves as much functionality from the kernel to the user space as possible. Communication between user modules takes place via messages passed through the kernel.

Microkernel structures are easier to extend and port to new architectures. They are also more reliable and secure, since less code runs in kernel mode. However, performance is lost due to added communication between the user space and the kernel space, as well as between user modules through the kernel.

6.14.5 Modular Approach

Many modern operating systems implement loadable kernel modules. This uses an object-oriented approach, in which each core component is separate, communicates with others through known interfaces, and is loaded as needed within the kernel. This is similar to the layered approach, but is more flexible. Linux and Solaris use the modular approach.

6.14.6 Hybrid Approach

Most modern operating systems do not use one pure model. Hybrid operating systems combine multiple approaches to address performance, security and usability. Linux and Solaris are actually monolithic (the OS works entirely in kernel space, similar to more complex or non-simple structures), but use modularity for the dynamic loading of functionality.

6.15 User Classes and Interfaces

Almost all operating systems have a user interface (UI). This interface can take several forms.

- Command-line interface (CLI)
- Batch interface (commands entered into files that are executed)
- Graphical user interface (GUI)

Any UI requires a software link to the hardware, which may be buried under other software.

Types of user:

- Programmers
 - System programmers (creators of operating systems, compilers, device drivers etc)
 - Application programmers
- Administrators (concerned with provision, operation and management of computing facilities)
- End-users (who apply software to some problem area)

Some end-users may be unaware that they are interacting with a computer, whereas others may have a substantial understanding of computers.

6.15.1 System Call Interface

All interaction with hardware must go through system calls. The OS provides a layer of subroutines known as an API.

6.15.2 Command-Line Interface

Most operating systems provide an interactive terminal into which commands may be entered. This can be used to imitate programs or perform housekeeping control routines on the system. Unix provides shell programs.

6.15.3 Job Control Language Interface

Job control language defines requirements for work submitted to a batch system. This is used in database administration.

6.15.4 Graphical User Interface (GUI)

Interaction proceeds via windows and usually a mouse-driven environment. There are desktops, icons and GUI APIs.

From the programmer perspective, this is slightly more complex than shell scripts, but can be more rewarding. Event-driven programming is used so that the interface is responsive to user actions.

From the user perspective, this can be friendlier. However, there is an increased processing load.

6.15.5 Touchscreen Interface

Sometimes computer mice are not available or not desired. Actions and selections are controlled by gestures. There may be a virtual keyboard for text entry.

6.15.6 Time Sharing

The CPU can execute multiple jobs simultaneously by switching between the jobs stored in memory. Switches occur so frequently that users can interact with each program as it is running. CPU scheduling is used to decide which job is brought to memory to be executed when space is an issue. Reasonable response time is of the utmost importance.

Processes can be swapped between main memory and the disk. 'Virtual memory' allows for the execution of a process that is not entirely in memory. The process can use both virtual and physical memory as 'logical' memory. This allows users to run programs that require more than the available physical memory and frees programmers from concern over memory limitations.

7 Process Management and Scheduling

7.1 Processes

A process is a program in execution. A program is a passive entity stored on the disk, whereas a process is an active entity. A program becomes a process when its executable file is loaded into memory.

A program may be started through a GUI or CLI. One program may consist of several processes. For example, there may be one process for each user or instance.

Processes are represented by three main components.

- Executable program code (also known as the 'text segment')
- Data related to the program
- Execution context
 - process ID, group ID, user ID
 - stack pointer, program counter, CPU registers
 - file descriptors, locks, network sockets

The execution context is essential for process switching.

A process can be an execution environment for other code. For example, the JVM executes as a process that interprets loaded Java code and performs instructions on behalf of it.

7.2 Process Structure

A process is more than just the program code. It also includes current activity such as the value of the program counter and the contents of CPU registers. The process stack contains temporary

data, such as function parameters, return addresses and local variables. The data segment contains global variables and a heap — memory allocated dynamically at runtime.

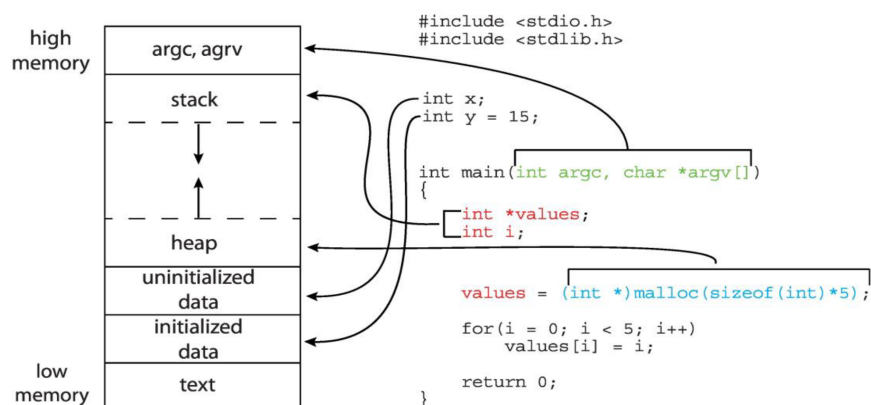
The layout of the memory of a process is known as a 'process image'. A process image is divided into segments.

- Stack segment
 - function parameters, return addresses and local variables
- Data segment
 - static variables and constants
 - memory allocated dynamically from the heap
- Text segment
 - program code — shared between processes

max	Stack	Stack segment
	↓ ↑	
	Heap	Data segment
	Data	
	Program code	Text segment
0	Kernel	Protected

The process image.

Each invoked program results in the creation of a separate process with its own image. Each image appears to cover the entire address space. In reality, these are virtual addresses mapped to physical addresses.



Process memory Layout

7.3 Process Control Block (PCB)

The information associated with each process is stored as a process control block (PCB), which is also known as a task control block (TCB). This information includes

- process state,

- program counter (PC) — address of next instruction to be executed,
- CPU registers,
- CPU scheduling information — priorities and scheduling queue pointers,
- memory-management information,
- accounting information — CPU usage, time elapsed and time limits, and
- I/O status information — I/O devices allocated to the processes and list of open files.

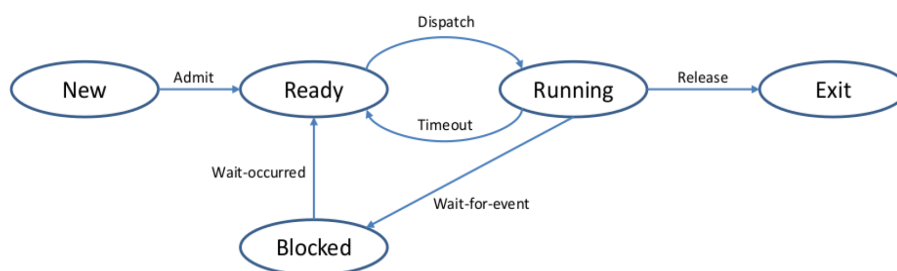
A process with multiple threads of execution has instructions at multiple addresses executing at once. This requires multiple program counters. These additional thread details are stored in the PCB.

7.4 Process States

As a process is executed, it changes state. Possible states for a process are

- new — the process is being created,
- ready — the process is waiting to be assigned to a processor,
- running — the process is executing instructions,
- waiting (blocked) — the process is waiting for an event to occur, and
- terminated — the process has finished execution.

When a process is admitted to the system, its state changes from 'new' to 'ready'. When it is dispatched to a CPU, its state changes to 'running'. From there, it may time out, in which case its state reverts to 'ready', or it may need to wait for an event to occur, in which case its state changes to 'blocked'. Alternatively, it may be released from execution, in which case its state changes to 'terminated'. A process in the 'blocked' state moves to the 'ready' state once the event it is waiting for occurs.



Process memory Layout

7.5 Process Scheduling

Process scheduling is used to maximise CPU usage and minimise response time by quickly switching processes onto the CPU for time sharing. A process will 'give up' the CPU if it sends an I/O request or after a certain period of time as elapsed. When a process gives up the CPU, it is added to the ready queue. The process scheduler selects available processes in the ready queue for the next execution.

The OS maintains various scheduling queues.

- Job queue — all processes in the entire system
- Ready queue — all processes in main memory ready and waiting to execute
- Device queues — all processes waiting for an I/O device

Processes migrate between queues. Queuing diagrams are used to represent flows of processes and resources.

7.5.1 Types of Scheduler

A short-term scheduler (or CPU scheduler) selects which process should be executed next and allocates a CPU. Sometimes this is the only scheduler in a system. It is invoked frequently and must be fast.

A long-term scheduler (or job scheduler) selects which processes should be brought into the ready queue. This is invoked infrequently and may be slow. The long-term scheduler controls the degree of multiprogramming.

7.5.2 Types of Process

An I/O-bound process spends more time performing I/O than computations. It uses many short CPU bursts.

A CPU-bound process spends more time performing computations. It uses a few long CPU bursts. The long-term scheduler attempts to schedule a good process mix.

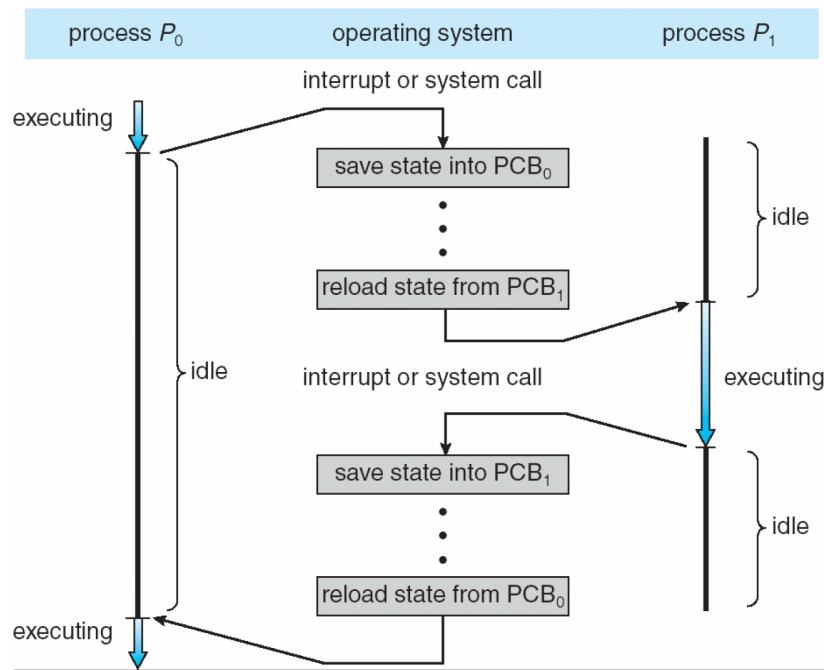
7.5.3 Context Switching

When the CPU switches to another process, the system saves the state of the old process and loads the saved state of the new process via a context switch. The context of a process is represented by its PCB.

The time required to perform a context switch is pure overhead; the system does no useful work while switching. More complex operating systems and process control blocks require more time for a context switch. The time required is dependent upon hardware support. Some hardware provides multiple sets of registers per CPU. This allows multiple contexts to be loaded simultaneously.

7.5.4 Dispatch Events

Context switches (dispatch events) are triggered by clock interrupts. These occur after a set time interval, typically between 3 ms and 10 ms. The execution of processes is interrupted and control returns to the OS. The current process is moved to the ready queue. The frequency of such an interrupt is an important system parameter used to balance the overhead of context switching and the responsiveness of the system.



CPU Switch from Process to Process

Context switches are also triggered by I/O interrupts. When an I/O event occurs and data has been loaded into memory, the current process is interrupted and all blocked processes waiting for the I/O event are moved to the ready queue. The dispatcher must decide whether to continue execution of the interrupted process.

Memory faults or page faults also trigger context switches. These occur when an executing process refers to a virtual memory address that has not been allocated a physical memory location (the data requested is still on the disk). The current process is interrupted and an I/O request for the data is issued. The current process is moved to the blocked state and execution switches to another process from the ready queue. When the I/O has completed, the blocked process returns to the ready queue.

7.5.5 Dispatcher

A dispatcher is an OS function that allocates processes to the CPU and switches the CPU between processes. The weakness of a dispatcher with a single device queue is that all processes waiting for an event must be transferred from the device queue to the ready queue when an event occurs. A dispatcher with multiple queues for different device events moves only the processes waiting for that particular event to the ready queue.

7.5.6 Virtualisation and Swapping

A computer system may appear to allow an unlimited number of processes to occur concurrently. Not all of these processes can fit into physical memory. It is useful, therefore, to achieve medium-term scheduling and reduce the degree of multiprogramming by removing a process from memory.

Moving a process between memory and storage is known as 'swapping'. This requires two additional process states: 'ready-suspended' and 'blocked-suspended'.

A process in the 'ready-suspended' or 'blocked-suspended' state is activated to the 'ready' or 'blocked' state when it is swapped into memory. It is suspended when it is swapped out of memory. A process in the 'blocked-suspended' state is moved to the 'ready-suspended' state when the event for which it is waiting has occurred. A process in the 'running' state may be suspended.

7.6 Process Creation

The system must provide mechanisms for process creation and process termination.

Processes are created at system boot, when an existing process spawns a child process, when the user requests the creation of a process, or when a batch system executes its next job. When a parent process creates a child process, a tree of processes is formed. A process is identified and managed via its process identifier (PID).

When a process is created, it is assigned a unique PID, memory is allocated for its process image, its PCB is initialised, and it is added to the ready queue.

7.6.1 Parent and Child Processes

A child process may share all the resources of its parent, a subset of those resources or none of them. A parent and child process may execute concurrently, or the parent may wait for its children to terminate.

A child process is a duplicate of the address space of its parent. The child process may load a new program into its address space.

7.6.2 Process Creation in Unix

A process creates new processes with the kernel system calls `fork()` and `exec()`.

- A new slot is allocated in the process table.
- A unique PID is assigned to the child process.
- The process image of the parent is copied, apart from the shared memory areas.
- The child process owns the same open files as the parent.
- The child process is added to the ready queue.

When `fork()` is called by the parent process, execution switches to kernel mode and the OS creates a copy of the parent process. This includes the program code in the text segment of the process image. Thus, the `fork()` call is also copied. The parent process continues execution at the return from the `fork()` call. The child process begins execution at the return from the `fork()` call.

In the parent process, the `fork()` call returns the PID of the child process. In the child process, the `fork()` call returns zero. This is used by the remainder of the program to identify whether it is running as a parent or a child. The program may perform different instructions in either of these cases.

For example, the child process may execute the `exec()` system call, which takes a program name as a parameter. This system call replaces the content of the child process image with a process image of the specified program. The `exec()` system call has a number of variations with different suffixes

```
int main()
{
    pid_t pid;
    /* fork a child process */
    pid = fork();

    if (pid < 0) { /* error occurred */
        fprintf(stderr, "Fork Failed");
        return 1;
    }
    else if (pid == 0) { /*child process */
        execlp("/bin/ls", "ls", NULL);
    }
    else { /* parent process */
        /* parent will wait for the child to complete */
        wait(NULL);
        printf("Child Complete");
    }
    return 0;
}
```



C program to create a separate process in UNIX

that describe what parameters are passed to the new process image. These variations are `execl()`, `execle()`, `execlp()`, `execv()`, `execve()` and `execvp()`.

- e — An array of pointers of environment variables is passed.
- l — Command line arguments are passed individually.
- p — The PATH environment variable is used to find the program to be executed.
- v — Command line arguments are passed as an array of pointers.

7.7 Process Waiting

A parent may use the system call `waitpid()` to wait for the termination of a child process.

7.8 Process Termination

There are a number of ways in which a process may be terminated.

- Normal termination (voluntary) or error termination (voluntary)
 - the process reaches regular completion, with or without an error code
- Abnormal termination (involuntary)

- OS or user intervenes with kill signal
- process attempts to access forbidden memory locations
- process times out
- process encounters an I/O error
- stack overflow
- parent process is terminated

Some operating systems do not allow a child process to exist if its parent has been terminated. This is known as 'cascading termination'.

A 'zombie process' is a process that has terminated but still has an entry in the process table. This occurs when a child process terminates. It allows a parent process to read the exit codes of its children.

7.9 Execution of the Operating System

An OS may be designed so that it executes

- separately from processes (non-process kernel),
- as part of the user process images (with mode switching), or
- as a set of processes (microkernel, with mode and context switching).

7.9.1 Non-Process Kernel OS

The kernel acts as a monitor for user processes. All processes are user processes. There is no concurrent execution of user processes and the kernel; the kernel may execute only during interruptions.

The OS code is placed in a reserved memory region and executes in privileged mode. It has its own system stack.

7.9.2 OS Execution within User Processes

The user address space includes kernel functions. System functions may be called from within a process by switching to kernel mode. No context switching is required as the system functions execute within the same process. The dispatcher executes context switches outside of processes.

7.9.3 Process-Based OS

Kernel functions run in kernel mode as separate processes. The dispatcher handles context switching separately. This allows concurrency within the kernel, with kernel processes scheduled together with user processes. This is useful in multiprocessor environments, as kernel functionality may be distributed amongst CPU cores.

7.10 Basic Concepts

7.10.1 CPU and I/O Burst Cycle

Process scheduling is the process of selecting the next process in the ready queue to be executed by the CPU. Scheduling decisions take place when events interrupt the execution of a process. Such events include Clock Interrupts (i.e. process running → ready state) I/O Interrupts (i.e. process waiting → ready state) Operating System Calls (read, write, process ready → waiting) Signals (e.g. Semaphores)

Process execution is a cycle of CPU execution and I/O waiting. Processes on a batch system are CPU-bound, whereas processes on an interactive system are I/O-bound with very short CPU bursts. When a process is in an I/O burst, the CPU becomes idle and the short-term scheduler must select another process from the ready queue. The ready queue is not necessarily a FIFO queue; it may be a priority queue, tree or unordered linked list.

7.10.2 Dispatcher

The dispatcher is the module that gives control of the CPU to the process selected by the short-term scheduler. This involves switching context, switching to user mode and jumping to the correct location stored in the PCB of the process in order to resume execution. The dispatcher must be as fast as possible since it is invoked at every process switch. The time taken by the dispatcher to stop one process and start another is known as 'dispatch latency'.

7.10.3 Levels of Scheduling

The long-term scheduler is responsible for admitting new processes to the ready queue. The short-term scheduler is responsible for selecting the next process to be given control of the CPU. The mid-term scheduler is responsible for moving processes between suspended and non-suspended states.

A non-preemptive scheduler allows an executing process to continue execution until it terminates, releases the CPU voluntarily (cooperative scheduling) or is blocked due to an event such as an I/O interrupt. A preemptive scheduler may stop an executing process due to a clock interrupt (time slice expired) or when a process with a higher priority becomes ready.

7.11 Scheduling Criteria

Scheduling algorithms are designed to maximise or minimise different parameters. OS parameters to be maximised include

- CPU utilisation, and
- throughput — the number of processes that complete execution per unit time.

User concerns to be minimised include

- turnaround time — the total time to execute a process to completion,

- waiting time — the time a process spends in the ready queue, and
- response time — the time between the submission of a new process and its first CPU burst (also known as latency).

7.12 Scheduling Algorithms

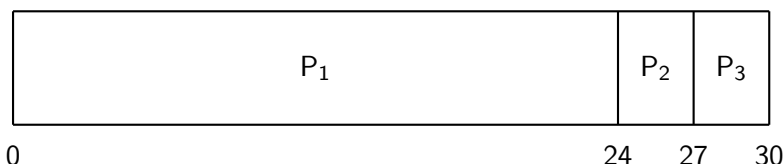
7.12.1 First Come First Serve (FCFS)

First come first serve (FCFS) is the simplest scheduling algorithm. Processes are given control of the CPU in the order that they request it. This is implemented with a FIFO ready queue. When a process enters the ready queue, its PCB is pushed to the tail. When the CPU becomes free, it is allocated to the process whose PCB is popped from the head of the queue.

Using this algorithm, the average process waiting time can vary greatly. If a long process arrives before a short process, the waiting time for the second process will be long. If the short process arrives first, the second process will have a short waiting time.

Suppose the processes arrive in the order: P1 , P2 , P3. Gantt Chart as follows:

Process	Burst Time
P ₁	24
P ₂	3
P ₃	3



Waiting time for P1 = 0; P2 = 24; P3 = 27 Average waiting time: $(0 + 24 + 27)/3 = 17\text{ms}$

7.12.2 Shortest Job First (SJF)

Using the shortest job first (SJF) scheduling algorithm, each process is associated with an estimate of the length of its next CPU burst. When the CPU becomes available, it is assigned to the process that has the shortest estimated next CPU burst. If two processes are estimated to have next CPU bursts of the same length, FCFS is used to choose between them. A more accurate name for this algorithm would be 'shortest next CPU burst first'.

The SJF scheduling algorithm can be proven to be optimal. It gives the minimum average waiting time for a given set of processes. This works because scheduling a short process before a long process

decreases the waiting time of the short process more than it increases the waiting time of the long process. Thus, the average waiting time decreases.

However, the algorithm is difficult to implement since there is no way to know the exact length of the next CPU burst. Instead, the length of the next CPU burst of a process is estimated by an exponential average of the measured lengths of its previous CPU bursts. The predicted length τ_{n+1} of the next $((n + 1)$ th) CPU burst is given by the following iterative formula in terms of the measured length t_n of the previous (n th) CPU burst and a weighting factor α .

$$\tau_{n+1} = \alpha t_n + (1 - \alpha) \tau_n$$

Commonly, the weighting factor α is set to 0.5. Thus, if the predicted and actual lengths of the n th CPU burst are 10 ms and 6 ms, respectively, the predicted length of the $(n + 1)$ th CPU burst would be 8 ms.

7.12.3 Shortest Remaining Time First (SRTF)

SJF can be either preemptive or non-preemptive. If the length of the next CPU burst of a newly submitted process is shorter than the remaining CPU burst of the currently executing process, the preemptive SJF would switch the two processes to minimise the average waiting time, whereas the non-preemptive SJF would allow the current process to finish its CPU burst.

The preemptive SJF scheduling algorithm is known as the 'shortest remaining time first' (SRTF) scheduling algorithm.

7.12.4 Priority Scheduling

The SJF scheduling algorithm is a special case of the general priority scheduling algorithm. A priority is associated with each process and the CPU is allocated to the process with the highest priority. Priorities are indicated by a fixed range of numbers. Lower numbers are used for higher priorities. Processes with equal priorities are scheduled using FCFS.

Priorities can be defined by internal criteria, such as time limits, memory requirements and number of open files, or external criteria, such as process importance, funds paid for computation or political factors.

Preemptive priority scheduling will switch processes if a newly submitted process has a higher priority than the current process. Non-preemptive priority scheduling would simply place that process at the head of the priority queue.

Priority scheduling can result in the indefinite blocking or starvation of low-priority processes if the system receives a steady stream of processes with higher priority. These processes may eventually

run after a very long waiting time, or may be lost if the system crashes before that point. This can be solved by increasing priority with age.

7.12.5 Round-Robin (RR)

The round-robin (RR) scheduling algorithm is designed especially for time-sharing systems. It is similar to FCFS, but uses preemption to enable the switching of processes. A circular ready queue and small 'time quantum' or 'time slice' are used. The CPU is allocated to each process in the queue for a time interval of up to one time quantum. The ready queue is treated as a FIFO queue.

The scheduler sets a timer to interrupt the CPU after one time quantum and dispatches the first process in the ready queue. If the process has a CPU burst length of less than one time quantum, the process will release the CPU voluntarily. Otherwise, the process is interrupted after one time quantum, resulting in a context switch to the second process. The first process is pushed to the tail of the ready queue.

If there are n processes in the ready queue and the time quantum is q , each process is given $\frac{1}{n}$ of the CPU time in bursts of at most q . Each process must wait no longer than $(n - 1)q$ until its next CPU burst. For example, in a queue of five processes with a time quantum of 20 ms, each process is given up to 20 ms of CPU time every 100 ms or less, resulting in a waiting time of no more than 80 ms between CPU bursts.

The performance of the RR scheduling algorithm depends heavily on the size of the time quantum. If the time quantum is longer than the process length, the algorithm would be no different from FCFS. If the time quantum is too short, there would be a large number of expensive context switches. Ideally, the time quantum should be much larger than the context switch time. If the context switch time is $n\%$ of the time quantum, approximately $n\%$ of CPU time would be devoted to context switching. Modern systems typically have time quanta between 10 ms and 100 ms and context switches of less than 10 μ s.

7.12.6 Appendix

Introduction to Operating Systems: Definition and role of an OS in managing hardware and software resources. Components of a computer system (Hardware, OS, Application, Users). Operating System Services & Evolution: Key functions like process management, memory management, I/O operations, file systems, and security. Transition from Uniprogramming to Multiprogramming, and Time-Sharing systems.

Computer Architecture & OS Structures Bootstrapping process, Main Memory and I/O structure. Overview of how I/O devices interact with the system. Types of I/O mechanisms and their role in performance. Process Management & Scheduling The concept of processes, their lifecycle (process states), and operations on processes. Scheduling criteria: efficiency, fairness, response time, throughput, etc. Common Scheduling Algorithms including FCFS, SJF, SRTF, Priority & Round-Robin.

8 Complexity of Algorithms

8.1 Algorithm Design and Analysis

An algorithm is an effective method for solving a problem expressed as a finite sequence of instructions. Often, multiple algorithms exist for the same task. There are many criteria for choosing a suitable algorithm for a particular task, including efficiency, simplicity, clarity, elegance and proof. It is also necessary to consider whether the algorithm is always correct, always terminates or whether the problem can even be solved with an algorithm.

In general, efficiency is achieved by finding a balance between many conflicting parameters, such as run time, response time, memory usage, bandwidth usage and power consumption. One very important issue is understanding how problems and algorithms grow in complexity. Although a solution may work very well for small problems, it is important to know how its performance may be affected if the problem becomes larger.

It may be possible to improve code so that it runs quickly or uses less memory, but it may be the case that an algorithm will inherently perform worse as the amount of data increases. It may even be the case that the problem itself is inherently difficult and becomes more difficult as the size of the problem grows. Sometimes the problem can be modified so that it still satisfies the overall requirements, but behaves in a more manageable way.

Algorithm efficiency can be considered in terms of time and space.

Space (or bandwidth) complexity is not considered in the same way as computational complexity, since memory and bandwidth are limited by practical constraints. The bandwidth of a system can often be increased if necessary.

Instead, computational complexity is usually considered in terms of 'time complexity' — a measure of how much computation is involved in solving a problem. It is not necessary to know quantitatively exactly how many times harder a problem will grow. Often a qualitative indication of complexity will suffice.

8.2 Time Complexity

There are two ways to determine the run time of an algorithm.

The first method is to measure empirically. This involves benchmarking the algorithm using a representative set of inputs. This does not measure the complexity of the algorithm, rather the complexity of an implementation of the algorithm on a particular piece of hardware.

The second method is to analyse theoretically the worst case scenario, by identifying the operations of the algorithm — its time complexity.

Time complexity is a measure of the number of operations that an algorithm must execute in terms of the size of the input or problem. Since it is the algorithm that is analysed, and not the implementation, time complexity is analysed using pseudocode. The time complexity $T(n)$ of an algorithm is a function of the size n of its input.

The following algorithm has time complexity $T(n) = 2n^2$. It is quadratic because it has two nested loops that go through every element. It has a coefficient of 2 because there are two operations in the nested statement.

Matrix-vector multiplication of size n .

```
for i=0...n-1:
    for j=0...n-1:
        x[i] = x[i] + A[i][j] * b[j]
```

8.3 Complexity Class and Big-O Notation

Usually, it is not necessary to know the exact time complexity. It is sufficient to know the complexity class, which ignores constant factors or overheads, and expressions of lower orders. This focusses on performance for large n .

Class complexity is expressed using 'big-O notation'. For example, $O(n^2)$ is "of the order of n^2 ".

Corresponding time complexities and complexity classes.

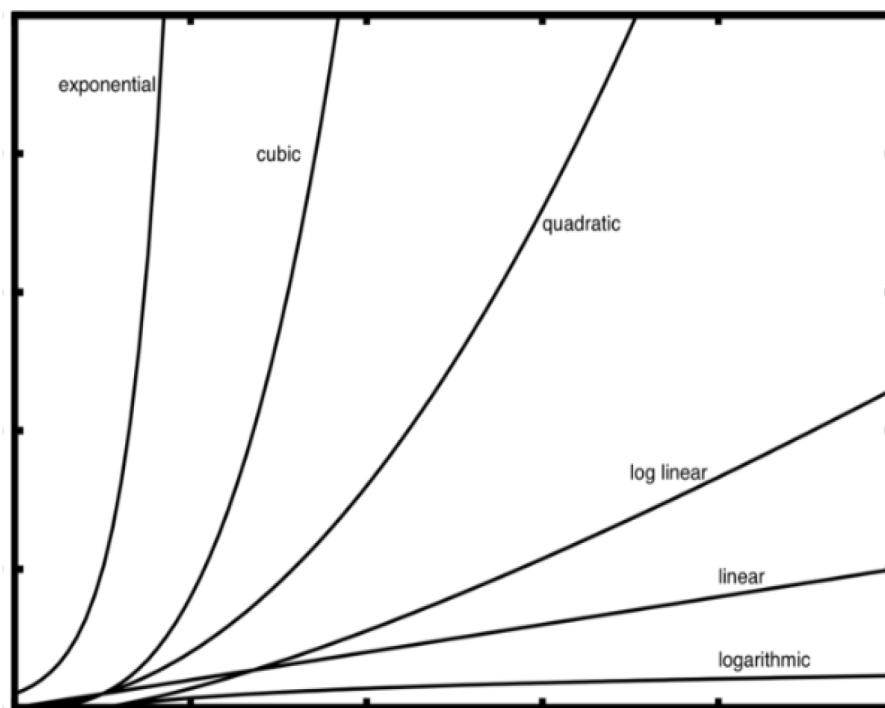
Time complexity	Complexity class
$T(n) = n$	$O(n)$
$T(n) = n + 2$	$O(n)$
$T(n) = 2n^2$	$O(n^2)$
$T(n) = 10n^3$	$O(n^3)$
$T(n) = 5(n + 2)$	$O(n)$
$T(n) = 1000$	$O(1)$
$T(n) = n^2 + n + 1$	$O(n^2)$

To determine complexity class, it is often sufficient to count the number of loops and the number of times they are executed.

Polynomial classes are described as 'tractable'. Exponential classes are described as 'intractable' — they can execute with small amounts of data, but cannot with large amounts.

Common complexity classes.

Complexity class	Name
$O(1)$	Constant
$O(\log_2 n)$	Logarithmic
$O(n)$	Linear
$O(n \log_2 n)$	Log Linear
$O(n^2)$	Quadratic
$O(n^3)$	Cubic
$O(2^n)$	Exponential



graphical

The binary search algorithm (on a sorted array) has class $O(\log_2 n)$ because the size of the loop is halved on each iteration.

Binary search algorithm.

```
left = 0; right = n-1
while left < right:
    mid = (left + right) / 2
    if x[mid] < v:
        left = mid + 1
    else:
        right = mid
if x[left] == v:
    return left
else:
    return -1
```

The total algorithm complexity is determined by the complexities of its components. Sequential algorithm phases are summed, and the maximum term is considered. Function or method calls are multiplied.

For example, this following altered matrix-vector multiplication algorithm has two sequential phases and a total complexity of $O(n) + O(n^2) \Rightarrow O(n^2)$.

Initialisation and matrix-vector multiplication of size n .

```
for i=0...n-1:
    x[i] = 0
for i=0...n-1:
    for j=0...n-1:
        x[i] = x[i] + A[i][j] * b[j]
```

The following iterative binary search algorithm uses a call to a method of logarithmic complexity within a fixed loop. It has complexity $O(n) \times O(\log_2 n) \Rightarrow O(n \log_2 n)$.

Iterative binary search.

```
for i=0...n-1:
    binary_search(x, v[i])
```

8.4 Algorithm and Problem Complexity

Algorithm complexity is the worst-case run time of an algorithm. This an upper bound and is the most informative complexity.

A problem has a complexity class of $O(f(n))$ if there exists an algorithm of complexity class $O(f(n))$ to solve it. Sometimes lower bounds are considered.

Problems solvable in polynomial time (PTIME or P) are assumed to be tractable. If a solution can be guessed and then checked in polynomial time, the problem is solvable in non-deterministic polynomial time (NP).

10 Computer Architecture and Operating System Structures

11 Concurrency and Synchronisation

11.1 Introduction to Concurrency

11.1.1 Motivation

A computer system typically has multiple users or user applications working independently. The OS must protect the memory of one process from interference from another process. It also interleaves the execution of processes to maximise the resource utilisation and responsiveness of the system. To do so, **it must preserve the state of each process so that its execution may resume.**

Concurrent execution of processes was originally intended solely for the maximisation of utilisation due to the great cost of computer systems, but nowadays it is intended to satisfy the expectations of users who wish to execute multiple tasks concurrently.

The problem is that these processes do not always work independently. They may compete for access to resources, such as devices, files and data. They may cooperate through messages, shared memory and files. They may also be distributed such that millions of users are reading and updating information at the same time. This is the case for social networking, banking and online shopping systems.

It is also necessary that applications can interleave multiple independent but potentially conflicting tasks. An application with a graphical user interface (GUI) should not freeze when it is performing a time consuming task. Similarly, a user of a banking system may not want to wait in a queue in order to perform a transaction. Nevertheless, it is still necessary to maximise the efficiency of the system.

11.1.2 Multiple Processes

OS design is concerned with the management of processes and threads.

- Multiprogramming — multiple independent processes on a single processor
- Multiprocessing — multiple processes on multiple processors
- Distributed processing — multiple processes on multiple machines

Concurrency may be employed in the context of sharing resources amongst multiple applications, sharing resources between processes in a single application, or sharing resources between processes and threads of the OS itself.

11.1.3 Key Terms

atomic operation A function or action implemented as a sequence of one or more instructions that appears to be indivisible; that is, no other process can see an intermediate state or interrupt the operation. The sequence of instructions is guaranteed to execute as a group, or not execute at all, having no visible effect on the system state. Atomicity guarantees isolation from concurrent processes.

critical section A section of code within a process that requires access to shared resources and that must not be executed while another process is in a corresponding section of code.

deadlock A situation in which two or more processes are unable to proceed because each is waiting for one of the others to do something.

livelock A situation in which two or more processes continuously change their states in response to changes in other processes without doing any useful work.

mutual exclusion The requirement that when one process is in a critical section that accesses shared resources, no other process may be in a critical section that accesses any of those shared resources.

race condition A situation in which multiple threads or processes read and write a shared data item and the final result depends on the relative timing of their execution.

starvation A situation in which a runnable process is overlooked indefinitely by the scheduler; although it is able to proceed, it is never chosen.

A critical section must behave as an atomic operation. Mutual exclusion occurs in a critical section. A race condition must be managed by mutual exclusion.

11.1.4 Principles of Concurrency

Interleaving and overlapping can be viewed as examples on concurrent processing. They both present the same problems.

In a uniprocessor system, the relative execution of processes cannot be predicted because it depends on the activities of the processes, how the OS handles interrupts and the scheduling policies of the OS.

11.1.5 Difficulties of Concurrency

Difficulties arise in the sharing of global resources, the optimal allocation of resources by the OS, and, most importantly, in the coordination of parallel and asynchronous processes so that they behave correctly. They must produce the same results if they are executed concurrently as they would if they were executed consecutively. The debugging of concurrent programming errors is difficult because observed behaviour is non-deterministic (dependent upon timings of other processes on the machine). It is therefore difficult to identify and reproduce.

11.2 OS Concerns and Process Interaction

In order to manage the issues raised by concurrency, the OS must be able to keep track of various processes, allocate and deallocate resources for each active process, protect the data and physical resources of each process from interference by other processes, and ensure that the processes and outputs are independent of 'processing speed' (the timing of completion of different operations).

Relationships between concurrent processes.

Degree of awareness	Relationship	Influence of one process on the other	Potential control problems
Processes unaware of each other	Competition	• Results of one process independent of the other • Timing of processes may be affected	• Mutual exclusion • Deadlock (reusable resource) • Starvation
Processes indirectly aware of each other (e.g. shared object)	Cooperation by sharing	• Results of one process may depend on information obtained by other • Timing of processes may be affected	• Mutual exclusion • Deadlock (reusable resource) • Starvation • Data coherence
Processes directly aware of each other (have communication primitives available to them)	Cooperation by communication	• Results of one process may depend on information obtained from other • Timing of processes may be affected	• Deadlock (consumable resource) • Starvation

11.3 Resource Competition

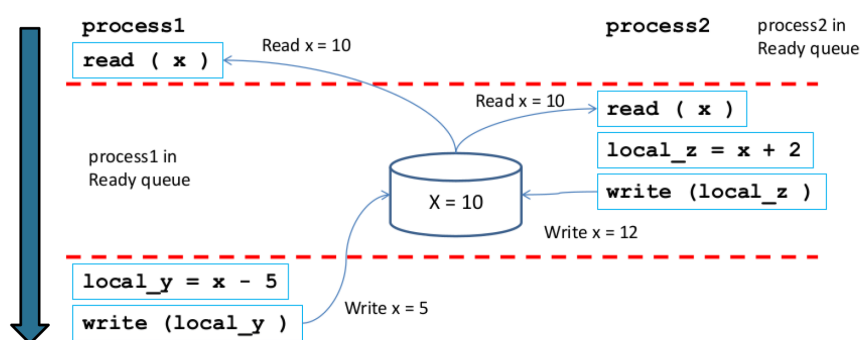
Concurrent processes come into conflict when they are competing for use of the same resource, such as I/O devices, memory, processor time and the clock. In the case of competing processes, three control problems must be faced.

- Mutual exclusion — to ensure correct behaviour
- Deadlock
- Starvation

A race condition occurs when multiple processes or threads read and write shared data items. The processes “race” to perform their read/write actions. The final result depends on the order of execution. The “loser” of the race is the process that performs the last update and determines the final value of the shared data item.

Race conditions can occur due to

- order of execution — whenever the state of a shared resource depends on the precise order of execution of the processes,
- scheduling — context switches at arbitrary times during execution, or
- outdated information — processes or threads operating with stale or dirty copies of memory values in registers or local variables.



Synchronization of Processes

Context switches may occur at any time Process 2 has its result overwritten by process 1 Process 1 operates with outdated information

11.4 Critical Section Problem

11.4.1 Critical Section

In order to avoid race conditions, it is necessary to control the concurrent execution of critical sections. **This is achieved through strict serialisation, or mutual exclusion**, causing the critical sections to behave as atomic operations.

An entry protocol is executed before entering a critical section. The process requests permission to enter the critical section. The process may have to wait for entry to be granted. It must communicate that it has entered the critical section.

A process is able to complete the execution of its critical section, even if it is preempted or interrupted. An exit protocol is executed after exiting a critical section. The process communicates to other processes that it has left the critical section.

It is important that the scope and length of critical sections is kept as small as possible. Large critical sections can be detrimental to the efficiency and throughput of a system.

11.4.2 Deadlock and Starvation

Enforcing mutual exclusion creates the problems of deadlocks (processes waiting forever for each other to free resources) and starvation (a process waiting forever to be granted entry to its critical section). Implementations of mutual exclusion must account for these problems. Critical sections must be as small as possible and processes must spend as little time as possible in a critical section.

11.4.3 Requirements for Solutions to the Problem

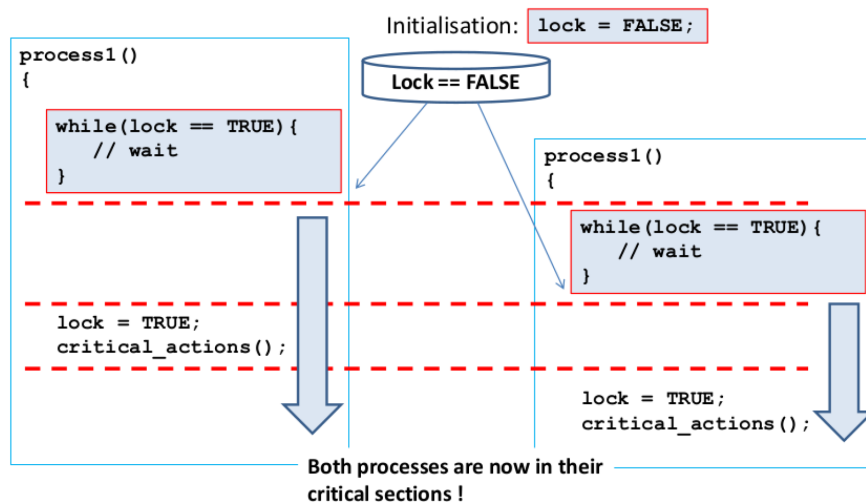
1. Serialisation of access
 - Only one process at a time is allowed in the critical section for a resource
 2. Bounded waiting (no starvation)
 - A process waiting to enter a critical section must be guaranteed entry within some defined limited waiting time
 - The scheduling algorithm must guarantee that the process is eventually scheduled
 3. Progress (liveness, no deadlock)
 - A process that halts in its non-critical section must do so without interfering with other processes waiting to enter their critical section
 - Only processes currently waiting to enter their critical section are involved in the selection of the one process that may enter
 - A process remains inside its critical section for a finite time only
-
1. Software solutions
 - Shared lock variables (busy-waiting)
 - Polling and spinning or strict alternation
 2. Hardware solutions
 - Disabling interrupts
 - Special instructions
 3. Higher-level OS constructs
 - Semaphores, monitors or message passing

11.5 Software Solutions

11.5.1 Lock Variables

Critical sections must be protected by some form of 'lock'. A lock is a shared data item. Processes must acquire a lock before entering a critical section, and release the lock when exiting the critical section. A lock is also known as a 'mutex'.

A two state lock may be implemented as a shared variable. A process will wait while the lock is true, then when it is false, set the lock to true and execute its critical actions. When its critical section is complete, it will set the lock to false. However, in this implementation, the lock itself is a shared resource and a race condition may occur on it. Atomic instructions must be used for a shared lock.

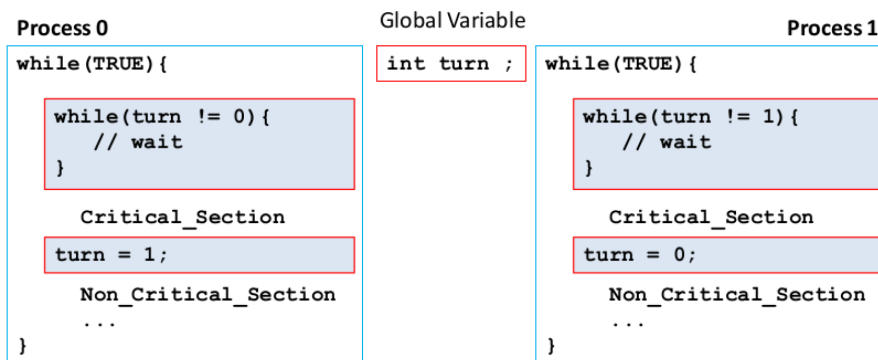


Shared Lock Problem

11.5.2 Busy-Waiting (Polling and Spinning)

In this solution, a process continuously evaluates whether a lock has become available. The lock is represented by a data item held in shared memory. This causes the process to consume CPU cycles without any progress. A process busy-waiting may prevent another process holding the lock from executing and completing its critical section and from releasing the lock. [Spin locks are used at kernel level \(special HW instructions\).](#)

11.5.3 Busy-Waiting (Strict Alternation)



Busy Waiting

This solution imposes a strict alternation between two processes; a process waits for its turn. A token is used as a shared variable. This is usually a process ID set by the previous process that indicates which process is next to enter. A process busy-waits until the token is its own process ID. When the process exits the critical section, it sets the token to the next process ID.

This solution guarantees mutual exclusion because there is no longer a race condition on the lock. However, there is still a problem of liveness and progression since multiple processes depend on the

change of the token. If one of the processes is delayed in its non-critical section, it cannot enter its critical section when it is its turn, and other processes will be blocked.

11.6 Hardware Solutions

11.6.1 Disabling Interrupts

In uniprocessor systems, concurrent processes cannot be overlapped, only interleaved. Thus, disabling interrupts guarantees mutual exclusion, but significantly reduces the efficiency of execution. This approach does not work on a multiprocessor system.

11.6.2 Special Hardware Instructions

These are applicable to any number of processes on either uniprocessor or multiprocessor systems sharing a main memory. They are simple and easy to verify and can be used to support multiple critical sections. However, they use busy-waiting and may still cause starvation and deadlocks.

11.7 Abstractions

Some systems may execute critical sections regardless, then detect whether problems have occurred. If so, the changes are undone. Sometimes such problems do not occur very often, and doing a lot of work to fix the problems when they do occur is overall more efficient than doing a little work every time to avoid them. This approach is used in database systems.

In other cases, it may be acceptable to have errors caused by concurrency.

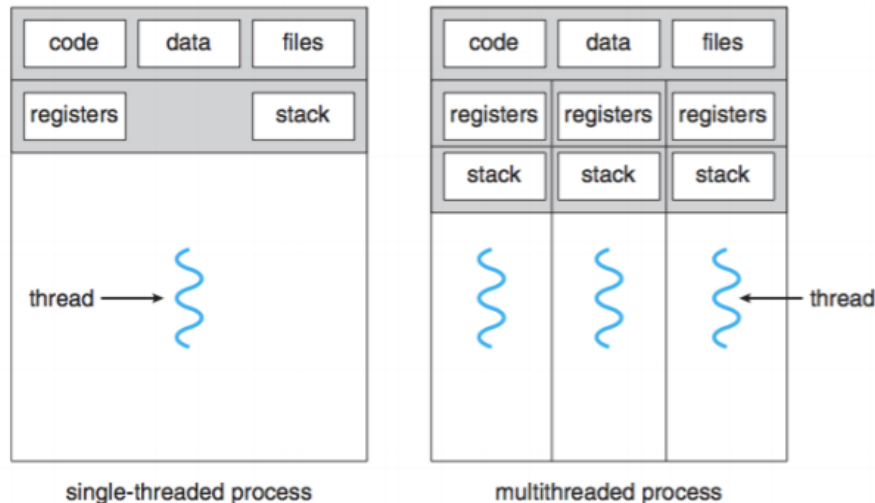
Many systems, such as databases and Java, implement their own concurrent mechanisms that work correctly and efficiently, and provide abstractions of these mechanisms to programmers. Therefore, it is often not necessary to worry about deadlocks in practice.

11.8 Parallelism and Concurrency

The process model assumes that a process is an executing program with a single thread of control. However, all modern operating systems provide features to enable a process to contain multiple threads of control. Multithreading is the ability of an OS to support multiple concurrent paths of execution within a single process.

A thread is part of a process that contains

- a thread ID (TID),
- a program counter (PC),
- a register set, and
- a stack.



Single and multithreaded processes.

All threads within the same process share a code section and a data section (containing objects, open files and network connections). A process with multiple threads can perform more than one task at a time.

Most applications are multithreaded. Applications are suitable for multithreading if they need several threads of control. Examples include web browsers with separate threads for each tab, extension or native utility, as well as any GUI application, so that user events can still be handled while the main thread is performing a computation.

Multithreading is also used in systems that provide services, such as web servers and databases. When the server receives a request, a new thread is created to service the request, while the main thread continues to listen for new requests.

Threads improve responsiveness, resource sharing and economy — multithreading results in a smaller memory footprint and greater efficiency of switching between threads. Threads also improve scalability by enabling the use of parallel processing in multithreaded processors. They also reduce program complexity by breaking problems into smaller, independent tasks that can execute in parallel.

- A system is parallel if it can perform more than one task simultaneously.
- A concurrent system supports more than one task by allowing each task to make progress.
- Concurrency can be achieved without parallelism.

11.9 Multicore Systems and Multithreading

Amdahl's law describes the ratio of the performance of a process running on a single processor to its performance running on multiple processors in terms of the proportion f of its code that is parallelisable and the number N of parallel processors. This ratio is known as 'speedup'. The difference $(1 - f)$ is the proportion of the code that is inherently serial.

$$\text{speedup} = \frac{\text{time to execute on a single processor}}{\text{time to execute on } N \text{ parallel processors}} = \frac{1}{(1 - f) + \frac{f}{N}}$$

Even a small amount of serial code has a noticeable impact on the overall performance. There are also overheads to parallelism that are not considered by Amdahl's law. Latency is introduced by memory access and cache loading, for example.

Parallel programming introduces a number of challenges.

Pressure is placed on system designers to make better use of multiple cores and to write scheduling algorithms that allow parallel execution. It is difficult both to modify existing programs and to design new programs to make use of multithreading. This involves identifying tasks that can be divided into separate, concurrent and, ideally, independent tasks, balancing tasks to achieve efficiency, splitting data, and identifying data dependencies. It is also difficult to test and debug parallel execution since it is non-deterministic.

11.10 Multithreading Models

~~Threads are used through a thread library that exists on the OS or virtual machine. Operating systems and programming languages provide APIs for creating and managing threads. Threads may exist at either the user or kernel levels. User threads are managed above the kernel. Kernel threads are managed by the OS directly. Threads can be mapped to processes using three models.~~

11.10.1 Many-to-One Model

~~The many-to-one model maps many user-level threads to one kernel thread. Thread management is handled by the thread library in user space, so it is efficient. However, only one thread can access the kernel at a time since a system call blocks other threads. This model lacks parallelism on multicore systems and is, therefore, not widely used.~~

11.10.2 One-to-One Model

~~The one-to-one model maps each user thread to a kernel thread. This overcomes the issue of blocking, and improves concurrency and parallelism. However, for each user thread, there must be a kernel thread.~~

11.10.3 Many-to-Many Model

~~A software developer can create as many user threads as necessary in a thread pool. Corresponding kernel threads can run in parallel on a multiprocessor. If a thread blocks, the kernel can schedule another thread for execution.~~

11.11 Thread Interference and Memory Consistency

~~Threads communicate by sharing data. If multiple threads reference the same object, thread interference can occur. This happens when two functions operating on the same data interleave. This~~

is unpredictable and difficult to debug. Memory consistency errors occur when different threads have inconsistent views of the same data. This can be solved through synchronisation, which creates a happens-before relationship between methods and statements. This guarantees that memory written by one statement is visible to another. Two invocations of a synchronised method on the same object cannot be interleaved. All other threads are blocked until the synchronised thread is complete. When a synchronised method exits, it automatically establishes a happens-before relationship with any subsequent invocation of a synchronised method on the same object. Synchronisation is based on intrinsic locks. These enforce exclusive access and establish happens-before relationships. Threads must acquire locks before they can do anything.

12 Introduction to Networks

12.1 The Internet

The Internet consists of billions of connected computing devices. End systems are also known as 'hosts' and run network applications. Computers are connected via communication links, which can be fibre optic, copper, radio or satellite. End systems and different networks are connected to routers and packet switches that forward chunks of data known as 'packets'.

The Internet is a network of networks and interconnected Internet service providers (ISPs). The sending and receiving of messages is controlled by protocols such as TCP, IP, HTTP, Skype and 802.11. A set of standards exists to help manage the Internet. Proposals for standards and protocols are published as requests for comments (RFCs), which are made available to key stakeholders to be reviewed and standardised. The Internet Engineering Taskforce (IETF) is the governing body that oversees the development of the Internet.

The Internet is also an infrastructure that provides services to applications. These services and applications include web, VoIP, email, games, e-commerce and social networks. It also provides a programming interface to applications. Web hooks allow applications to connect to the Internet.

Residential, institutional and mobile access networks each have end systems connected to an edge router. The bandwidth of a connection is affected by the hardware and connection used. Radio links are shared, whereas an Ethernet cable is dedicated.

- Network Edge:
Consists of the hosts/endsystems, both clients and servers. Hosts send packets of data to their access network.
- Access networks:
The physical wired/wireless communication links that connect edge systems to the core. These are the first routers that see the data packets and forward them to the network core.
- Network core:
The central network of networks: a mesh of interconnected routers. The core routes and forwards packets of data from their source to their destination.

Hosts break application messages into smaller chunks known as 'packets' and transmit them to the access network. The speed at which the packet is transmitted to the access network is the link transmission rate, which is also known as 'link capacity' or 'link bandwidth'. The packet transmission delay when a packet of length L bits is transmitted into a link with transmission rate R bits per second is $\frac{L}{R}$ seconds.

The network core forwards packets from one router to the next across links that form a path between source and destination. Each packet is transmitted at full link capacity. Routing algorithms in the network core determine the source to destination route taken by packets. Routers forward packets from their input to their appropriate output.

12.2 Distributed Systems

A distributed system is one in which hardware or software components located at networked computers communicate and coordinate their actions by only passing messages. Distributed networks allow for high concurrency. There is no global notion of the correct time in a distributed network; there is only relative time between computers. Each component in a network may fail independently. The failure of one component does not affect the other components.

12.3 Internet, Intranet and Firewalls

The Internet is a distributed system that enables users all over the world to make use of its services. Some highly connected links in the Internet are known as backbones. These have high link capacity and can be connected via satellite link.

An intranet is part of the Internet that is separately administered and uses a firewall to enforce its own local security policies. Users in an intranet share data by means of file services.

A firewall is a network security system that monitors and controls incoming and outgoing network traffic according to predetermined security rules. A firewall establishes a barrier between a trusted internal network and an untrusted external network, such as the Internet. Firewalls may be network-based or host-based, including network-layer or packet filters and application-layer firewalls.

12.4 Network Principles

Networking is concerned with sending messages over a carrier.

- Latency is the term given to any kind of delay that occurs during data communication over a network.
- Bandwidth is the transmission capacity of a computer network or telecommunication system.
- Speed is the rate at which data is able to move.

A logical unit of data transmitted via a network is known as a message. A message of arbitrary length is divided into packets before transmission. A packet is a bit stream of restricted length. It includes not only the data, but also relevant addressing information.

12.5 Switching Schemes

A network is a set of nodes connected by circuits. In order to transmit information between two nodes, a switching scheme is required.

- Broadcast — no switching, data is transmitted throughout the network
- Circuit switching — source and destination are connected through a switch at an exchange
- Packet switching — a store-and-forward network with a computer at each end

12.5.1 Circuit Switching

End-to-end resources are allocated to create a reserved connection between source and destination. The resources are dedicated; there is no sharing. Full bandwidth circuit-like performance is guaranteed. Circuit segments remain idle if not used for a connection. This transmission scheme is traditionally used in telephone networks.

12.5.2 Packet Switching

Packet switching is a store-and-forward transmission scheme that allows more users to use the network. Packets are transmitted to a router, where they are stored in a queue, waiting for an output link. The entire packet must arrive at the router before it can be transmitted through the next link.

The delay for transmission of an L bit packet through an R bits per second link is $\frac{L}{R}$ seconds. This is the 'one-hop transmission delay'. Assuming no propagation delay, the end-to-end delay for transmission from source to destination through one router is $2\frac{L}{R}$. In general, the delay is $N\frac{L}{R}$, where N is the number of links and, therefore, $N - 1$ is the number of routers.

If the arrival rate to a router exceeds its transmission rate for a period of time, the packets will queue, waiting to be transmitted through a link. Additionally, packets can be dropped (lost) if the memory (buffer) is filled.

12.6 Internet Structure (Network of Networks)

End systems connect to the Internet via access ISPs. These may be residential, company or university ISPs. In turn, access ISPs must be interconnected so that any two hosts can communicate. The resulting network is very complex. Its evolution has been driven by economics and national policies.

Connecting each access ISP to every other access ISP is not feasible. This would require $O(n^2)$ connections. An alternative is to connect each access ISP to a global ISP. There would be an economic agreement between customer and provider ISPs. However, if one global ISP is a viable business, there will exist competitors, which must in turn be interconnected.

Such global ISPs are connected to each other through Internet exchange points (IXPs) or peering links. Regional networks also exist to connect groups of access networks to global ISPs. Large

Internet companies, such as Google and Microsoft, run their own content delivery networks to bypass connections and provide services and content closer to end systems.

The result is small number of well-connected large networks. These are 'Tier 1' commercial ISPs that offer national and international coverage, and private content delivery networks that connect data centres to the Internet to bypass Tier 1 and regional ISPs.

12.7 Packet Loss and Delays

When the arrival rate of packets at a router exceeds the output link capacity, the packets wait in a buffer queue for transmission. A packet is dropped (lost) if there is no available buffer space in the queue when it arrives. A lost packet may be retransmitted by the previous node, the end system source or not at all. Nodal delay d_{nodal} is the delay of packets at a node in a network.

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

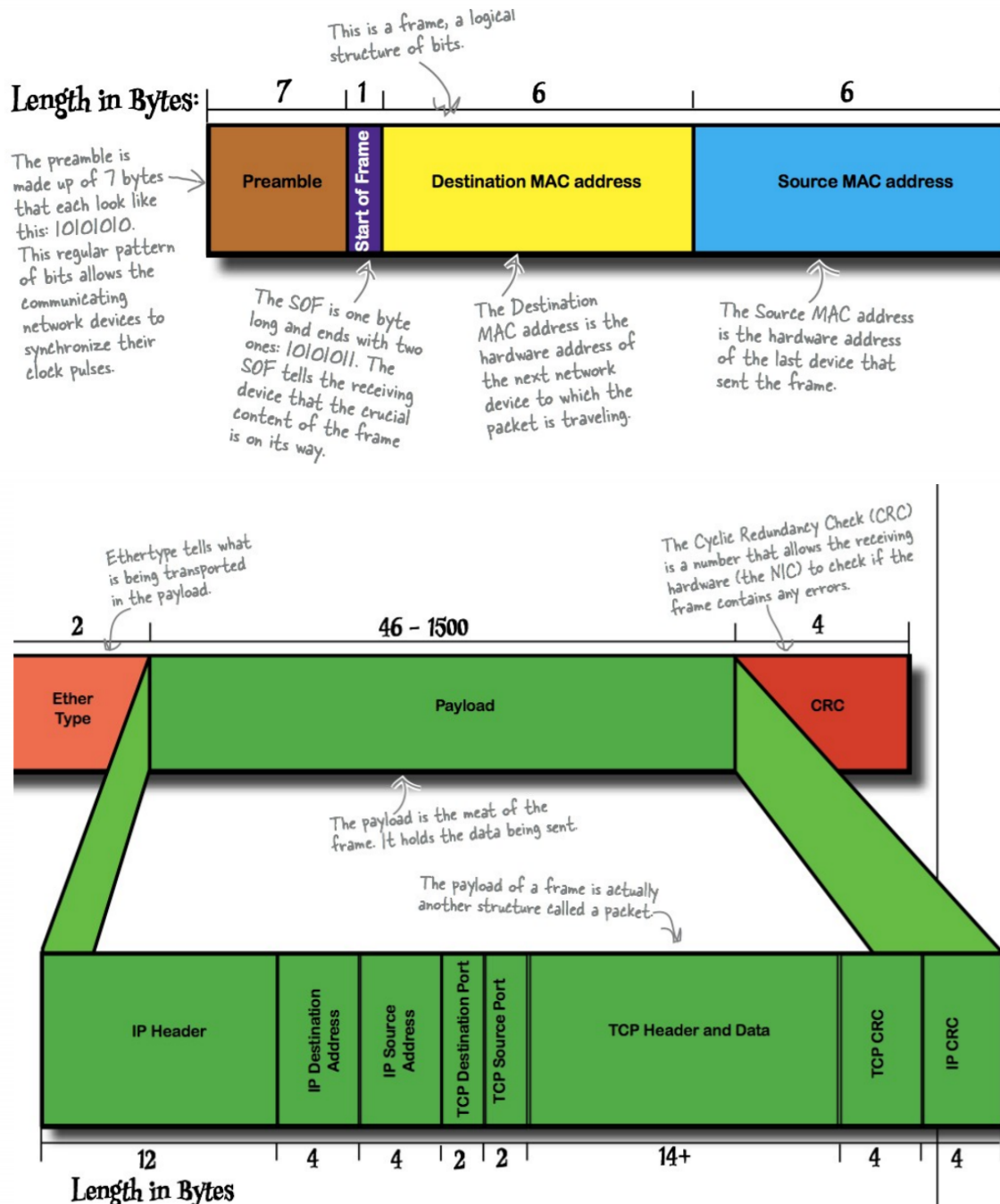
- Nodal processing delay (d_{proc})
 - Time spent checking bit errors and determining output link
 - Typically less than a millisecond
- Queueing delay (d_{queue})
 - Time spent waiting at output link for transmission
 - Dependent upon congestion level at link/router
- Transmission delay (d_{trans})
 - Time spent placing packet onto output link
 - For a packet of length L bit and link of bandwidth R bit s⁻¹, $d_{\text{trans}} = \frac{L}{R}$ s
- Propagation delay (d_{prop})
 - Time spent travelling through output link to next node
 - For a link of length d m and a propagation speed through the link medium of s m s⁻¹, $d_{\text{prop}} = \frac{d}{s}$ s

Throughput is the rate at which bits are transferred between sender and receiver. Throughput can be measured as instantaneous throughput at a specific point in time or average throughput over a longer period of time. The average end-to-end throughput is determined by the link with the lowest capacity. A link on an end-to-end path that constrains the throughput is known as a 'bottleneck link'.

12.8 Protocols

Network protocols define a set of guidelines that allow network devices to communicate effectively. A protocol defines the sequence of messages that should be exchanged and the format of the data in the messages. Protocols are implemented by pairs of software modules in the sending and receiving systems.

A transport protocol transmits a message from a sender to a receiver. A process that wishes to send a message passes it to the transport protocol module. The software divides the message into packets, which are then transmitted using the network protocol. Inverse operations are performed by the receiver to reconstruct the message.



The structure of a packet frame.

12.9 Protocol Layers and Encapsulation

Network software is arranged in a hierarchy of layers, in which each layer presents an interface to the layer above. Each layer accepts an item of data in a specified format from the layer above. It applies a transformation to the data in order to encapsulate it. The data is passed to the layer

below. Layers communicate with those directly above and below through procedure calls. Every computer in a network must have these layers.

A complete set of protocol layers is known as a 'protocol suite' or 'protocol stack'. Each layer adds a header to the data before it is sent.

12.10 Open Systems Interconnection (OSI) Model

The Open Systems Interconnection (OSI) model is a conceptual protocol stack model. Its layers from highest to lowest are

- application,
- presentation,
- session,
- transport,
- network,
- (data) link, and
- physical.

Data are passed down through the layers in the source end system and across the network via physical connections. Switches and routers use the data link and network layers to route the data through the network. When the data arrive at the end system, they are passed up through the layers and are reconstructed.

The layers of the OSI model.

<i>Layer</i>	<i>Description</i>	<i>Examples</i>
Application	Protocols that are designed to meet the communication requirements of specific applications, often defining the interface to a service.	HTTP, FTP , SMTP, CORBA IIOP
Presentation	Protocols at this level transmit data in a network representation that is independent of the representations used in individual computers, which may differ. Encryption is also performed in this layer, if required.	Secure Sockets (SSL),CORBA Data Rep.
Session	At this level reliability and adaptation are performed, such as detection of failures and automatic recovery.	
Transport	This is the lowest level at which messages (rather than packets) are handled. Messages are addressed to communication ports attached to processes, Protocols in this layer may be connection-oriented or connectionless.	TCP, UDP
Network	Transfers data packets between computers in a specific network. In a WAN or an internetwork this involves the generation of a route passing through routers. In a single LAN no routing is required.	IP, ATM virtual circuits
Data link	Responsible for transmission of packets between nodes that are directly connected by a physical link. In a WAN transmission is between pairs of routers or between routers and hosts. In a LAN it is between any pair of hosts.	Ethernet MAC, ATM cell transfer, PPP
Physical	The circuits and hardware that drive the network. It transmits sequences of binary data by analogue signalling, using amplitude or frequency modulation of electrical signals (on cable circuits), light signals (on fiber optic circuits) or other electromagnetic signals (on radio and microwave circuits).	Ethernet base- band signalling, ISDN

Protocol layering simplifies and generalises the software interfaces that access communication services. However, they reduce the performance of the network. A total of N protocol layers results in N control transfers for each message at an end system, and N copies of the data due to encapsulation. Thus, the data transfer rate may be much slower than the available network bandwidth.

The five-layer protocol stack used by the Internet differs from the seven-layer OSI model. Its layers from highest to lowest are

- application,
- transport,
- network,
- link, and
- physical.

The application, presentation and session layers of the OSI model are not clearly distinguished in the Internet protocol stack, and are instead all part of a single application layer. Applications are responsible for deciding how to handle these protocols. The session layer is integrated with the transport layer.

12.11 Large Messages and Message Consistency

An Ethernet frame can only hold 1500 B of data. Messages larger than this limit must be broken into packets. Message consistency and reliable data transfer are achieved by communicating via the Transmission Control Protocol (TCP). If there are errors in the packets, the receiver notifies the sender, and those packets are sent again. If the message is a single long packet, there may be issues due to poor connection.

Packets may not be received in the same order that they are sent. Each packet contains a sequence number so that they can be reordered correctly.

12.12 Ports and Addressing

The responsibility of the transport layer is to provide a network-independent message transfer service between pairs of network ports. The destination ports at a host computer are defined by software and are attached to processes. Port 25 is used for email.

The transport layer delivers messages to destinations specified by transport addresses. A transport address comprises a network address and a port number. In the Internet, every host system is assigned an Internet Protocol (IP) address that identifies it and the subnet to which it is connected.

12.13 Datagram Packet Delivery

In datagram packet delivery, it is the responsibility of the receiver to indicate if there is a problem with transmission. The network retains no information after it has been delivered. The sequence of packets may take different routes from host to destination, so they may arrive out of sequence. Datagrams contain the full network addresses of the source and destination. The network layer of the Internet uses datagram delivery.

A link-layer switch accesses the data link layer of a packet to determine the next link to which the packet must be sent. A router is a three-layer switch. It accesses the network layer of the packet in order to route the packet to the destination host.

13 Application Layer

13.1 Principles of Network Applications

Network applications run on different end systems and communicate over a network. For example, a browser communicates with web server software. There is no need to write network applications for network-core devices. Applications run only on end systems. This allows for rapid application development and propagation. Network applications can follow either client-server or peer-to-peer (P2P) architecture.

The Application Layer is the highest layer on the TCP/IP Stack. The protocols of this layer define communication for network applications. Example network applications:

- World Wide Web
- Email
- P2P File Sharing
- Remote Login
- Video Streaming
- VoIP
- Social Networking
- Search

A process is part of a program or application running within a host machine:

Within the same host, processes communicate using inter-process communication protocols defined by the operating system.

Within different hosts, processes communicate by exchanging messages on the network (i.e., by using network protocols). A client process is a process that initiates communication. A server process is a process that waits to be contacted. P2P applications use both client processes and server processes.

13.1.1 Client-Server Architecture

P2P applications use both client processes and server processes. In a client-server architecture, the server is an always-on host with a permanent IP address and data centres for scaling. Clients communicate with the server, and may have intermittent connections and dynamic IP addresses. Clients do not communicate with each other directly.

13.1.2 Peer-to-Peer (P2P) Architecture

In a peer-to-peer architecture, there is no always-on server. Arbitrary end systems communicate directly. Peers request and provide services from and to other peers. Such an architecture is self-

scalable; new peers bring additional service capacity as well as additional service demands. Peers are intermittently connected and have dynamic IP addresses. This requires a complex management of connections.

13.1.3 Process Communication and Sockets

These are the process-to-network interface between the Application Layer and the Transport Layer. A process sends/receives messages to/from its socket, which communicates with the transport protocol.

A process sends and receives messages to or from its socket. A sending process relies on the transport infrastructure on the outside of its socket to deliver its message to the socket at the receiving process.

In order to receive messages, each process must have an identifier. Every host has a unique 32 bit IP address. The process identifier comprises the host IP address and the port number associated with the process on the host. Example port numbers include 25 for mail servers and 80 for HTTP servers.

Every host has a unique 32-bit IP address (e.g.: 194.80.24.5)

Every process uses a unique port number (e.g.: 80 for HTTP)

Socket Address = IP Address : Port Number (e.g.: 194.80.24.5:80)

13.2 Transport Services

Network applications require different transport services.

- Data integrity
 - Some apps, such as file transfer and online transactions, require reliable data transfer.
 - Other apps, such as audio streaming, can tolerate some loss.
- Timing
 - Some apps, such as Internet telephony and games, require low delay to be effective.
- Throughput
 - Some apps (e.g. videos) require a minimum amount of throughput
 - Other apps (e.g. email) make use of whatever they get
- Security
 - Some apps require encryption and other security services.

13.2.1 Transmission Control Protocol (TCP)

- Connection-oriented, reliable data transfer
- Flow and congestion control

The Transmission Control Protocol (TCP) provides reliable transport between sending and receiving processes; the data received is identical to the data sent. The protocol also provides flow control (so that the sender does not overwhelm the receiver) and congestion control (so that the sender is

throttled when the network is overloaded). The protocol does not provide guarantees for timing, minimum throughput or security — these must be handled by the application layer. TCP is a connection-oriented protocol; a connection must be set up between the client and the server before communication can occur.

13.2.2 User Datagram Protocol (UDP)

- Connectionless,unreliable data transfer
- No flow or congestion controls

The User Datagram Protocol (UDP) provides unreliable data transfer between sending and receiving processes. It does not provide reliability, flow control, congestion control, timing, throughput, security or connection setup.

UDP exists both because it is a very simple protocol to use and because it allows a lot more data to be transferred across a network than TCP.

Data integrity is sacrificed for greater throughput.

This is useful for Internet telephony applications.

Application and transport protocols of common applications.

Application	Application-Layer Protocol	Underlying Transport Protocol
Electronic mail	SMTP [RFC 5321]	TCP
Remote terminal access	Telnet [RFC 854]	TCP
Web	HTTP [RFC 2616]	TCP
File transfer	FTP [RFC 959]	TCP
Streaming multimedia	HTTP (e.g., YouTube)	TCP
Internet telephony	SIP [RFC 3261], RTP [RFC 3550], or proprietary (e.g., Skype)	UDP or TCP

13.3 Hypertext Transfer Protocol (HTTP)

A web page consists of a base HTML file that may include several referenced objects, such as images, applets or media files. Each object is addressable by a Uniform Resource Locator (URL), which consists of a host name and a path.

The Hypertext Transfer Protocol (HTTP) is the application layer protocol of the web. The web uses a client-server architecture. A web browser is a client that requests, receives and displays web objects. The web server sends objects in response to requests.

HTTP uses the TCP protocol. HTTP itself is a stateless protocol. The server does not retain any information about past client requests.

13.3.1 Non-Persistent HTTP

1. The HTTP client initiates a TCP connection to the HTTP server at the requested host address on port 80.
2. The HTTP server accepts the connection and notifies the client.
3. The HTTP client sends an HTTP request message (containing a URL to the requested object) through its TCP socket.
4. The HTTP server receives the request message, forms a response message containing the requested object and sends it through its TCP socket.
5. The HTTP server waits until its message is received by the client, then closes the TCP connection.
6. The HTTP client receives the response message and displays the HTML file.
7. If object references are found in the HTML file, the above process is repeated for each object.

The round-trip time (RTT) is the time it takes for a small packet to travel from the client to the server and back again. The HTTP response time consists of one RTT to initiate the TCP connection, another RTT for the HTTP request to be sent and the HTTP response to start being received, and the file transmission time for the object being served.

$$\text{HTTP response time} = 2\text{RTT} + \text{file transmission time}$$

- 1 RTT to initiate TCP connection
- 1 RTT for HTTP request and first bytes of HTTP response to return
- File transmission time (FTT)

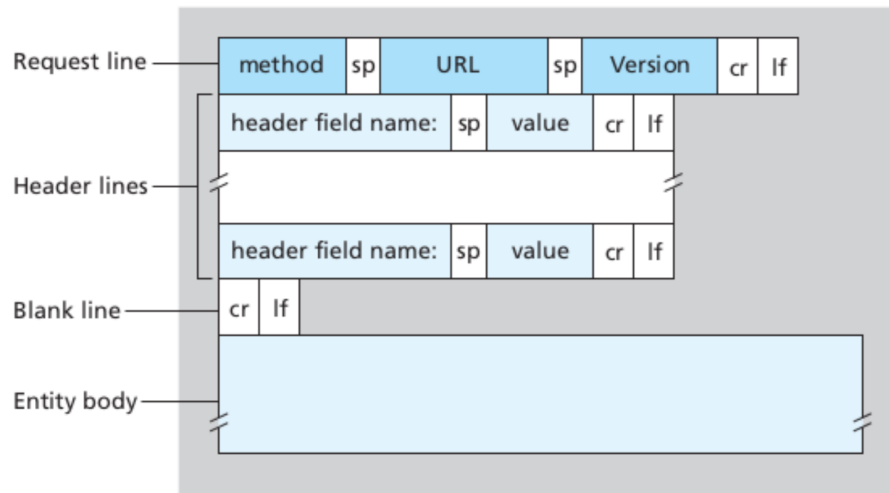
13.3.2 Persistent HTTP

The use of non-persistent HTTP leads to a number of issues. It takes two round-trips for each object and causes an OS overhead for each TCP connection. Browsers often open parallel TCP connections to fetch referenced objects.

In persistent HTTP, the server leaves the connection open after sending a response. Subsequent HTTP messages between the client and server are sent over an open connection. This allows the client to send a request as soon as it encounters a referenced object. This uses only one round-trip for each referenced object.

13.3.3 HTTP Request Message

HTTP messages are written in a human-readable ASCII format. A request message begins with a request line, which contains the HTTP method (GET, POST or HEAD, for example). The request line is followed by header lines. The end of the header is indicated by a carriage return and line feed (CRLF) at the start of a line.



General format of a HTTP request message.

```

Request Method      URL      Protocol      Carriage return
                  and line feed

Request line: GET /somedir/page.html HTTP/1.1 \r\n
Header Lines      Host: www.bham.ac.uk \r\n
(most optional)   Connection: close \r\n
                  User-agent: Mozilla/5 \r\n
Blank line:      \r\n
Optional         ...
message body     ...
(e.g. form data) ...

```

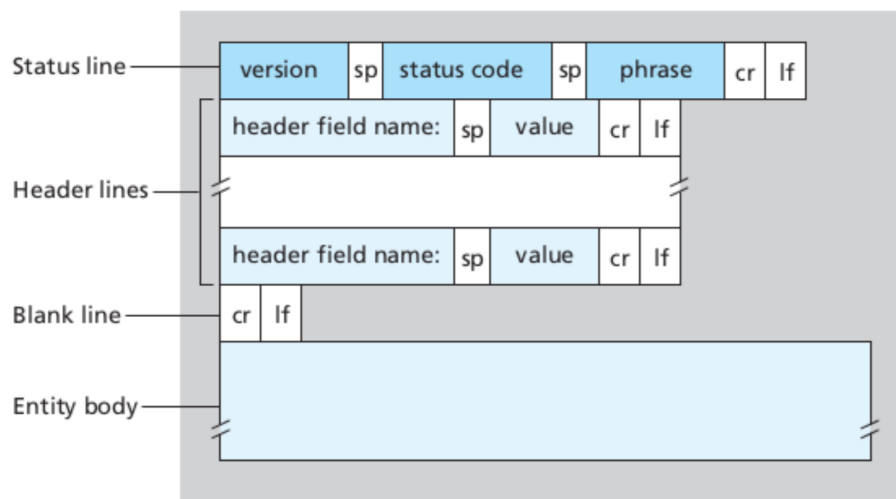
http request message

13.3.4 HTTP Response Message

A response message begins with a status line, which contains the status code and status phrase. Possible status codes and phrases include

- 200 (OK) — request succeeded, requested object contained in this message,
- 300 (Moved Permanently) — requested object has moved to a new location specified in this message,
- 400 (Bad Request) — request message not understood by server,
- 404 (Not Found) — requested document not found on the server, and
- 505 (HTTP Version Not Supported).

The status line is followed by a header that ends with a CRLF at the start of a line. The requested data follows the header.



General format of a HTTP response message.

13.4 Simple Mail Transfer Protocol (SMTP)

Three major components to email:

- User Agents “Mail Readers” Used for composing/editing/reading email messages For example, Outlook, Thunderbird, iPhone mail app
- Mail Servers Where incoming/outgoing messages are stored Mailbox contains incoming messages for user Message queue of outgoing mail messages
- SMTP Simple Mail Transfer Protocol between mail servers to send email messages

A user agent (UA) is an email client that is responsible for composing, editing and reading messages. Incoming and outgoing messages are stored on a server.

A mail server contains mailboxes for each user that contain incoming messages and a message queue of outgoing messages. The Simple Mail Transfer protocol (SMTP) is used to send email messages between mail servers. A mail server takes on the role of client when it is sending emails, and the role of server when it is receiving emails.

SMTP uses TCP to reliably transfer emails directly from client to server at port 25. An email transfer consists of three phases: handshake, transfer and closure. SMTP uses a command and response interaction similar to that of HTTP. Commands and responses are written in ASCII text and responses contain status codes and phrases. SMTP commands include HELO, MAIL FROM, RCPT TO, DATA and QUIT.

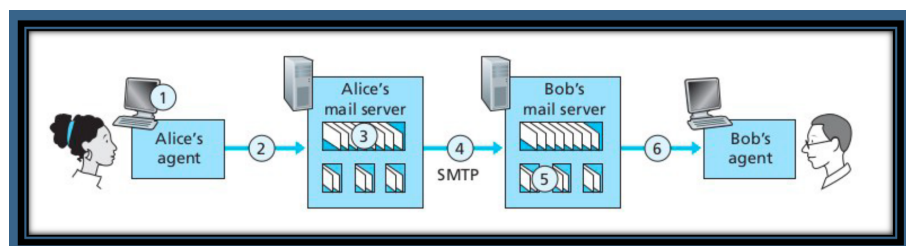
SMTP uses persistent connections. It requires that messages are composed in 7 bit ASCII. The end of a message is indicated by a CRLF followed by a period and another CRLF. Whilst HTTP uses a “pull” interaction, SMTP uses a “push” interaction. In HTTP, each object is encapsulated in its own response message. In SMTP, multiple objects can be sent in multipart messages.

SMTP is the Application Layer protocol for sending email between mail servers

1. Architecture: Client-Server The “client” is the sending The “server” is the receiving
2. Default Port: 25
3. Transport Protocol: TCP

A Scenario

1. Alice uses her user agent to compose a message to bob@bham.ac.uk
2. Alice user agent sends a message to her mail server
3. Alice 's message is placed in mail queue
4. Client side of SMTP opens a TCP connection with Bob's mail server, and then sends Alice's message over the TCP connection
5. Bob's mail server places the message in his mailbox
6. Bob uses his user agent to read Alice's message



SMTP Scenario

- Three phases of transfer: Handshaking, Transfer of messages, Closure
- Command/Response interaction like HTTP
 - Commands: ASCII text
 - Response: Status code and phrase
- Uses persistent connections
- In HTTP, each object has its own response message
 - In SMTP, multiple objects are sent in multipart messages

13.5 Mail Access Protocols

Whilst SMTP is used for delivering and storing messages to the mail server of a receiver, a mail access protocol is used for retrieving messages from a server.

- POP3 (Post Office Protocol Version 3) — authorisation and download
- IMAP (Internet Mail Access Protocol) — more features, including manipulation of a stored message on a server, keeps user state across sessions
- HTTP (Hypertext Transfer Protocol) — For web email clients

POP3 uses an authorisation phase, in which client issues `user` and `pass` commands to declare a username and password, and a transaction phase, in which the client issues `list`, `retr`, `dele` and `quit` commands to list message numbers, retrieve a message by number, delete a message by number and quit the server.

POP3 download-and-delete mode deletes messages from the server once they are downloaded to the client. They cannot be downloaded again on another client. POP3 download-and-keep mode allows copies of messages on multiple clients. POP3 is stateless across sessions.

IMAP keeps all messages on the server and allows messages to be organised in folders. IMAP retains the user state across sessions. This includes the names of folders and the mappings of message identifiers to folder names.

13.6 Domain Name System (DNS)

Internet hosts and routers have 32 bit IP addresses used for addressing datagrams, and hostnames that are used by humans. A Domain Name System (DNS) is a distributed database that maps hostnames to IP addresses.

A DNS provides

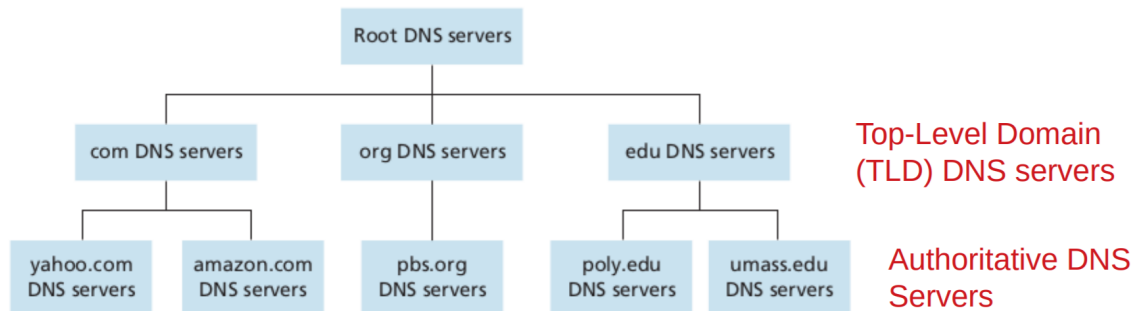
- hostname to IP address translation,
- host aliasing (allows a host with a single canonical name to have multiple alias names), mail server aliasing
- load distribution (web servers are replicated such that many IP addresses correspond to one host name).

DNS is an application layer protocol used by hosts and name servers to communicate when resolving names. DNS servers are end systems that exist at the network edge. Lookup is performed as close as possible to the source host using a local DNS server.

A single centralised DNS does not exist as this would introduce a [single point of failure](#). It would also be impossible to implement as all Internet [traffic volume](#) would have to pass through a single server. There would be long delays for end systems far from the server, and performing [maintenance](#) on the server would cause all Internet services to stop.

13.6.1 The DNS Distributed Hierarchical Database

DNS is implemented using a hierarchy of many name servers.



A portion of the hierarchy of DNS servers.

There are several root DNS servers worldwide. Each is actually a network of replicated servers. A top-level domain (TLD) server is responsible for `com`, `org`, `net`, `edu`, `gov` and all top-level country domains. An authoritative server is a DNS server owned by an organisation. It provides authoritative hostname to IP mappings for the named hosts of the organisation. These can be maintained by the organisation itself or a service provider on its behalf.

Each ISP has its own local DNS server, which is also known as a 'default name server'. Local DNS servers do not strictly belong to the DNS hierarchy. When a host makes a DNS query, the query is sent to the local DNS server, which maintains a cache of recent name to address mappings (that may be out of date). It also acts as a proxy and forwards the query into the hierarchy if necessary.

13.6.2 DNS Resolution

With an iterative DNS query, if a contacted server does not know the mapping of a host name, it will respond with the name of server that might know. For example, a local DNS server may query the root DNS server, which may respond with the name of the TLD server. The local DNS server then queries the TLD server and receives the name of an authoritative server. When the local DNS server queries the authoritative server, it will return the address of the requested host.

With a recursive query, each contacted server resolves the name by querying the next server instead of returning a reference to it. This may not be desirable as it can place a heavy load on the upper levels of the DNS hierarchy.

1. Client queries Root DNS to find `.com` DNS
 2. Client queries `.com` DNS to find `amazon.com` DNS
 3. Client queries `amazon.com` DNS. and gets `amazon.com`'s IP address
- Top-Level Domain (TLD) Servers
 - Responsible for `com`, `org`, `net`, `edu`, `gov`
 - Responsible for all top-level country domains: `uk`, `fr`, `ca`, `jp`, etc.

- Different registry services responsible for different TLD Servers E.g.: Verisign responsible for com, Educase responsible for edu
- Authoritative Domain Servers
 - Organization's own DNS server(s)
 - Provide authoritative hostname to IP mappings for organization's named hosts
 - Can be maintained internally by the organization or externally by a service provider

13.6.3 Local DNS Servers

- Each ISP (residential, company, university) has its own local DNS server
 - These do not strictly belong to the hierarchy
- When a host makes a DNS query, the query is sent to its local DNS server
- The local DNS server acts as a proxy, and forwards the query into the hierarchy
 - It also has a cache of recent name-to-address translation pairs
 - This may be out of date! Amazon changes its default IP every few hours

13.6.4 DNS Caching and Records

Once a name server learns a mapping, it stores that mapping in cache. Cache entries timeout after a period of time known as a 'time-to-live' (TTL). TLD mappings are typically cached in local DNS servers. This means that root servers are not visited often. A cached entry may be out of date. If a host changes its IP address, it may not be known Internet-wide until all of its cached mappings have expired.

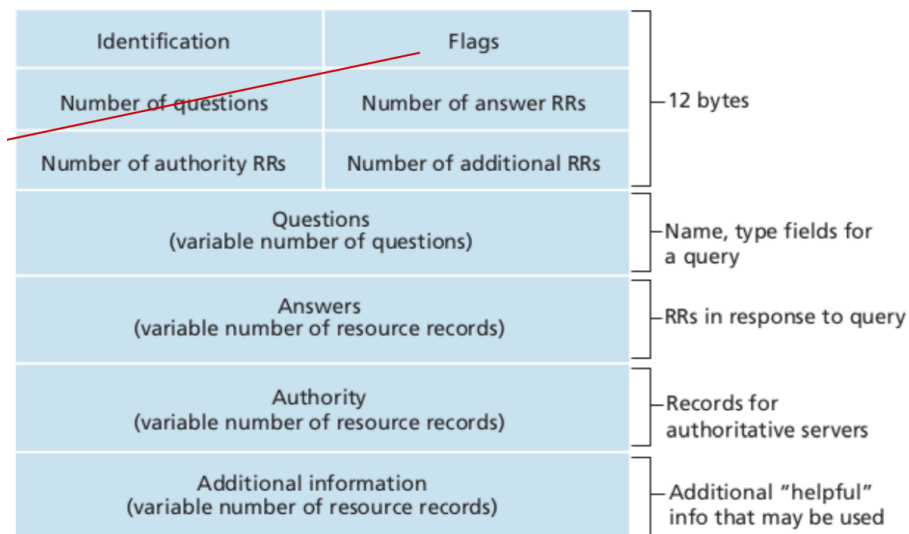
DNS records are stored in the distributed database as resource records (RR), which consist of a name, value, type and TTL.

- Type A
 - Name is the host name
 - Value is the IP address
- Type CNAME
 - Name is an alias for a canonical name
 - Value is the canonical name
- Type NS
 - Name is a domain name
 - Value is the host name of the authoritative server for the domain
- Type MX
 - Value is the name of the mail server associated with the name

13.6.5 DNS Protocol Messages

DNS queries and replies follow the same format. The message header contains a 16 bit identification number for the query. The same number is used for a reply to that query. The flags state whether

the message is a query or reply, whether recursion is desired, whether recursion is available and whether the reply is authoritative.



The general format of a DNS query or response message.

13.6.6 DNS Registration

When an organisation registers its host name `example.com` with a DNS registrar, the registrar inserts two RRs into the relevant TLD server. These are a type NS record for the name of the authoritative server of the organisation (`example.com`, `dns.example.com`, NS) and a type A record for the IP address of the authoritative server (`dns.example.com`, `111.111.111.1`, A). The authoritative server contains type A records for its subdomains and a type MX record for its mail server.

14 Transport Layer

14.1 Transport Layer Services and Protocols

The transport layer services provide logical communication between application processes running on different hosts. The protocols on this layer may provide transport services (such as congestion/flow control) or they may not... But they do not provide any speed, delay or bandwidth guarantees! Whilst the network layer provides logical communication between hosts, the transport layer provides logical communication between processes. It therefore relies on and enhances the network layer services. Transport protocols run on network edge devices (hosts/end-systems)

- Sender:
 - Gets message (arbitrary size) from the Application Layer (via socket) ,breaks it into logical segments (specific size) , passes it on to the Network Layer
- Receiver:
 - Gets segments from the Network Layer,reassembles them into a message,passes it on to the Application Layer (via socket)

The Internet uses TCP and UDP transport protocols.

TCP provides reliable ordered delivery with congestion control, flow control and connection setup. connection oriented protocol. UDP provides unreliable unordered delivery. It is a “best-effort” extension of IP, meaning that it provides no guarantee of delay or bandwidth; delay and packet loss are dependent upon traffic in the network.

14.2 Multiplexing and Demultiplexing

- Multiplexing (sender) allows multiple data streams from different applications on a single host sharing the same network connection to reach their intended processes. Each stream is split into transport layer segments. At this point:
 - A transport header is added to each segment.
 - The header includes the source and destination port numbers, which are essential for proper delivery at the receiver.
- Demultiplexing (receiver) When IP datagrams arrive, each contains:
 - A source IP address and a destination IP address.
 - A transport layer segment with source and destination port numbers.

The receiving host uses this information to determine which socket the segment belongs to. The correct segment is then passed to the corresponding application process, where the original data is reassembled.

In demultiplexing, the host uses port numbers (and IP addresses) to direct TCP/UDP segments to the appropriate socket

- UDP is connectionless
 - UDP sockets have the form dest ip, dest port
 - When a host receives a UDP segment (sometimes called a datagram), it checks the destination port and directs the segment to the socket with that port
 - Segments with the same destination port will be directed to the same destination socket, even if they are from different sources
- Meanwhile, TCP is connection-oriented
 - TCP sockets: source ip, source port, dest ip, dest port
 - Servers will therefore have a different socket for each connection
 - Segments are therefore sent to sockets that depend on both source and destination – i.e. they depend on the overall connection

14.2.1 Connectionless Demultiplexing

When a UDP socket is created, a host-local port is assigned to it. When a datagram is created to send to a UDP socket, the destination IP address and port number must be specified. When a host receives a UDP segment, it directs the segment to the socket with the given destination port number. IP datagrams with the same destination port number will be directed to the same destination socket regardless of the source IP address or port number.

14.2.2 Connection-Oriented Demultiplexing

A TCP socket is identified by a tuple of the source IP address, source port number, destination IP address and destination port number. The demultiplexer uses all four values to direct the segment to the appropriate socket. The server may support many simultaneous TCP sockets each identified by its own tuple. Web servers have a different socket for each connecting client. Non-persistent HTTP web servers have a different socket for each request.

14.3 User Datagram Protocol (UDP)

UDP is connectionless; there is no handshaking between a UDP sender and receiver, and each UDP segment is handled independently. UDP segments may be lost or delivered out of order. UDP is used in multimedia streaming applications, which are loss-tolerant and rate-sensitive, and DNS. Reliable transfer over UDP can be achieved by adding reliability functionality at the application layer, but this requires application-specific error recovery. This is used in the Quick UDP Internet Connections (QUIC) protocol.

The underlying transport protocols of common Internet applications.

Application	Application-Layer Protocol	Underlying Transport Protocol
Electronic mail	SMTP	TCP
Remote terminal access	Telnet	TCP
Web	HTTP	TCP
File transfer	FTP	TCP
Remote file server	NFS	Typically UDP
Streaming multimedia	typically proprietary	UDP or TCP
Internet telephony	typically proprietary	UDP or TCP
Network management	SNMP	Typically UDP
Routing protocol	RIP	Typically UDP
Name translation	DNS	Typically UDP

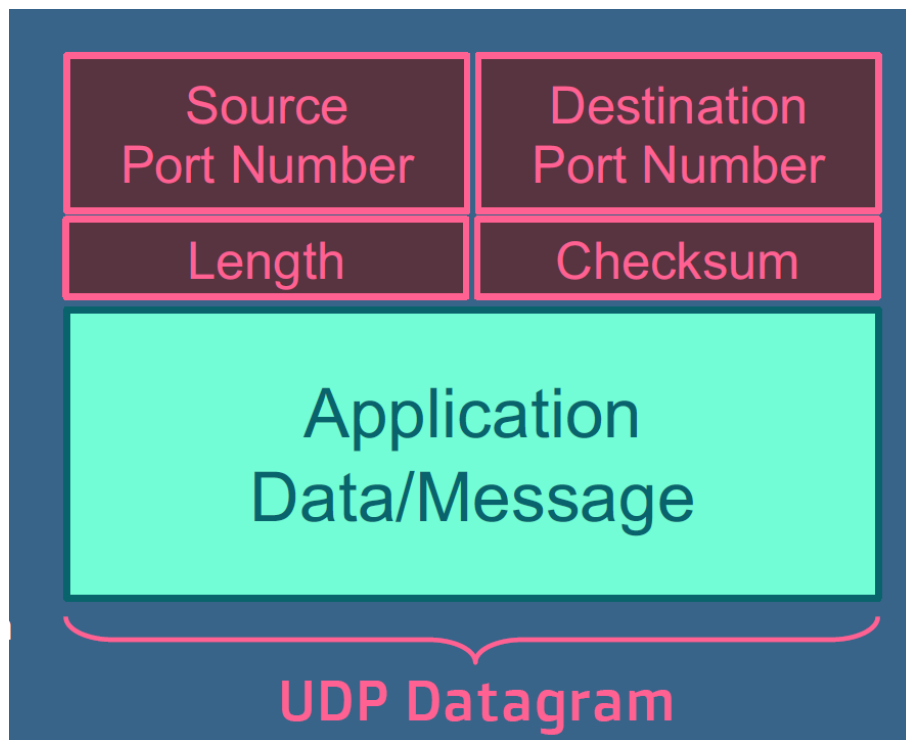
There are benefits to using UDP as a transport protocol:

- No connection establishment (which can add delay)
- Simplicity: no connection state at sender or receiver
- Smaller header size than TCP (only 8 bytes!)
- No congestion control, so UDP can send data as fast as it wants

A UDP segment contains the source port number, destination port number, length of the UDP segment (including the header) in bytes, a checksum and the application data message. UDP exists because it is simple.

14.3.1 UDP Checksum

A checksum exists to detect errors such as flipped bits in a transmitted segment. The sender treats segment contents, including header fields, as a sequence of 16 bit integers. It calculates a checksum



UDP Datagram

as the one's complement of the sum of the segment contents. The checksum is placed in the checksum field of the UDP segment.

The receiver computes the checksum of the received segment. If the computed checksum is equal to the checksum field value, no error has been detected. If the checksums are not equal, an error has been detected. Even though the checksums may be equal, it does not necessarily mean that there is no error.

This checksum is not foolproof: there can still be errors nonetheless. . .

Given two 16 bit integers 1111001100110011_2 and 1101010101010101_2 , their sum is the 17 bit integer 11011101110111011_2 . The overflow bit is removed from the sum and added to it as the least significant bit. The sum of 1_2 and 1011101110111011_2 is 1011101110111100_2 . The one's complement (inverse) of this sum is 0100010001000011_2 . This is the checksum for the two integers.

14.4 Principles of Reliable Data Transfer

Although the service provided by a reliable transfer protocol may be that of transfer through a reliable channel, it may actually be implemented through an unreliable network layer channel.

14.4.1 Reliable Data Transfer over a Perfectly Reliable Channel (RDT1.0)

If the underlying channel is perfectly reliable, i.e. there are no bit errors or loss of packets, reliable data transfer (RDT) can be accomplished by the sender transferring the data into packets to send through the unreliable channel. At the receiver, the packets are extracted into a message that is delivered to the application layer.

14.4.2 Reliable Data Transfer over a Channel With Bit Errors (RDT2.0)

If an underlying channel may flip bits in a packet, a checksum can be used to check bits. Acknowledgements (ACKs) are sent from the receiver to the sender to indicate that the packet was received correctly. Negative acknowledgements (NAKs) indicate that the received packet contained errors. The sender retransmits the packet if it receives a NAK.

14.4.3 Reliable Data Transfer With Corrupted Acknowledgements (RDT2.1)

The flaw with RDT2.0 is that an ACK or NAK may itself be corrupted. If the sender simply retransmits the packet when it receives a corrupted ACK/NAK, duplicate packets may be delivered. Duplicates can be handled by adding a sequence number to each packet before it is sent. The receiver discards a duplicate packet (one with a sequence number that has already been received).

This is a stop-and-wait implementation. The sender sends a packet then waits for a response. Thus, only two sequence numbers are required. This implementation requires the sender to check whether acknowledgements are corrupted, and the receiver to check whether a received packet is a duplicate. Both the sender and receiver must manage twice as many states as they must remember whether a packet should have a sequence number of 0 or 1.

14.4.4 Reliable Data Transfer Without Negative Acknowledgements (RDT2.2)

This achieves the same functionality as RDT2.1 using only ACKs. The receiver responds with an ACK containing the sequence number of the last correctly received packet. If the sender receives a duplicate ACK, it retransmits the current packet.

14.4.5 Reliable Data Transfer Over a Channel with Errors and Losses (RDT3.0)

Assuming the underlying channel can also lose data packets, both data and ACKs can be dropped, the use of checksums, sequence numbers, acknowledgements and retransmissions is not sufficient. The sender must wait a reasonable amount of time for an ACK. If no ACK is received in this time, the current packet is retransmitted. If the ACK was delayed or lost, the retransmitted packet will be a duplicate. This is already handled by sequence numbers. This implementation requires the added complexity of a countdown timer.

Although RDT3.0 works correctly, it has poor performance. For a 1 Gbit s^{-1} link with a 15 ms propagation delay, an 8000 bit packet will have a transmission delay of $d_{\text{trans}} = \frac{L}{R} = \frac{8000}{10^9} = 8 \times 10^{-6} \text{ s}$, and a sender utilisation of $U_{\text{sender}} = \frac{\frac{L}{R}}{RTT + \frac{L}{R}} = \frac{8 \times 10^{-6}}{2 \times 15 \times 10^{-3} + 8 \times 10^{-6}} = 0.00027$.

This protocol limits the use of physical resources. A solution is to use pipelined protocols. A sender allows multiple “in-flight” yet to be acknowledged packets. This requires the range of sequence numbers used to be increased. The sender and receiver must have a buffer.

- RDT 2.0 = RDT 1.0 + ACKs + NAKs
- RDT 2.1 = RDT 2.0 + Sequence Numbers + Stop and Wait

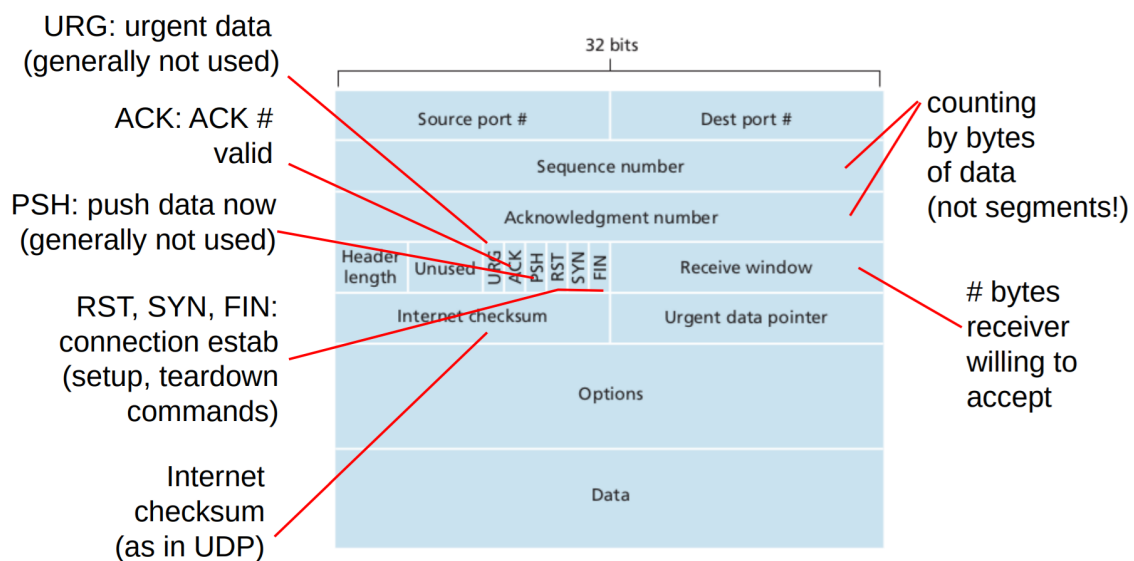
- RDT 2.2 = RDT 2.1 – NAKs
- RDT 3.0 = RDT 2.2 + Countdown Timer

RDT in Practice

- On the Network Layer, the IP protocol is an inherently unreliable, best-effort service
- On the Transport Layer, RDT 3.0 is assured by the TCP protocol
 - Pipelining
 - Cumulative ACKs (one ACK for several messages)
 - Single retransmission timeout (timer is reset each time a new one is set)
 - Retransmissions triggered by timeout and duplicate ACKs
- The QUIC (Transport Layer, 2012) and HTTP/2 (Application Layer, 2015) can be used hand-in-hand to add RDT support to UDP

14.5 Transmission Control Protocol (TCP)

TCP provides point-to-point communication with one sender and one receiver. It is a reliable protocol that sends an ordered byte stream. TCP is pipelined with congestion and flow control to set the window size of available sequence numbers. It allows full duplex data flow (bidirectional data flow in a single connection) and sets a maximum segment size (MSS). It is connection-oriented — handshaking initialises the sender and receiver states before data exchange.



The structure of a TCP segment.

The sequence number of a TCP segment is the index of the first byte of data it contains. The acknowledgement number of a TCP segment is the sequence number of the next segment expected from the other host. This is known as 'cumulative acknowledgement'. The TCP specification does not state how out-of-order segments must be handled. This is decided by the implementation. Typically, out-of-order segments are kept.

14.5.1 TCP Round-Trip Time and Timeout

The TCP timeout interval should be longer than the RTT. However, the RTT varies. A timeout interval that is too short results in premature timeout and unnecessary retransmissions. A timeout value that is too long will result in a slow reaction to segment loss.

RTT can be estimated using a sample RTT. This is the measured time between segment transmission and acknowledgement receipt, ignoring retransmissions. Since the sample RTT will vary, an exponential weighted moving average is used.

$$\text{estimated RTT} = (1 - \alpha) \times \text{estimated RTT} + \alpha \times \text{sample RTT}$$

The influence of past samples decreases exponentially. A typical value for the coefficient α is 0.125.

The timeout interval can be calculated as the sum of the estimated RTT and a safety margin. A large variation in the estimated RTT should result in a larger safety margin. Deviation of the sample RTT from the estimated RTT can be approximated.

$$\text{deviation} \approx (1 - \beta) \times \text{deviation} + \beta \times |\text{sample RTT} - \text{estimated RTT}|$$

A typical value for the coefficient β is 0.25.

$$\text{timeout} = \text{estimated RTT} + 4 \times \text{deviation}$$

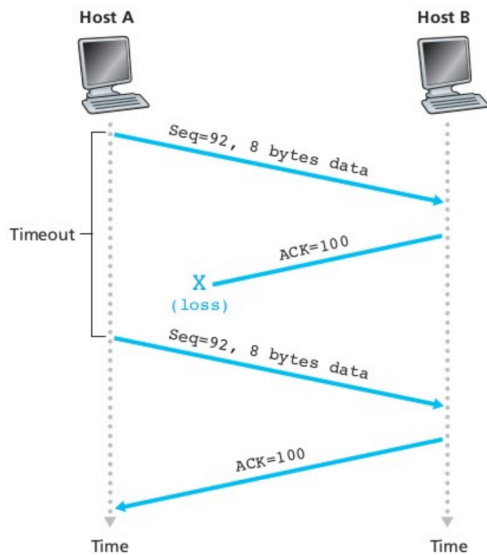
14.5.2 TCP Retransmission Scenarios

TCP creates a reliable data transfer service over the unreliable best-effort service of IP using pipelined segments, cumulative ACKs and a single retransmission timer. Retransmissions are triggered by timeout events and duplicate ACKs.

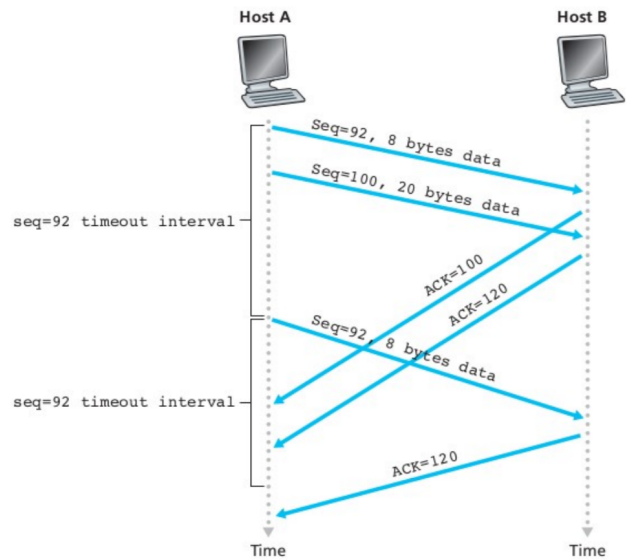
1. Create segment with sequence number — sequence number is byte stream number of first data byte in segment
2. Start timer if not already running — timer corresponds to oldest unacknowledged segment
3. On timeout, retransmit the segment and restart the timer
4. On acknowledgement, update which segments are known to be acknowledged and start timer if there remain unacknowledged segments

14.5.3 TCP Fast Retransmit

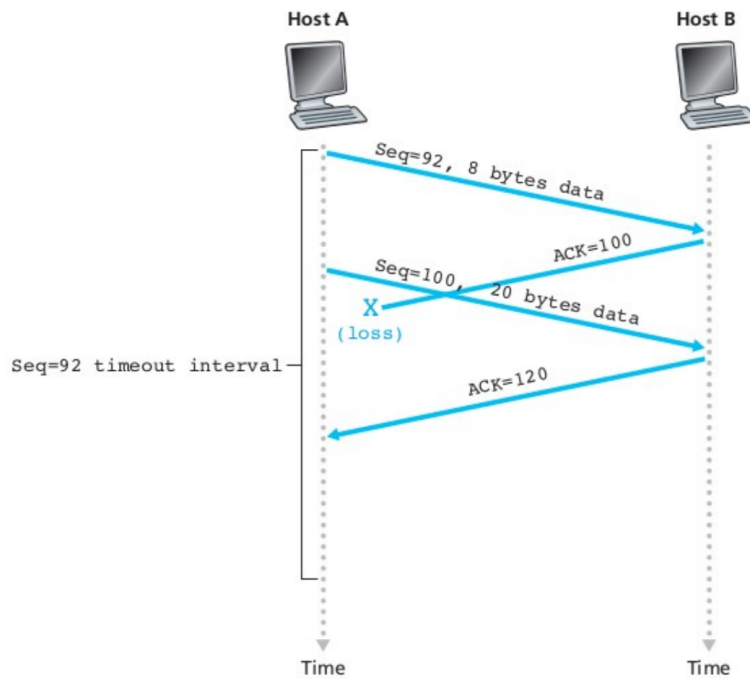
Since the timeout period is relatively long, there may be a long delay before resending a lost packet. Lost packets can be detected via duplicate ACKs. If a segment is lost, there will likely be many



Lost ACK Scenario



**Time-out Scenario;
Seg#100 not retransmitted**



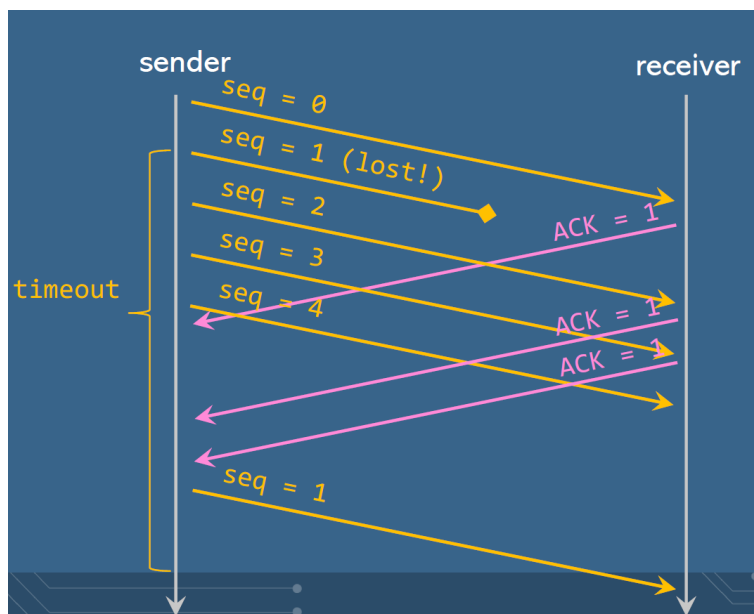
Cumulative ACK

TCP retransmission scenarios.

TCP acknowledgement generation.

Event at Receiver	TCP Receiver Action
Arrival of in-order segment with expected seq #. All data up to expected seq # already ACKed	Delayed ACK. Wait up to 500ms for next segment. If no next segment, send ACK
Arrival of in-order segment with expected seq #. One other segment has ACK pending	Immediately send single cumulative ACK, ACKing both in-order segments
Arrival of out-of-order segment higher-than-expected seq. #. Gap detected	Immediately send duplicate ACK , indicating seq. # of next expected byte
Arrival of segment that partially or completely fills gap	Immediate send ACK, provided that segment starts at lower end of gap

duplicate ACKs. If a sender receives three ACKs for the same data (“triple duplicate ACKs”), the unacknowledged segment with the smallest sequence number is resent. Since the next few responses will likely indicate that the unacknowledged segment was lost, there is no need to wait for the timeout.

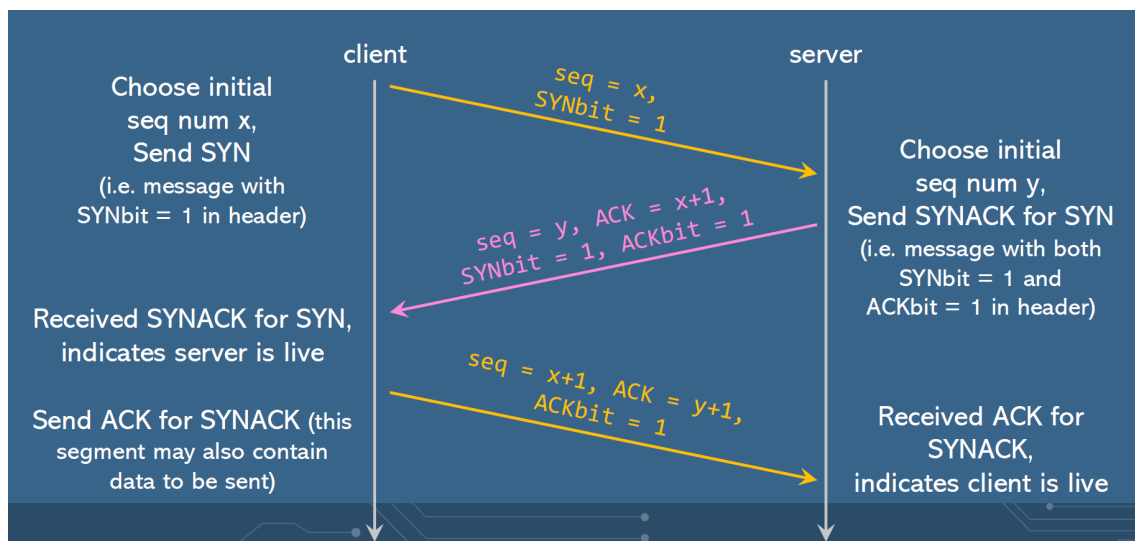


Retransmission Timeout

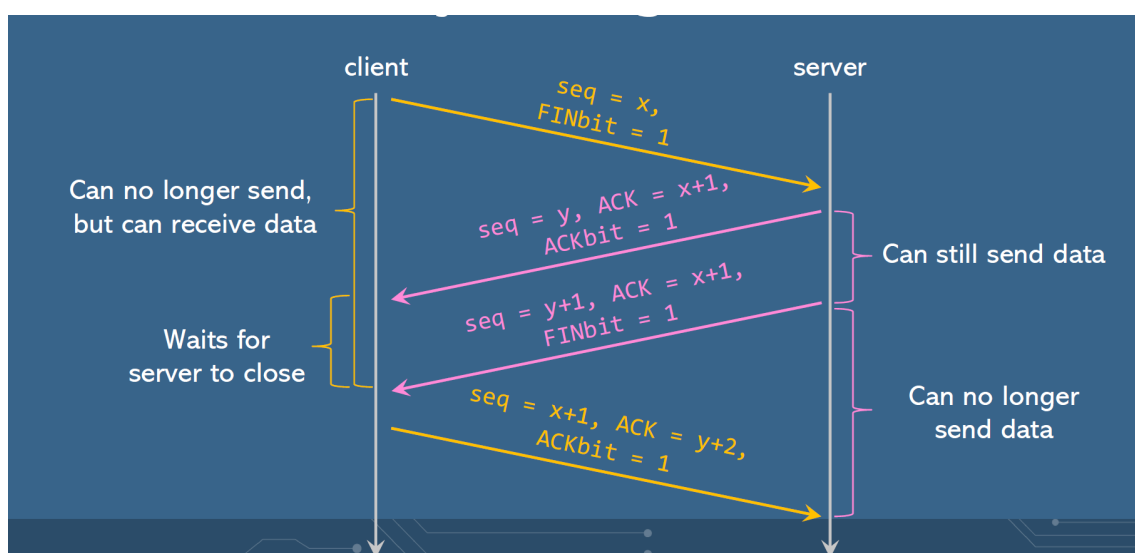
14.5.4 TCP Connection Management

Before exchanging data, the sender and receiver exchange a ‘handshake’ through which they agree to establish a connection and agree on connection parameters. TCP uses a 3-Way Handshake for establishing the connection, and a 4-Way Handshake for closing the connection.

When a TCP connection is closed, the client and server both close their side of the connection by sending a TCP segment with a FIN bit of 1. When a host receives a FIN message, it replies with an ACK. When the host is ready to close the connection it sends another FIN message and receives an ACK. The TCP connection is now closed. Simultaneous FIN exchanges can be handled.



The TCP three-way handshake.



Closing a TCP connection.

14.5.5 TCP Flow Control

The receiver controls the sender so that the sender does not overflow the buffer of the receiver by transmitting too much data too quickly. The receiver advertises its free buffer space by including a receive window size in the TCP header of receiver-to-sender segments. The receive buffer size is set via socket options. Many operating systems automatically adjust the receive buffer.

The receiver sets the receive window size to the difference between receive buffer size and the difference between the number of bytes received and the number of bytes read. The sender ensures that the difference between the last byte sent and the last byte acknowledged is less than or equal to the receive window size. This ensures that the sender only sends an amount of data that can be handled by the receiver.

- The receiver alters the window size of the pipeline
 - By changing the `window_size/rwnd` parameter in the TCP header
- This parameter is changed based on how much buffer space is free
- The window size is the amount of unACK'd ("in-flight") data that the sender is allowed to send
 - Decreasing the window size therefore limits the amount of incoming data

$$\text{receive window} = \text{receive buffer} - (\text{last byte received} - \text{last byte read})$$

$$\text{last byte sent} - \text{last byte acknowledged} \leq \text{receive window}$$

15 Network Layer

15.1 Forwarding and Routing

The Network Layer provides the host-to-host communication service for the Internet. The key protocol on this layer is the Internet Protocol (IP).

The Network Layer provides a 'Best Effort' service.

There are therefore no guarantees: no speed, delay or bandwidth guarantees; nor any in-order delivery guarantee and not even any guarantee of delivery itself!

On the sending side, it encapsulates segments into datagrams. On the receiving side, it delivers segments to the transport layer. Network layer protocols exist in every host and router. A router examines the header fields of all IP datagrams passing through it.

~~Channels for transporting datagrams can have different service models. Services for individual datagrams include guaranteed delivery or guaranteed delivery with less than a specified delay. Services for a flow of datagrams include ordered delivery, guaranteed minimum bandwidth and restrictions on changes in packet spacing.~~

The network layer (router) performs forwarding and routing.

- Forwarding is the movement of packets from the router input to the appropriate router output. A forwarding table determines local forwarding within a router.
 - Like the process of planning a road trip from source to destination which includes many interchanges/junctions
- Routing is the determination of the route taken by packets from their source to their destination. A routing algorithm determines the end-to-end path through the network.
 - Like the process of going through an interchange/junction and choosing an appropriate exit lane

Before datagrams can flow, the end hosts and intermediate routers establish a virtual connection.

15.2 Router Architecture

- Input Port – Forwarding
 - Using the IP packet header's destination IP address. . .
 - . . . and the Forwarding Table in the input port's memory. . .
 - . . . lookup the appropriate router output port
- Input Port - Queueing
 - Goal: Complete input port processing as fast as packets arrives (at 'line speed')
 - But if packets arrive faster than the forwarding rate, they will queue in the input port buffer.
- Switching Fabrics
 - Transfer packets from the input buffers to the appropriate output buffer
 - Various different architectures, with different advantages and disadvantages
- Switching Rate: The rate at which packets can be transferred from inputs to outputs
 - Often measured in terms of the line rate
 - For n inputs, a switching rate of n times the line rate is desirable.
- Output Port – Queueing
 - But if packets arrive faster than the transmission rate, they will queue in the output port buffer.
 - A scheduling discipline chooses among the queued datagrams whom is next for transmission.
- Output Port – Packet Loss
 - As we have previously discussed, queueing/buffers lead to packet loss
 - This can happen in both the output buffer and the input buffer (i.e. if the input buffer fills up because the switching fabric is slower than the input ports)

The input port of a router consists of a physical layer that receives the data stream from the network and a data link layer that strips the Ethernet header and forwards the remaining IP datagram to the network layer. Within the network layer, the input port acts as a decentralised switch by performing lookup, forwarding and queueing. The output port corresponding to the datagram destination is obtained from the local forwarding table. The goal is to complete input port processing at 'line

speed' (the input link speed). If datagrams arrive at a faster rate than the forwarding rate into the switch fabric, queueing occurs.

As there are billions of IP addresses, the forwarding table maps ranges of addresses (aggregate table entries) to output ports. The corresponding table entry for a given address is the one with the longest matching address prefix. Packets are transferred from the input buffer to the appropriate output buffer via a switching fabric. The switching rate is the rate at which packets can be transferred from input to output and is often expressed as a multiple of the input or output line rate. For N inputs, a switching rate of no more than N times the line rate is desirable.

Input port queueing can occur when the fabric is slower than the input ports combined. Packets may be lost due to buffer overflow. Head-of-line (HOL) blocking occurs when a datagram at the head of a queue prevents other datagrams from moving forward.

15.2.1 Longest Prefix Matching

When looking up a forwarding table entry for a given IP destination address, we use Longest Prefix Matching

○ e.g. IP destination = 11001000.00010111.00011000.00101000

<i>Destination IP Address</i>	<i>Output Port #</i>
11001000.00010111.00010000.00000000	2
11001000.00010111.00011000.00000000	<u>3</u>
11001000.00010111.00011001.00000000	2
otherwise	1

Figure 1. Longest Prefix Matching

15.3 Router Switching Fabrics

The three types of switching fabric are

1. memory,
2. bus, and
3. crossbar.

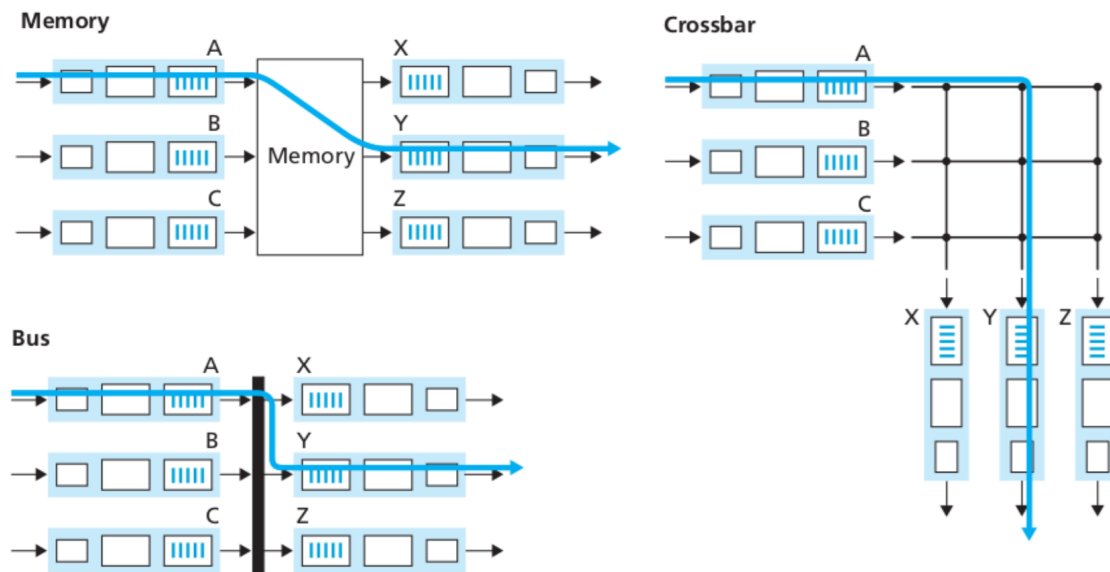


Figure 2. The three types of switching fabric.

15.3.1 Memory Switching Fabric

Memory switching fabrics were used by early routers and traditional computers with switching under direct control of the CPU. The packet is copied to system memory and then to the output. Speed is limited by memory bandwidth and two bus crossings are required for each datagram.

15.3.2 Bus Switching Fabric

Bus switching fabrics are used in routers for access and enterprise networks. A datagram is transferred from input port memory to output port memory via a shared bus without intervention by the routing processor. The input port prepends a switch-internal label (header) and transmits the packet onto the bus. Header matching occurs at the output ports to ensure only the relevant port receives the packet. Switching speed is limited by bus bandwidth. This may lead to bus contention.

15.3.3 Crossbar Switching Fabric

Crossbar switching fabrics are used in high-bandwidth interconnection network routers. To overcome bus bandwidth limitations, a crossbar with m inputs and n outputs requires $m + n$ buses. Crossbars support the forwarding of multiple packets in parallel since a crossbar switch is non-blocking. Crossbars require an advanced design that allows datagrams to be fragmented into cells of fixed length that are transmitted through multiple switching fabrics.

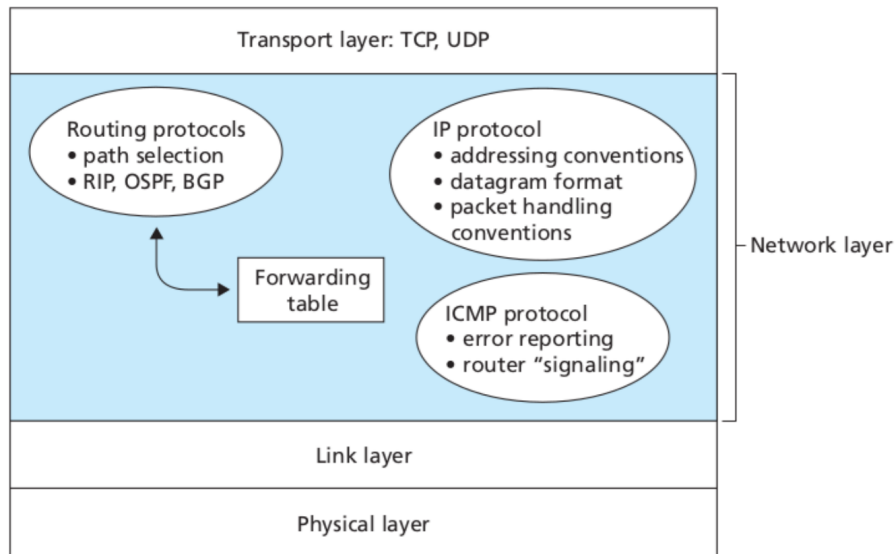
15.4 Router Output Ports

Buffering is required when datagrams arrive from the switching fabric faster than the transmission rate. A scheduling discipline chooses among the queued datagrams for transmission. Datagrams (packets) can be lost due to congestion and a lack of buffers.

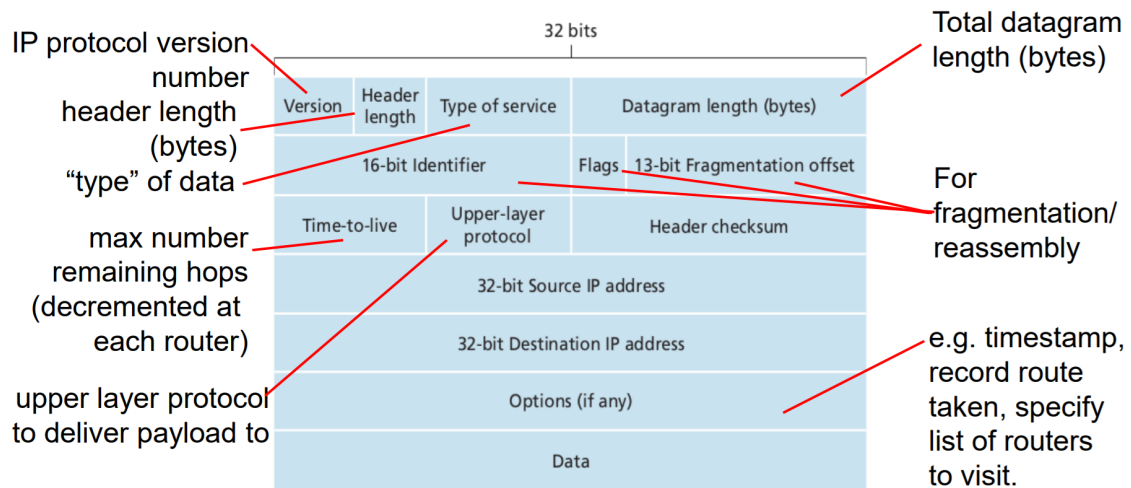
An old recommendation was that average buffering should be equal to the product of the typical RTT and the link capacity C . However, an updated recommendation is that the average buffering should be equal to the quotient of that value and the square root of the number N of TCP flows.

$$\frac{RTT \times C}{\sqrt{N}}$$

15.5 IP Datagram Fragmentation



Host and router network layer functions.



The IP datagram format.

The header of an IP datagram is 20 B long. The total overhead is the sum of 20 B of IP header, 20 B of TCP header and the application layer overhead.

Network links have a maximum transmission unit (MTU). This is the largest possible link-level frame. Different types of link along a path may have different MTUs. A large datagram is fragmented within

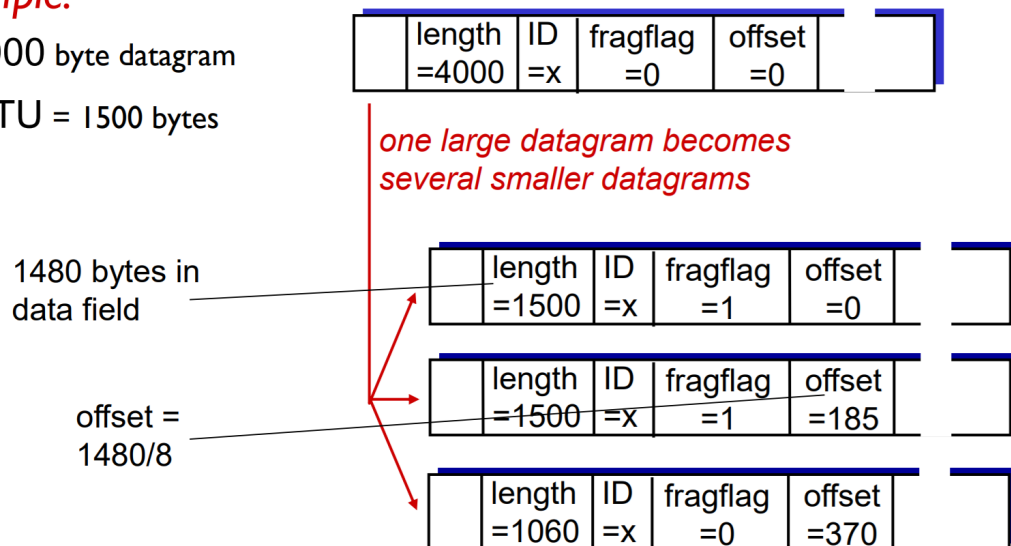
a network and reassembled at its destination. IP header bits are used to identify and order related fragments.

IP datagrams contain fields for their length, ID, fragmentation flag and offset. The fragments of an IP datagram have their fragmentation flags set to 1, except for the final fragment, which has a fragmentation flag of 0. The offset of a fragment is the quotient of the size of the data (the difference between the MTU size and the header size of 20 B) and the number 8.

Example:

◆ 4000 byte datagram

◆ MTU = 1500 bytes



IP datagram fragmentation.

15.6 IP Addressing

An IP address is a 32 bit identifier for the host and router interface. More specifically, they identify each individual host or router interface

An interface is a connection between the host or router and the physical link. A router typically has multiple interfaces. A host can have multiple interfaces for different connection devices, such as Ethernet and wireless. Each interface has its own IP address.

An IP address consists of a high-order subnet portion and a low-order host portion. A subnet is a group of interfaces with the same subnet portion of their IP addresses. Hosts in a subnet can communicate without intervention by the router.

15.6.1 Classless Interdomain Routing (CIDR)

Classless Interdomain Routing (CIDR) produces addresses of the format a.b.c.d/x, where x is the number of bits in the subnet portion of the address. This allows the subnet portion of the address to be an arbitrary length.

in the subnet part of the address

- Remember: 32 bits in total!

- So, host part has 32 - x bits

223.1.3.2
subnet part host part

A host can be given a fixed IP address by the system admin, or a dynamic IP address from a DHCP server.

15.7 Dynamic Host Configuration Protocol (DHCP)

The goal of the Dynamic Host Configuration Protocol (DHCP) is to allow a host to dynamically obtain its IP address from the network server when it joins. A host can renew its lease on the address in use, or the address can be reused for a different host that connects.

- The host broadcasts a DHCP discover message (optional).
- The DHCP server responds with a DHCP offer message (optional).
- The host requests an IP address with a DHCP request message.
- The DHCP server sends the address in a DHCP acknowledge message.

A DHCP server can also return the address of the first-hop router for the client, the name and IP address of the DNS server, and the network mask, which indicates the subnet and host portions of the address.

The subnet portion of an IP address is allocated as a portion of the address space of its ISP. ISPs are allocated address spaces by the Internet Corporation for Assigned Names and Numbers (ICANN), which also manages DNS and assigns domain names.

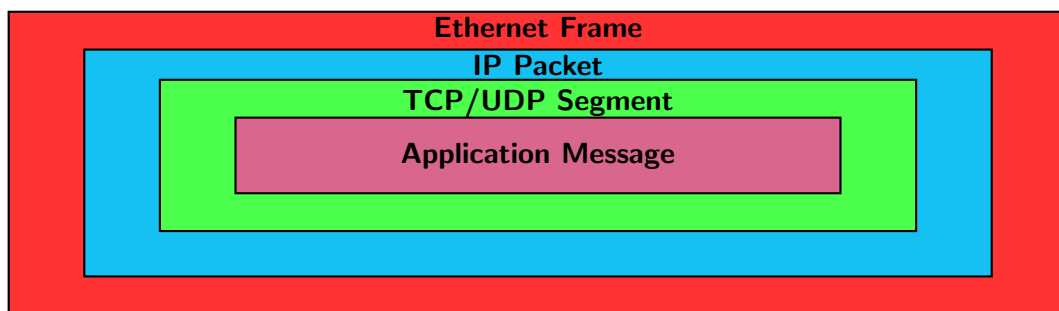
15.8 Network Address Translation (NAT)

Network Address Translation (NAT) ensures that all datagrams leaving a local network have the same single source NAT IP address, but different source port numbers. This allows a local network to use just one IP address as an interface to the Internet and to change its ISP without having to change the addresses of its local hosts. Additionally, IP addresses can be changed within the local network without affecting their addresses to the rest of the Internet, and devices within the local network are not explicitly addressable or visible to the outside world. Only one address is needed from the ISP for all hosts in the network.

A NAT router must replace the source IP address of every outgoing datagram with the NAT IP address and a new port number, store every translation pair in a NAT translation table, and replace the NAT IP address and port number of every incoming datagram with the corresponding source IP address and port number.

- As far as the outside world is concerned, all the hosts in a single local network use the same IP Address
 - So only one IP Address is needed from the ISP for all devices on the network
 - Local network devices can have their IP Addresses changed by the router without notifying the outside world
 - And we can even change ISPs without changing the addresses of devices in the local network
- Due to NAT, devices inside a local network are not explicitly addressable or even visible to the outside world – a security plus

NAT uses a 16 bit port number to allow many simultaneous connections with a single IP address. NAT is controversial since a router should only process up to the network layer. This violation of the end-to-end agreement must be taken into account by application designers. The address shortage could instead be solved by using IPv6 addresses.



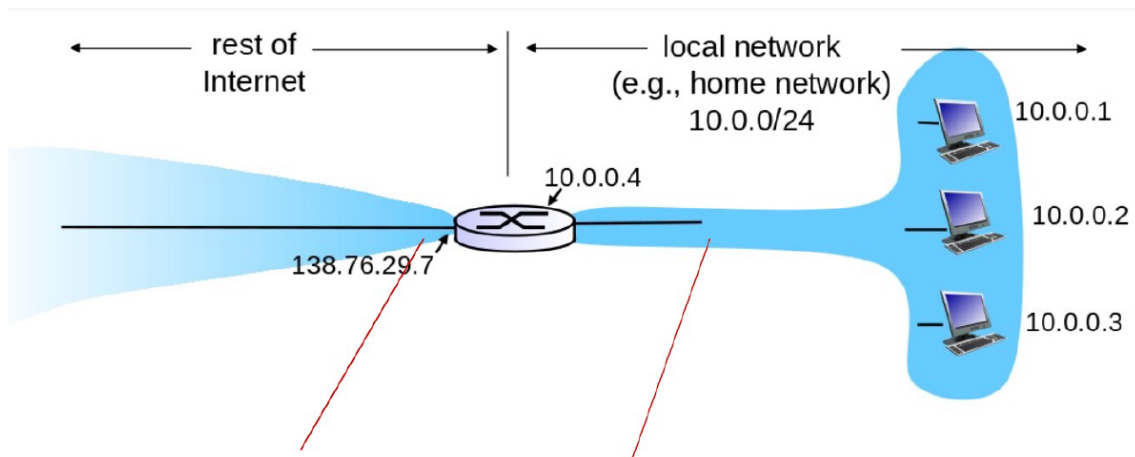
15.9 Network Layer Routing Algorithms

A graph $G(N, E)$ comprises a set of routers N and a set of links E between those routers. The cost of a link between routers x and y is expressed as $c(x, y)$. The cost of a link could always be 1, or could be inversely related to bandwidth or congestion. A routing algorithm finds the least-cost path between two routers.

15.10 Classification of Routing Algorithms

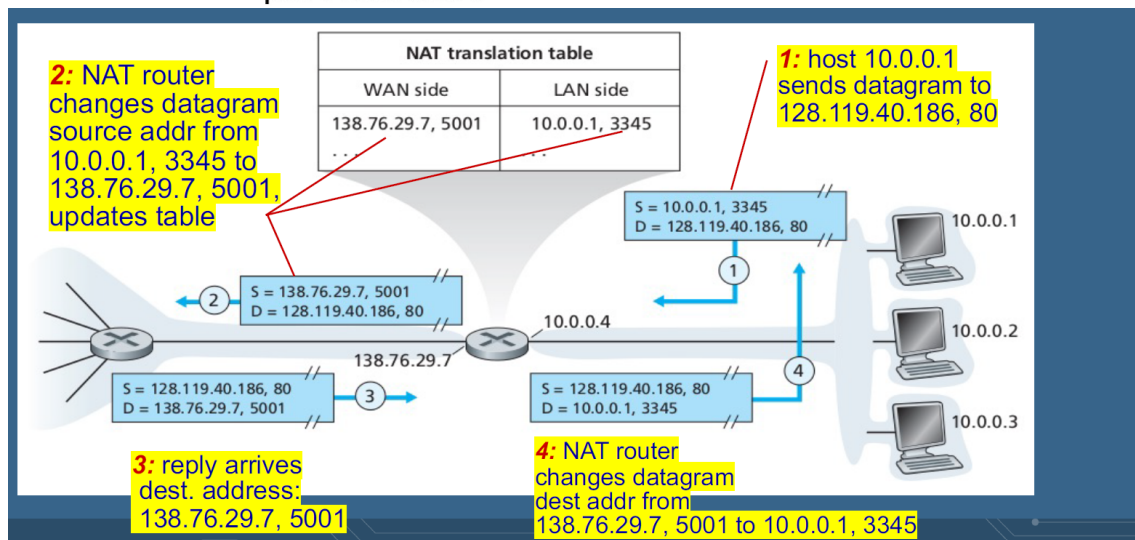
A routing algorithm in which topology and link cost information is shared globally between all routers is known as a 'link-state' algorithm. A decentralised routing algorithm in which a router knows only the link costs to its neighbours and communicates information only with its neighbours is known as a 'distance-vector' algorithm.

A routing algorithm in which routes change slowly over time is a static algorithm, whereas an algorithm in which routes change quickly with periodic updates and responses to link cost changes is a dynamic algorithm.

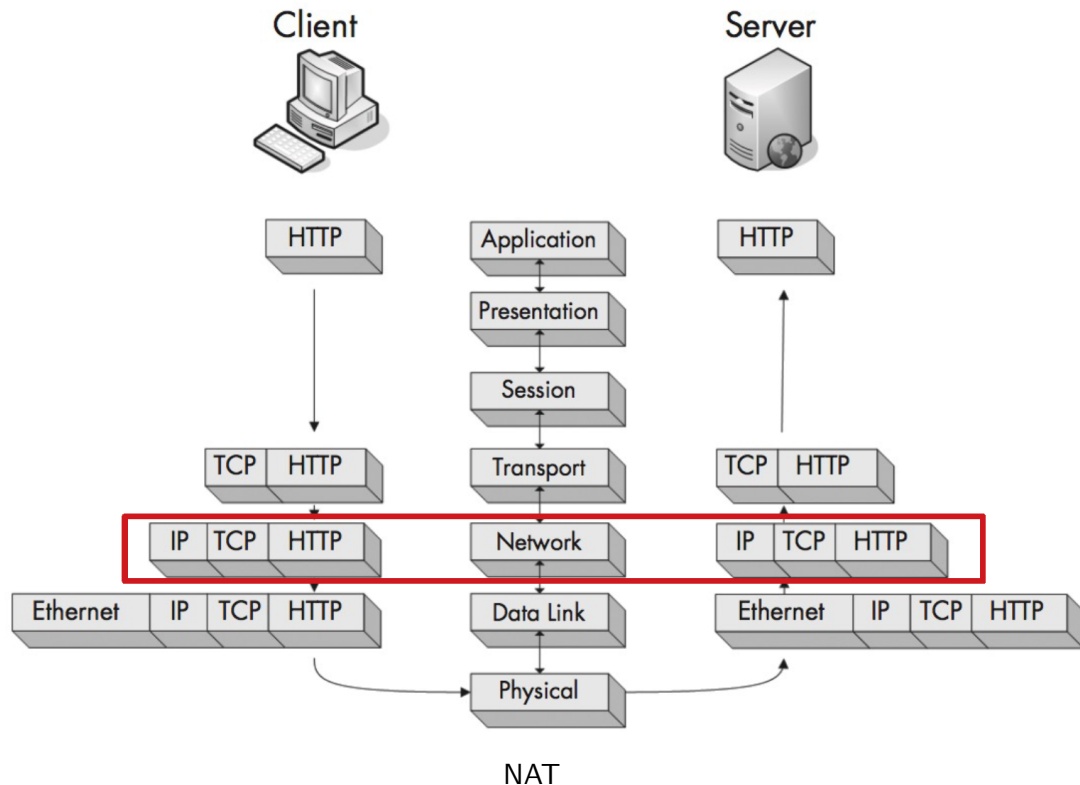


all datagrams *leaving* local network have *same* single source NAT IP address: 138.76.29.7, different source port numbers

Datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)



NAT



15.11 Link-State Routing (Dijkstra's Algorithm)

Dijkstra's algorithm is a link-state routing algorithm in which the network topology and link costs are known to all nodes. This is accomplished via link-state broadcast. All nodes have the same information. The least-cost paths from the source node to all other nodes is computed to provide a forwarding table for that node. The algorithm is iterative; after k iterations, the least-cost paths to k destinations are known.

- $c(x, y)$ — link cost from node x to node y , infinite cost if not direct neighbours
- $D(v)$ — current cost of path from source to node v
- $p(v)$ — predecessor of node v along path from source to node v
- N' — set of nodes whose least-cost paths are definitively known

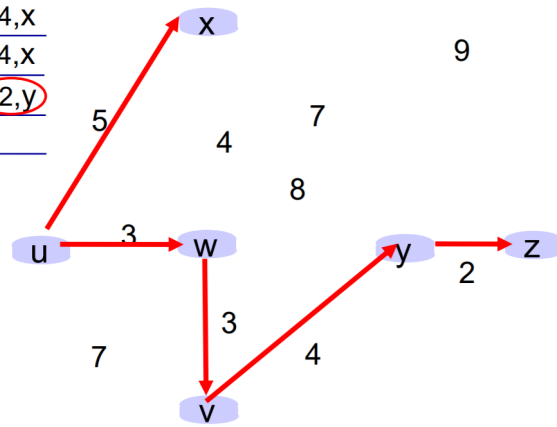
1. Add source node u to N'
2. For all nodes v , if v is adjacent to u , set $D(v) = c(u, v)$, else set $D(v) = \infty$
3. Find the node w not in N' with the least $D(w)$
4. Add node w to N'
5. Update $D(v)$ for all nodes v adjacent to w and not in N' such that $D(v) = \min(D(v), D(w) + c(w, v))$
6. Repeat from step 3 until all nodes are in N'

Each iteration requires all nodes not in N' to be checked. This is $n(n+1)$ comparisons. Thus, the algorithm has complexity class $O(n^2)$, although more efficient implementations are possible.

Step	N'	D(v) p(v)	D(w) p(w)	D(x) p(x)	D(y) p(y)	D(z) p(z)
0	u	7,u	3,u	5,u	∞	∞
1	uw	6,w		5,u	11,w	∞
2	uwx	6,w			11,w	14,x
3	uwxv				10,v	14,x
4	uwxvy					12,y
5	uwxvyz					

Notes:

- ◆ construct shortest path tree by tracing predecessor nodes
- ◆ ties can exist (can be broken arbitrarily)



Example of Dijkstra's algorithm.

In a network in which link cost is determined by traffic, Dijkstra's algorithm can result in oscillations. If all routers simultaneously direct all traffic along the least-cost paths, those paths will become the greatest-cost paths. The routers would then direct traffic along the next least-cost paths, continuing the cycle.

To solve this problem, routers could employ the algorithm at different times rather than simultaneously. However, routers will tend to synchronise over time. A better solution is not to base link cost solely on network traffic.

15.12 Distance-Vector Routing (Bellman-Ford Algorithm)

Distance-vector routing can be employed using the Bellman-Ford equation. The cost of the least-cost path from node x to node y is the minimum, taken over all neighbours v of x , of the sum of the cost to neighbour v and the cost from neighbour v to destination y .

$$d_x(y) = \min_v (c(x, v) + d_v(y))$$

The node achieving the minimum is the next hop in the shortest path used in the forwarding table.

The function $D_x(y)$ is an estimate of the least cost from x to y . Node x knows the cost to each neighbour v and maintains its distance vector $\mathbf{D}_x = D_x(y)$ for all y in N . It maintains a copy of the distance vectors of its neighbours. For each neighbour v , x maintains a distance vector $\mathbf{D}_v = D_v(y)$.

Periodically, each node sends its own distance vector estimate to its neighbours. When a node x receives a new estimate from its neighbour, it updates its own distance vector using the Bellman-Ford equation. Under natural conditions, the estimate $D_x(y)$ converges to the actual least cost $d_x(y)$.

The distance-vector algorithm is iterative and asynchronous. Each local iteration is caused by a local link cost change or a distance-vector update message from one of its neighbours. The node updates its routing info and recalculates its distance vector. It notifies its neighbours only when its distance-vector changes.

One drawback of the distance-vector algorithm is that an increase in link cost does not trigger a notification message. The network continues to operate as if cost had not increased.

Message Complexity

- ◆ **LS:** with N nodes, E links, $O(NE)$ messages sent
- ◆ **DV:** exchange between neighbours only
 - convergence time varies

Speed of Convergence

- ◆ **LS:** $O(N^2)$ algorithm requires $O(NE)$ messages
 - may have oscillations
- ◆ **DV:** convergence time varies
 - may be routing loops
 - count-to-infinity problem

Robustness: what happens if router malfunctions?

LS:

- node can advertise incorrect **link** cost
- each node computes only its own table

DV:

- DV node can advertise incorrect **path** cost
 - ▶ each node's table used by others
 - ▶ **error propagate through the network**



Comparison of link-state and distance-vector algorithms.

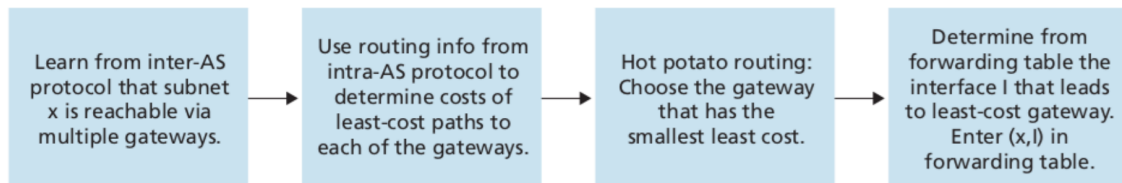
15.13 Hierarchical Routing

In reality, not all routers are identical and there are millions of destination address that cannot all be stored in routing tables. The exchange of so many routing tables would significantly harm the performance of the Internet. Additionally, the administrators of each network in the Internet may want to control routing within their own network.

Routers are aggregated into regions known as autonomous systems (AS). Routers within the same AS run the same routing protocol. This is known as the 'intra-AS' routing protocol. Routers in different autonomous systems can run different intra-AS routing protocols. A gateway router is a router at the edge of its AS that has a link to another AS.

The forwarding table of a router is configured by both intra-AS and inter-AS routing protocols. Entries for internal destinations are set by the intra-AS protocol. Entries for external destinations are set by both the intra-AS and inter-AS protocols.

It is the job of the inter-AS protocol to learn which destinations are reachable through each gateway router, and to propagate this information within its AS. If an external destination is reachable through more than one gateway, the inter-AS protocol may use "hot-potato routing" to send the packet to the closest gateway.



Multiple gateways and hot-potato routing.

16 Quiz Answer Shee

16.1 Topic 1: Data Representation – Exercise Sheet [Solutions]

Q#1: We have discussed Binary, Denary (aka Decimal), and Hexadecimal number systems in class. Complete the following table, doing the necessary conversions (without using a calculator and with a little-bit of self learning using the notes given below).

gray!20 Decimal	Binary	Octal	Hexadecimal
175	10101111	257	AF
204	11001100	314	CC
503	111110111	767	1F7
3995	111110011011	7633	F9B

Notes: The **octal number system** is a base-8 numeral system that uses digits from 0 to 7. It is commonly used in computing as a more compact way to represent binary numbers, with each octal digit corresponding to a group of three binary digits. Octal is less commonly used today but was historically useful in systems like UNIX file permissions and in early computing architectures. For example, the binary number 110010 can be grouped as 110 010 and represented as 62 in octal. For more details, please read Chapter #2, Section 2.2.4 of Forouzan.

Q#2: In a positional number system with base b , the largest integer number that can be represented using k digits is $b^k - 1$. Find the largest number in each of the following systems with **six** digits.

- (a) Binary $2^6 - 1 = 63$
- (b) Decimal $10^6 - 1 = 999999$
- (c) Hexadecimal $16^6 - 1 = 16777215$
- (d) Octal $8^6 - 1 = 262143$

Q#3: Without converting, find the minimum number of digits needed in the destination system for each of the following cases:

- a) Five-digit decimal number converted to binary
- b) Four-digit decimal converted to octal
- c) Seven-digit decimal converted to hexadecimal

- a) A five-digit decimal number can range between 00000-99999. We will consider 99999 as an example value to be represented in binary. Taking $\log_2 (99999) = \log_{10} (99999) / \log_{10} (2) = 16.61 = 17$ bits.
- b) A four-digit decimal number can range between 0000-9999. We will consider 9999 as an example value to be represented in octal. Taking $\log_8 (9999) = \log_{10} (9999) / \log_{10} (8) = 4.43 = 5$ digits.
- c) A seven-digit decimal number can range between 0000000-9999999. We will consider 9999999 as an example value to be represented in hexadecimal. Taking $\log_{16} (9999999) = \log_{10} (9999999) / \log_{10} (16) = 5.81 = 6$ digits.

Q#4: A company has decided to assign a unique bit pattern to each employee. If the company has 900 employees, what is the minimum number of bits needed to create this system of representation? How many patterns are unassigned? If the company hires another 300 employees, should it increase the number of bits? Explain your answer.

For 900 employees, the minimum number of bits needed will be: $\log_2 (900) = 9.81 = 10$ bits. Using 10 bits, we can represent $2^{10} = 1024$ different patterns. Assuming we don't use the first pattern (0000000000)₂ (as it doesn't look good to call someone as employee zero!) and start with (0000000001)₂, we will still have $1023 - 900 = 123$ available patterns. If the company hires 300 more employees, they will require $\log_2 (1200) = 10.23 = 11$ bits, hence they will need to increase the pattern size by 1 more bit.

Q#5: An audio signal is sampled 8000 times per second. Each sample is represented by 256 different levels. How many bits per second are needed to represent this signal?

For each sample, we have 256 different levels, therefore, we need $\log_2 (256) = 8$ bits / sample. Sampling rate is 8000 times/second, therefore, we need $8000 * 8 = 64000$ bits/second = 64kbps.

16.2 Topic 1: Data Representation – Exercise Sheet #2 [Solutions]

Q#1: Convert the following negative decimal integers into sign-and-magnitude and two's complement representations (using 16 bits):

Decimal Number	Sign-and-Magnitude Representation	Two's Complement Representation
-153	1000 0000 1001 1001	1111 1111 0110 0111
-421	1000 0001 1010 0101	1111 1110 0101 1011
-3276	1000 1100 1100 1100	1111 0011 0011 0100

-36317	1000 1101 1101 1101 (Out of range, represents -3549)	0111 0010 0010 0011 (Out of range, represents 29219)
--------	---	---

Q#2:

Convert the following binary numbers (using 16 bits) into decimal equivalent using sign-and-magnitude and two's complement interpretations:

Binary Representation	Decimal using Sign-and-Magnitude	Decimal using Two's Complement
0000 0000 1011 1101	189	189
1000 1100 1011 1001	-3257	-29511
1000 1111 0000 1010	-3850	-28918
1111 1111 1111 1111	-32767	-1

Q#3:

Calculate the range of numbers that could be represented using different representations:

No of Bits	Unsigned Numbers (Min to Max)	Sign-and-Magnitude (Min to Max)	Two's Complement (Min to Max)
8	0 to 255	-127 to +127	-128 to +127
16	0 to 65535	-32767 to +32767	-32768 to +32767
32	0 to 4294967295	-2147483647 to +2147483647	-2147483648 to +2147483647
64	0 to $1.844674407 \times 10^{19}$	$-9.223372037 \times 10^{18}$ to $+9.223372037 \times 10^{18}$	$-9.223372037 \times 10^{18}$ to $+9.223372037 \times 10^{18}$
128	0 to $3.402823669 \times 10^{38}$	$-1.701411835 \times 10^{38}$ to $+1.701411835 \times 10^{38}$	$-1.701411835 \times 10^{38}$ to $+1.701411835 \times 10^{38}$

Alpha	Beta	Gamma
0	2	4
1	3	5

Q#4:

What would be the result of adding a number to its one's complement? What would be the result of adding an integer to its two's complement?

When you add a number to its one's complement, the result is always a bit pattern where all the bits are set to 1. The value of this number will depend on how we interpret it. For example, when using sign-and-magnitude representation (11111111_2) will be equal to $(-127)_{10}$, and in two's complement representation it will be equal to $(-1)_{10}$. In general, for an n -bit number, the result of adding a number to its one's complement will be $2^n - 1$.

When you add an integer to its two's complement, the result is always zero. When you add the number x and its two's complement, the binary result overflows, leaving a sum of zero. This is because the two's complement is effectively the negative representation of the original number in the binary system.

Q#5:

Normalize the following binary floating-point numbers. Explicitly show the value of the exponent after normalization.

a) 1100.0111 b) 10111.0011×2^{-5} c) 0.0000010101 d) 1101.001101×2^7

The normalized representations are given below:

- a) 1.1000111×2^3
- b) 1.01110011×2^1
- c) 1.0101×2^6
- d) $1.101001101 \times 2^{10}$

Q#6:

Convert the following decimal real numbers into fixed point representation, both in sign-and-magnitude and two's complement representations (using 16 bits). For fixed point sign-and-magnitude, use 10 bits (including sign bit) for the integer part and 6 bits for the fractional part.

Decimal Number	Fixed Point Sign-and-Magnitude Representation (10.6 format)	Fixed Point Two's Complement Representation (10.6 format)
11.40625	0000001011.011010	0000001011.011010
141.8125	0010001101.110100	0010001101.110100
-412.21875	1110011100.001110	1001100011.110010
-465.09375	1111010001.000110	1000101110.111010

Note: For 2's complement representation of a negative real number, we initially assume the number to be positive, convert to binary in the same way as fixed point conversion and apply two's complement on the binary representation to get the final answer.

Q#9:

Given the following numbers in IEEE 754 (32-bit) floating point format with 8-bit exponent, calculate the decimal real numbers.

IEEE 754 Floating Point Representation (32-bit)	Decimal Number
---	----------------

0 1000 0111 101 1100 1100 0000 0000 0000	Sign bit: 0 (+ve) Biased Exponent: $(10000111)_2 = 135$ Real Exponent: $135 - 127 = 8$ Stored Mantissa: $(.101110011000000000000000)_2$ Actual Mantissa: $(1.101110011000000000000000)_2$ Normalized representation: 1.101110011×2^8 Binary number: 11011100.11 Decimal value: 441.5
0 0111 1000 011 0000 0000 0000 0000 0000	Sign bit: 0 (+ve) Biased Exponent: $(01111000)_2 = 120$ Real Exponent: $120 - 127 = -7$ Stored Mantissa: $(.011000000000000000000000)_2$ Actual Mantissa: $(1.011000000000000000000000)_2$ Normalized representation: 1.011×2^{-7} Binary number: 0.0000001011 Decimal value: 0.010742188
1 1000 1010 101 0101 0000 0100 0000 0000	Sign bit: 1 (-ve) Biased Exponent: $(10001010)_2 = 138$ Real Exponent: $138 - 127 = 11$ Stored Mantissa: $(.101010100000100000000000)_2$ Actual Mantissa: $(1.101010100000100000000000)_2$ Normalized representation: $1.1010101000001 \times 2^{11}$ Binary number: 11010101000001.00 Decimal value: -3408.25
1 0111 1010 100 0000 0000 0000 0000 0000	Sign bit: 1 (-ve) Biased Exponent: $(01111010)_2 = 122$ Real Exponent: $122 - 127 = -5$ Stored Mantissa: $(.100000000000000000000000)_2$ Actual Mantissa: $(1.100000000000000000000000)_2$ Normalized representation: 1.1×2^{-5} Binary number: 0.000011 Decimal value: -0.046875

17 The Java Virtual Machine and Java Bytecode

17.1 Execution on the Java Virtual Machine

The javac compiler compiles Java source code in .java files to intermediate-level Java bytecode in .class files. As long as there is a JVM on another platform, the bytecode can be run on that platform without recompilation. This feature is known as 'portability'.

The JRE loads Java bytecode files, verifies their internal consistency and executes their methods using the JVM. Different CPUs or operating systems can run the same Java bytecode but use different JREs.

17.2 Stack Frames

Every time a method is called in Java, a stack frame is constructed for it. Each frame has

- a reference to the frame of its calling method,
- space for local variables,
- space for an operand stack
- a PC and SP (for the JVM, not the CPU), and
- other items.

17.2.1 Variable and Operand Storage

All local variables and operand stack entries are stored as 4 bytes. The `long` and `double` data types, which need 8 bytes, are stored as two consecutive entries.

Storage of data types.

Data type	Size		Slots
	/ bit	/ B	
<code>bool</code>	1		1
<code>byte</code>		1	1
<code>char</code>		2	1
<code>short</code>		2	1
<code>int</code>		4	1
<code>float</code>		4	1
<code>reference</code>		4	1
<code>long</code>		8	2
<code>double</code>		8	2

17.2.2 Local Variables

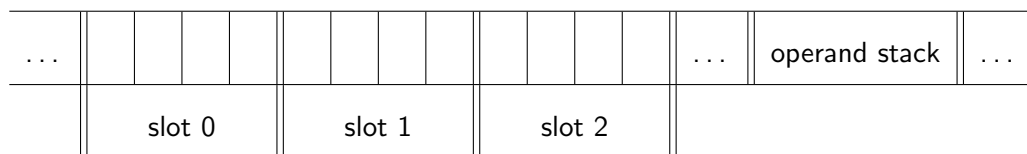
In the context of Java bytecode, local variables include:

1. `this` reference (for non-static methods only)
 - a reference to the object on which the method was called
 - stored as local variable in slot 0
2. Parameters of the method
 - stored from slot 0 onwards for static methods
 - stored from slot 1 onwards for non-static methods
3. Variables declared inside the method
 - stored in slots immediately after parameters

Instance variables and class (static) variables are not stored as local variables since they are defined outside of methods.

17.2.3 Slot Numbers

Java bytecode does not refer to local variables by their identifiers in the source code. Instead, it refers to them by their slot numbers. A local variable of the `long` or `double` data types takes the number of the first of its pair of slots.



Local variable slots in the stack frame.

17.2.4 Stack Overflow

The operand stack can never overflow. The Java compiler calculates exactly how much space is required and the loader checks each method to verify that no operand stack underflow or overflow is possible. Operand stack overflow is different from stack overflow, which occurs when a new frame is needed, but there is not sufficient memory.

17.3 Java Bytecode Instructions

Each Java bytecode instruction takes up at least one byte (for its opcode). It may also have one or more operand bytes. For each opcode, there is a corresponding mnemonic.

A single operand may consist of multiple bytes. In this case, the operand is stored with its most significant byte first (in the smaller address). This order is known as 'big endian'.

17.3.1 Arithmetic

Each arithmetic instruction has a prefix letter that denotes the data type on which it operates. For example, the `iadd` instruction pops two entries from the operand stack, adds them as if they are of the type `int`, and pushes them back onto the stack.

Mnemonic prefixes.

Data type	Prefix	Addition
byte	b	
short	s	
int	i	<code>iadd</code>
long	l	<code>ladd</code>
float	f	<code>fadd</code>
double	d	<code>dadd</code>
reference	a	

Errors are likely to occur if an instruction is used on the wrong data type. For example, calling `fadd` on `int` entries will result in the wrong answer since the instruction expects entries in floating point representation. If there is only one operand on the stack and the instruction expects two, operand stack underflow will occur. This can occur when `ladd` is called on `int` entries.

No checks are performed during execution. The JRE verifies the bytecode when the class is loaded. This includes checking that types are used consistently.

17.3.2 Pushing

The `bipush <1_byte_operand>` instruction has two prefixes: `b` for `byte` and `i` for `int`. This instruction sign extends its byte operand (1 B) to an `int` value (4 B) and pushes the result to the operand stack. Similarly, there exists an `sipush <2_byte_operand>` instruction to push a `short` to the operand stack as an `int`.

There exist seven instructions for pushing common integer constants to the operand stack that do not require an operand. Instructions that take no operand help to save space in the bytecode.

Load instructions of the form `iload <slot_number>` push local variables (indicated by their slot numbers) onto the operand stack. There are special load instructions that take no operand for pushing values stored in common slots to the operand stack (`iload_0`, `iload_1`, `iload_2`, `iload_3`).

Instructions for pushing common integer constants.

Instruction	Value pushed
<code>iconst_m1</code>	-1
<code>iconst_0</code>	0
<code>iconst_1</code>	1
<code>iconst_2</code>	2
...	...
<code>iconst_5</code>	5

17.3.3 Popping

Store instructions pop entries from the operand stack into local variable slots. For example, `istore <slot_number>` or `istore_2`.

17.3.4 Jumps

In Java bytecode, the source and destination of a jump must be within the same method. It is not possible to jump to another method.

Unconditional jumps are achieved with the `goto <2_byte_offset>` instruction. The operand is a 2 B offset that is added to the address of the instruction to calculate the address of the next instruction to execute. There exists a version of this instruction that takes a 4 B operand to allow for larger jumps: `goto_w <4_byte_offset>`.

Conditional jumps are achieved with instructions of the form `ifeq <offset>` and `if_icmpeq <offset>`.

17.4 Call and Return

17.4.1 The Stack

If method A calls a static method B(p0, p1) that returns a result:

1. method A calculates the arguments on its operand stack,
2. a stack frame is constructed for method B,
 - the arguments are popped from the operand stack of method A,
 - the values are used to initialise the local variables p0 and p1 of method B,
3. method B calculates the result on its stack frame
4. the result is pushed to the operand stack of method A, and

5. execution returns to method A, discarding the stack frame of method B.

On the operand stack of method A, this has the effect of

$$\dots, p0, p1 \rightarrow \dots, \text{result}$$

Each frame has its own PC. While method B is under execution, its PC is used. When execution returns to method A, it resumes with its old PC value. The local variables of method A are unchanged by method B.

Linked frames have the effect of a return stack. A chain of linked frames in the JVM is officially known as a stack.

17.4.2 Runtime Constant Pool

The class variables, instance variables and methods of a class may be used by other classes. Since classes are compiled separately, the compiler does not know the address of these items in other classes. Hence, the class, source file and bytecode file must share the same name to create a 'symbolic reference'.

In addition to class variables, instance variables and method definitions, each class in Java has a runtime constant pool that contains read-only constants used in the class. This includes literal constants, symbolic references to methods and their classes, types and more.

Each pool entry has an index. Each stack frame includes a pointer to the constant pool of its class in its space for 'other items'.

17.4.3 Static Method Calls

Static methods are called using the `invokestatic <2.byte_index>` instruction. The operand is an index in the runtime constant pool of the class of the current method. The indexed constant pool entry is a symbolic reference to the requested class and method.

The JVM uses this information to find the address of the class, the size needed for the new frame, and the address of the bytecode for the method.

17.4.4 Method Returns

The return instruction for void methods is `return`. This has no effect on the operand stack of the caller.

For methods that return a result, the instruction is of the form `ireturn`. This pops the result from the operand stack of the current frame and pushes it to the operand stack of the caller.

After either of these return instructions, the current frame is discarded and execution continues on the frame of the caller, resuming from its PC.

17.5 Java Disassembler

The Java 'disassembler' (javap) converts Java bytecode to mnemonics with the `-c` option. For example, `MyClass.class` can be viewed with `javap -c MyClass`. Jump instruction operands in disassembler output show the destination instruction address rather than the offset used in the bytecode.

18 Deadlocks

18.1 Reusable and Consumable Resources

A reusable resource is used by only one process at a time and is not depleted by that use. Reusable resources include processors, I/O channels, main and secondary memory, devices, and data structures such as files, databases and semaphores. Processes obtain reusable resources and later release them for use by other processes. A deadlock may occur if each process holds one such resource and requests another.

A consumable resource is created (or produced) and later destroyed (or consumed). Consumable resources include interrupts, signals, messages and information in I/O buffers. A deadlock may occur if two processes request information from each other at the same time. Neither can send the required information because both are waiting for a response.

18.2 Deadlock Conditions

A deadlock is a permanent blocking of a set of processes competing for resources. It is a circular resource conflict between a set of processes that occurs when each process currently holds a resource requested by another process and, therefore, each process is also waiting for another process to release its resource.

There exist four conditions that lead to a deadlock. The first three are preconditions that, when all hold true, may lead to a deadlock. If the fourth condition occurs in addition to the first three, it is inevitable that a deadlock shall occur.

1. Mutual exclusion — only one process may use a resource at a time
2. Hold-and-wait — a process can hold a resource while it is waiting for other resources to become available
3. No preemption — processes cannot be interrupted to free their resources by force
4. Circular wait — processes wait for each other to release resources they wish to acquire, leading to a closed chain of processes

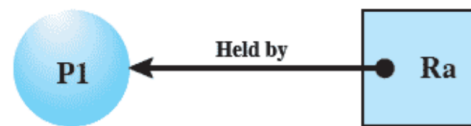
A deadlock occurs when a circular wait occurs and cannot be resolved. A circular wait cannot be resolved if the three preconditions are true. Thus, if the first three conditions occur, there is a possibility of a deadlock, and if all four conditions occur there is an inevitable deadlock.

18.3 Resource Allocation Graphs

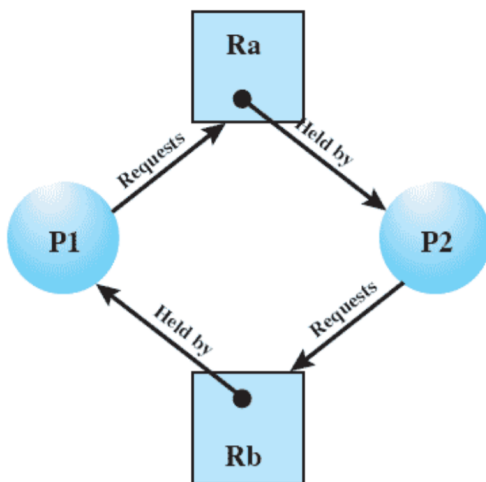
A resource allocation graph is a directed graph that depicts the state of a system of resources and processes.



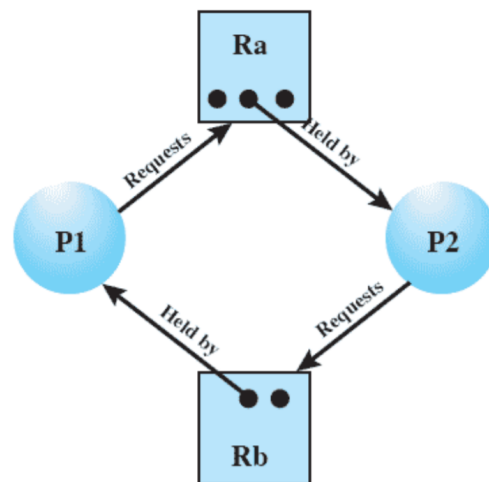
(a) Resource is requested



(b) Resource is held



(c) Circular wait



(d) No deadlock

18.4 Deadlock Handling

Deadlocks can be handled through three methods.

- Deadlock prevention
 - Avoid at least one of the four deadlock conditions
- Deadlock avoidance
 - Requests for resources whose reservation may lead to deadlock are not granted
- Deadlock detection and recovery
 - No restriction on resource allocation
 - Periodic check for deadlocks
 - When a deadlock is detected, recovery mechanisms are employed

18.5 Deadlock Prevention

In a deadlock prevention system, the OS is designed to prevent deadlocks from occurring. The avoidance of the three preconditions is known as 'indirect deadlock prevention'. The avoidance of the final condition is known as 'direct deadlock prevention'.

18.5.1 Mutual Exclusion Prevention

The mutual exclusion precondition cannot be avoided, as a lack of mutual exclusion may lead to race conditions.

18.5.2 Hold-and-Wait Prevention

The hold-and-wait precondition may be avoided by preallocating all required resources in advance and blocking a process until all its resources become available. This results in long delays for processes, as well as low concurrency and inefficient utilisation of resources. It is also difficult to achieve since a process may not know in advance what resources it may require.

18.5.3 No Preemption Prevention

The avoidance of this condition may be achieved in different ways. If a process is holding a resource and is denied a request for another resource, it may be preempted. Alternatively, the process holding the other resource may be preempted. These methods are only possible if the state of the resources can be saved and restored so that the preempted process may resume execution.

18.5.4 Circular Wait Prevention

One method of direct deadlock prevention is to associate an index to each resource. A process that requires multiple resources must acquire them in ascending index order. For example, if two processes are competing for resource 1 and resource 2, the first process will acquire resource 1 and is free to acquire resource 2, since the second process cannot acquire either; the second process cannot acquire resource 2 before resource 1, and cannot access resource 1 because it is locked. This method of resource ordering to prevent circular waits is usually inefficient because processes that use common resources are forced to execute in series rather than parallel.

18.6 Deadlock Avoidance

In a deadlock avoidance system, the decision of whether to grant a resource allocation is made dynamically depending on whether it is at all possible that such a request may lead to a deadlock. This method requires knowledge of future resource requests. The three preconditions are not avoided. Instead, intelligent decisions are made when allocating resources.

- Process initiation denial — do not start a process if its demands may lead to deadlock
- Resource allocation denial — do not grant an incremental resource request to a process if the allocation may lead to deadlock

18.6.1 The Resource Matrix System

Resources and process that require them can be represented using vectors and matrices. The system resource vector $\mathbf{r}$ is a vector of the total number of instances of each resource in the system.

$$\mathbf{r} = [r_1, r_2, \dots, r_n]$$

The available resource vector $\mathbf{v}$ is a vector of all available resources (resources that are not in use).

$$\mathbf{v} = [v_1, v_2, \dots, v_n]$$

The claim matrix $\mathbf{C}$ represents the resource requirement of each process. An element c_{pr} is the number of instances of resource r required by process p . Each row $\mathbf{c}_p$ represents the total resource claim of process p .

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

The allocation matrix $\mathbf{A}$ represents the resource allocation of each process. An element a_{pr} is the number of instances of resource r allocated to process p . Each row $\mathbf{a}_p$ represents the total resource allocation of process p .

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

All the resources in the system are either available or allocated.

$$r_r = v_r + \sum_p a_{pr} \quad \forall r$$

No process may claim more instances of a resource than exist in the system.

$$c_{pr} \leq r_r \quad \forall p, r$$

No process is allocated more instances of a resource than it claims.

$$a_{pr} \leq c_{pr} \quad \forall p, r$$

18.6.2 Process Initiation Denial

A new process ($n + 1$) is initiated only if the sum of claims of the new process and all the currently running processes can be met by the system.

$$r_r \geq c_{(n+1)r} + \sum_{p=1}^n c_{pr} \quad \forall r$$

This deadlock avoidance scheme assumes the worst possible scenario — that all processes will demand their resources at the same time.

18.6.3 Resource Allocation Denial

Resource allocation denial is achieved through the 'banker's algorithm'. A process is only allocated a resource if the system will remain in a safe state — a state in which there exists at least one sequence of resource allocations that does not result in a deadlock.

A system is in a safe state if any of the participating processes can run to completion with the resources available, i.e. the difference between the requirement and allocation of any process can be met with the resources available.

$$c_{pr} - a_{pr} \leq v_r \quad \forall r$$

Deadlock avoidance is less restrictive than deadlock prevention since resources can be allocated out of order. However, the maximum resource requirement of all processes must be known in advance, processes under consideration must be independent (without synchronisation), the number of instances of resources in the system must remain constant, and no processes may exit whilst holding resources.

18.7 Deadlock Detection and Recovery

The algorithm for deadlock detection is as follows.

1. Unmark (assume deadlocked, mark with T) all processes and define a matrix $\mathbf{Q}$ that represents the future requests of each process.
2. Mark (assume complete, mark with F) all processes that have zero resources allocated to them. These processes have reached completion.
3. Initialise a temporary vector $\mathbf{w}$ equal to the available resource vector $\mathbf{v}$.
4. Find a process p that is unmarked (assumed deadlocked, marked with T) such that the future requests of the process can be met by the currently available resources ($q_{pr} \leq w_r \quad \forall r$).
5. If such a process is found, update each resource r in the currently available resources $\mathbf{w}$ as if the resources allocated to the process have been freed ($w_r \mapsto w_r + a_{pr} \quad \forall r$), and mark (assume complete, mark with F) process p . Repeat from step 4.
6. If no such process is found, continue to step 7.
7. All unmarked (marked with T) processes are deadlocked processes.

If a deadlock is detected, there are several methods to recover from it.

- Abort all deadlocked processes.
- Roll back each deadlocked process to a previous state and resume.
- Abort deadlocked processes one by one until the deadlock disappears.
- Release the resources by force via preemption until the deadlock disappears.

The order of preemption or abortion should be based on some criteria, such as increasing order of resource usage.

18.8 Integrated Deadlock Strategy

An integrated deadlock strategy is a combination of deadlock handling approaches. Resources can be grouped into resources classes. For example,

- swappable space — blocks of memory on secondary storage used for swapping processes,
- process resources — assignable devices or files, and
- main memory — assignable to processes in pages or segments.

Different deadlock handling strategies can be applied to each resource class.

- Swappable space — deadlock prevention (all required resources are allocated at once, maximum storage requirements are known)
- Process resources — deadlock avoidance (processes declare ahead of time the resources they will require) or deadlock prevention (resource ordering)
- Main memory — deadlock prevention by preemption

Summary of deadlock handling approaches.

Approach	Resource Allocation Policy	Different Schemes	Major Advantages	Major Disadvantages
Prevention	Conservative; undercommits resources	Requesting all resources at once	• Works well for processes that perform a single burst of activity • No preemption necessary	• Inefficient • Delays process initiation • Future resource requirements must be known by processes
		Preemption	• Convenient when applied to resources whose state can be saved and restored easily	• Preempts more often than necessary
		Resource ordering	• Feasible to enforce via compile-time checks • Needs no run-time computation since problem is solved in system design	• Disallows incremental resource requests
Avoidance	Midway between that of detection and prevention	Manipulate to find at least one safe path	• No preemption necessary	• Future resource requirements must be known by OS • Processes can be blocked for long periods
Detection	Very liberal; requested resources are granted where possible	Invoke periodically to test for deadlock	• Never delays process initiation • Facilitates online handling	• Inherent preemption losses

19 Network Security

19.1 Principles of Cryptography

- Confidentiality — only the sender and intended recipient should be able to understand the message content
- End-point authentication — the sender and receiver wish to confirm the identity of the other
- Message integrity — the sender and receiver wish to ensure the message is not altered without detection
- Access and availability — services must be accessible and available to users

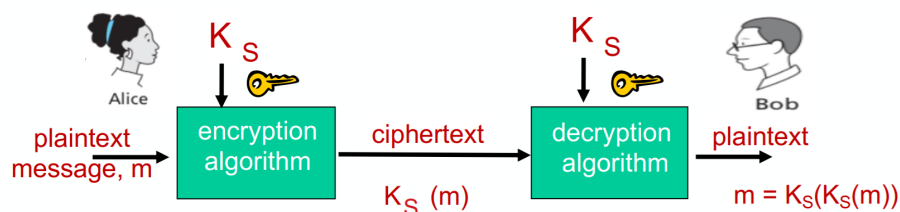
The sender and receiver may be users, applications, servers or routers. Malicious parties may attempt to eavesdrop (intercept messages), actively insert messages into the connection (man-in-the-middle attack), impersonate (fake or spoof the source address or any other field in a packet), hijack an ongoing connection by replacing the sender or receiver, or deny service (prevent it from being used by others by overloading resources, for example).

A plaintext message m is encrypted with a key K_A to produce a ciphertext $K_A(m)$. The ciphertext is decrypted with the key K_B to reproduce the plaintext message $m = K_B(K_A(m))$.

The two main types of encryption schemes are

- symmetric key systems — sender and receiver keys are identical and secret, and
- public key systems — one key is known to the whole world and the other key is known only to the sender or receiver (but not both).

19.2 Symmetric Key Cryptography



Symmetric key cryptography.

With symmetric key cryptography, Alice and Bob share the same symmetric key. The key may be a mapping for a substitution cipher. A monoalphabetic cipher is used to substitute each letter for another.

An example of symmetric key cryptography is the Data Encryption Standard (DES), which uses a 56 bit symmetric key and acts on 64 bit plaintext input. There is no known good analytic attack, but the key can be cracked through brute force typically in under a day. DES can be made more secure using 3DES (three DES encryptions using three different keys).

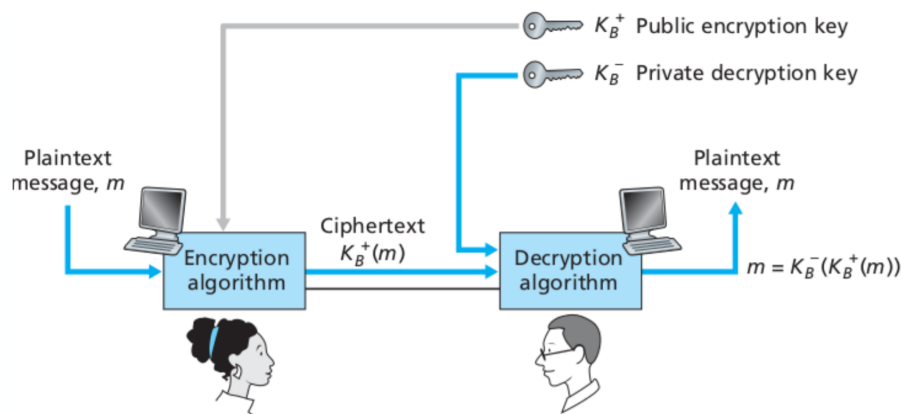
The DES encryption operations consists of an initial permutation followed by 16 rounds of function application that each use a different 48 bit of the key, and a final permutation.

DES was replaced by the Advanced Encryption Standard (AES), which processes data in 128 bit blocks and uses 128 bit, 192 bit or 256 bit keys. A brute force decryption taking a maximum of one second on a DES encryption would take a maximum of 149 trillion years on an AES encryption.

19.3 Public Key Cryptography

Symmetric key encryption requires the sender and receiver to know a shared secret key. This can be difficult to achieve, particularly if the sender and receiver never meet.

Public key encryption does not require the sender and receiver to share a secret key. The public encryption key is known to all, but the private decryption key is known only to the receiver.



Public key cryptography.

Public key encryption requires an encryption key K_B^+ and a decryption key K_B^- such that $K_B^-(K_B^+(m)) = m$. Given a public key K_B^+ , it should be impossible to compute the private key K_B^- .

19.3.1 Rivest, Shamir, Adelson (RSA) Encryption

An example of public key encryption is the Rivest, Shamir, Adelson (RSA) algorithm, which uses modular arithmetic.

$$((a \bmod n) + (b \bmod n)) \bmod n = (a + b) \bmod n$$

$$((a \bmod n) - (b \bmod n)) \bmod n = (a - b) \bmod n$$

$$((a \bmod n) \times (b \bmod n)) \bmod n = (a \times b) \bmod n$$

Thus, a similar identity can be constructed for $(a \bmod n)^d$.

$$(a \bmod n)^d \bmod n = a^d \bmod n$$

A message is simply a bit pattern that can be uniquely represented by an integer. Thus, the message m can be encrypted by encrypting its corresponding integer. The resulting integer is the ciphertext.

1. choose two large prime numbers p, q .
(e.g., 1024 bits each)
2. compute $n = pq$, $z = (p-1)(q-1)$
3. choose e (with $e < n$) that has no common factors with z (e, z are “relatively prime”).
4. choose d such that $ed - 1$ is exactly divisible by z .
(in other words: $ed \bmod z = 1$).
5. public key is (n, e) . private key is (n, d) .

The RSA public and private key pair creation algorithm.

To encrypt the message m to ciphertext c , compute $c = m^e \bmod n$. To decrypt the ciphertext c , compute $m = c^d \bmod n$.

$$m = (m^e \bmod n)^d \bmod n$$

- ◆ Must show that $c^d \bmod n = m$
where $c = m^e \bmod n$
- ◆ Fact: for any x and y : $x^y \bmod n = x^{(y \bmod z)} \bmod n$
 - where $n = pq$ and $z = (p-1)(q-1)$
- ◆ Thus,

$$\begin{aligned}
 c^d \bmod n &= (m^e \bmod n)^d \bmod n \\
 &= m^{ed} \bmod n \\
 &= m^{(ed \bmod z)} \bmod n \\
 &= m^1 \bmod n \\
 &= m
 \end{aligned}$$

Proof of the RSA algorithm.

RSA is secure since it is impossible to determine d of the private key from n and e of the public key. This would require finding factors of n without knowing the two factors p and q . Factoring a large number is very difficult.

The following property is *very* useful:

$$\underbrace{K_B^-(K_B^+(m))}_{\text{use public key first, followed by private key}} = m = \underbrace{K_B^+(K_B^-(m))}_{\text{use private key first, followed by public key}}$$

use public key first,
followed by
private key

use private key
first, followed by
public key

Result is the Same!

Associativity of RSA encryption.

Why $K_B^-(K_B^+(m)) = m = K_B^+(K_B^-(m))$?

Follows directly from modular arithmetic:

$$\begin{aligned} (m^e \bmod n)^d \bmod n &= m^{ed} \bmod n \\ &= m^{de} \bmod n \\ &= (m^d \bmod n)^e \bmod n \end{aligned}$$

Proof of the associativity of RSA encryption.

19.3.2 RSA Session Keys

The exponentiation required by RSA is computationally intensive. Thus, RSA is significantly slower than DES. Public key cryptography, such as RSA, is used to establish a secure connection, through which a symmetric session key is established for encrypting data. Once the sender and receiver have the symmetric key, it is used for communication.